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The OECD Start-ups
Database: A new lens on
the global entrepreneurial
ecosystems

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The OECD Start-ups Database

A new lens on the global entrepreneurial ecosystems

Marius Berger¹, Sara Calligaris¹, Antoine Dechezleprêtre¹, Hélène Dernis¹, Andrea Greppi¹, Dmitri Kirpichev¹, Alfonso Muñoz Alvarado¹

This paper introduces the OECD Start-ups Database, a new global micro-level dataset tracking nearly 4.5 million start-ups founded between 2000 and 2025. Integrating proprietary and public sources, it offers a rich and standardised view of entrepreneurial activity worldwide, with detailed information on financing rounds, intellectual property, founding teams, financial indicators, and exit events. The paper documents the database construction process, methodological choices, and validation procedures, and then presents stylised facts and cross-country trends in start-up activity. The OECD Start-ups Database is benchmarked against national and international datasets to assess coverage and reliability, promoting transparency on its scope, strengths, and limitations.

¹ OECD Directorate for Science, Technology and Innovation

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Executive summary

Start-up companies – although typically small – play a disproportionately large role in driving innovation, output growth, and employment. Despite their economic importance, studying these companies presents persistent challenges, most notably related to limitations in data availability. Their initially small scale, private ownership, and often short lifespan hinder systematic data collection and complicate cross-country and cross-sectoral comparisons. In addition, start-up dynamics do not occur in isolation but rather depend on an ecosystem that includes several different actors. Therefore, understanding start-ups more comprehensively requires data on these other actors, which further expands existing data gaps.

This paper introduces the OECD Start-ups Database, which is aimed at addressing these measurement challenges and better informing policymaking. This novel resource integrates two of the most comprehensive global datasets on start-up activity: Crunchbase and Dealroom. These platforms compile and integrate information from a range of sources – including user-generated content, public records, and contributions from ecosystem partners – to capture a rich and dynamic picture of the entrepreneurial ecosystem, including start-up companies, investors, founders, universities, and other relevant entities.

Unlike traditionally used databases, which largely focus on venture capital-backed firms in the U.S., the combination of Crunchbase and Dealroom captures a wider set of start-ups. It includes start-ups located in locations outside the U.S. and start-ups relying on alternative forms of funding, allowing for a more accurate and inclusive analysis across countries and industries. Likewise, it also captures information on other relevant players in entrepreneurial ecosystems. The OECD Start-ups Database is further enriched with data on patent and trademark filings (from the OECD STI Microdata Lab infrastructure), financial indicators (Orbis), exit transactions (Orbis M&A), and public and corporate venture capital (OECD survey and Eikon Refinitiv, respectively), resulting in one of the most comprehensive resources currently available for analysing entrepreneurial ecosystems worldwide.

The OECD Start-ups Database is benchmarked against the data of national and supra-national venture capital associations of OECD countries and partner economies. The OECD Start-ups Database consistently captures more deals and higher levels of venture capital funding than the benchmark sources. This is because it is built using data from Crunchbase and Dealroom, whose collection methodologies cover a broad range of companies and include both VC and non-VC deals. In contrast, the benchmark data are less targeted and do not systematically capture non-member VC firms or individual investors, such as angel investors, making them less comprehensive in representing the full scope of venture capital activity.

Finally, the paper demonstrates the ability of the combined dataset to perform in-depth analysis of different dimensions of the entrepreneurial ecosystem, including the rapid growth of start-ups outside North America and Europe and the heterogeneity in the importance of different funding instruments across locations. For example, while North American and European start-ups make up the majority of the observations in the data, the share of start-ups from the rest of the world still accounts for a non-negligible share, approximately 30% of the sample, and has grown at a much faster pace over the last years. In terms of funding, North American companies have received the majority of financing captured in the data over the past decade,

and Asian companies have been catching up at a high rate. While venture capital remains the dominant source of funding, other forms of financing also play a significant role, particularly in Europe.

By overcoming key limitations of traditional data, the OECD Start-ups Database provides a more accurate, inclusive, and globally representative view of the entrepreneurial ecosystem. Its ability to reveal new trends equips researchers and policymakers with critical insights for designing evidence-based innovation and entrepreneurship policies, making it a foundational tool for shaping policies that foster inclusive, innovation-driven growth worldwide.

1 Introduction

Start-ups are small players that can have a big impact on the economy. Their key role in fostering employment creation, output growth, and radical innovation has been widely studied in the literature (Haltiwanger, Jarmin and Miranda, 2013^[1]; Haltiwanger et al., 2017^[2]; Schnitzer and Watzinger, 2022^[3]). Nevertheless, the study of start-ups has historically faced hurdles, most prominently due to the limited availability of suitable data. The reason is largely due to the inherent characteristics of start-ups: they are small, privately held, and often short-lived, making consistent data collection a challenge. In addition, these features hinder international and cross-industry comparisons.

This paper aims to help overcome the abovementioned hurdles by presenting the OECD Start-ups Database. The OECD Start-ups Database represents an unprecedented effort by the OECD to provide a comprehensive database on the entrepreneurial ecosystem at a global scale. It links two of the most trusted proprietary data sources on the entrepreneurial ecosystem, Crunchbase and Dealroom.¹ It complements them with data on patent and trademark filings, balance sheet indicators, government and corporate venture capital, and other financial transactions of the entities in the database. The result is one of the richest and most comprehensive databases on the worldwide entrepreneurial ecosystem to date.

Prior literature has relied on different strategies to characterise and study the entrepreneurial ecosystem. Before the 2010s, most of the entrepreneurial literature relied on proprietary databases like ThomsonOne and Venture Source, which provided information on venture capital-backed start-ups (Kaplan and Lerner, 2016^[4]). Since then, reliance on these databases has declined in favour of others such as Preqin or Pitchbook, which nevertheless share the same drawbacks for studying start-ups (Da Rin, Hellman and Puri, 2013^[5]; Kaplan and Lerner, 2016^[4]). First, these databases are typically constructed from the investor's portfolio. This means that the focus of the data is not on the investee but on the investor, which generates problems in following investees (including start-ups) over time if they undergo name changes or other corporate restructurings. Second, they are focused on start-ups backed by venture capital. In some locations, the venture capital markets are still developing, and alternative funding sources are more prevalent (e.g. grants or assistance). Third, all these sources are centred around the United States, with a limited coverage of other countries and locations.

Alternatives, like survey data and databases that exploit administrative data, have also been used in the entrepreneurial literature (Jarmin and Miranda, 2002^[6]; Desnoyers-James, Calligaris and Calvino, 2019^[7]; Retterath and Braun, 2020^[8]). However, their country coverage is narrower, and the number of available variables is more limited.

Relative to existing alternatives, the OECD Start-ups Database offers an unparalleled combination of scope, depth, and flexibility for the analysis of entrepreneurial ecosystems. As mentioned, it mainly builds on Crunchbase and Dealroom, two commercial databases that get their information from several sources. Among others, these datasets build on data from communities and partners in the entrepreneurial ecosystem, publicly available information, and others, focusing on capturing innovative start-ups and their surrounding ecosystems (i.e. their investors, competitors, and other relevant entities) (Dalle, den Besten and Menon, 2017^[9]; Da Rin, Hellman and Puri, 2013^[5]; Breschi, Lassébie and Menon, 2018^[10]). These databases have relevant characteristics. First, they focus on start-ups rather than on their investors, drawing information not only from investor portfolios but also from public sources and partners in the entrepreneurial ecosystem, which facilitates the construction of historical records. Second, companies are

not required to have venture capital funding to appear in the data. This is relevant as, according to the National Venture Capital Association, less than 0.1% of companies in the United States had some venture capital funding in 2024 (National Venture Capital Association, 2025^[11]). According to the same source, the United States accounts for approximately 60% of the worldwide venture capital. This implies that venture capital-backed companies are even rarer occurrences in other locations. Going beyond venture capital-backed start-ups, the OECD Start-ups Database offers a more complete portrait of the worldwide entrepreneurial landscape. Hence, the combination of these two databases generates a valuable resource because of the complementary geographic coverage of Crunchbase and Dealroom, as the former is centred on the United States, while the latter is focused on Europe. This overcomes the traditional focus on the United States of the abovementioned databases.

Importantly, the objective of the database is not to achieve statistical representativeness of the start-up universe per se, but rather to preserve and harmonise the full breadth of information available in the two underlying sources. This design choice reflects three main reasons. First, Crunchbase and Dealroom capture not only what researchers typically understand as start-ups, but also a broad set of entities that interact with them. This includes relevant competitors, investors, corporates, government agencies, universities, and other actors in the entrepreneurial ecosystem. Second, definitions of start-ups vary substantially across research questions, and different empirical applications may legitimately require focusing on different subsets of entities. Third, as non-administrative sources, Crunchbase and Dealroom do not register firms at birth but rather when they reach observable milestones (e.g. raising funding or launching a product), implying registration lags, potential survivorship bias, and heterogeneity in coverage across countries and over time. In this context, aiming for strict representativeness at the outset would be neither feasible nor desirable.

For these reasons, the paper deliberately refrains from adopting a single definition of “start-up” at the outset. The introduction and initial sections refer more broadly to the *entrepreneurial ecosystem*, encompassing all entities included in the consolidated database. The database is, therefore, designed to leave researchers the flexibility to focus on different subsets of the data depending on their application. Researchers are advised to apply selection criteria to focus on particular subsets of the data that are representative of the populations they intend to capture. While Crunchbase and Dealroom undeniably capture start-ups and entities in their ecosystems, there is substantial heterogeneity in the available information for each company, which can complicate the identification of their precise role in these ecosystems. This paper highlights the available information that enables researchers to refine the database for their specific needs, illustrates how it can be flexibly leveraged across applications, and explicitly acknowledges the substantial grey area that makes a precise delineation of start-ups inherently application-specific.

This paper first overviews the different sources used to build the OECD Start-ups Database and the methodology to link them. The building blocks are Crunchbase and Dealroom. Both databases capture relevant information related to the entrepreneurial ecosystem and all its actors, including start-ups, their founders, employees, investors, and other relevant entities. The OECD Start-ups Database first links the entities across the two databases and then merges the available funding information. Next, these data on the entrepreneurial ecosystem are complemented with information on the patent filings and registered trademarks contained in the STI Microdata Lab infrastructure, balance sheet information from Orbis, information on financial transactions from Orbis M&A, corporate venture capital from Thomson Reuters, and government venture capital from a survey conducted with the Delegates of the Committee on Industry, Innovation, and Entrepreneurship of the OECD. All these complementary data allow the OECD Start-ups Database to track not only start-ups and their funding, but also their innovation activities, performance, exit strategies, and interactions with relevant public and private stakeholders.

The paper next benchmarks the OECD Start-ups Database against data from national and supra-national venture capital associations. The benchmarking exercise covers almost all OECD countries, as well as

some partner economies. Overall, the OECD Start-ups Database records a higher number of deals and larger transaction volumes than those reported by the associations. This finding can be explained by the different methodologies, criteria, and sources used by these associations to elaborate their aggregate statistics. Nevertheless, the benchmark exercise showcases the OECD Start-ups Database's unprecedented ability to track venture capital deals and companies across OECD and partner economies over time.

Finally, the paper delves into the trends in entrepreneurship that arise from the database, showcasing the ability of the OECD Start-ups Database to perform in-depth analysis of different dimensions of the entrepreneurial ecosystem. This analysis first distinguishes between start-ups and investors among the entities included in the data. In this section, the paper defines start-ups as all corporate entities born after 2000 that are not identified as investors. Regarding the start-ups, North American and European start-ups make up the majority of the observations in the data; however, the share of start-ups from the rest of the world still accounts for a non-negligible share, approximately 30% of the sample. In addition, the number of start-ups originating from the rest of the world has grown at a much faster pace than those in Europe and North America. In terms of funding, North American companies have received the majority of financing captured in the data over the past decade, with Asian companies catching up at a high rate. While venture capital (VC, henceforth) remains the dominant source of funding, other forms of financing also play a significant role, particularly in Europe.

The analysis further examines the investors within the entrepreneurial ecosystem. Investors and investees tend to be geographically concentrated, although North American targets also receive significant funding from other world locations. Across the world, VC firms account for the majority of venture capital investments.² Meanwhile, other sources of financing are more evenly distributed across private and public institutions. In these cases, the relative importance of different investor types varies more widely across locations than in the case of venture capital. Finally, the paper illustrates how to leverage the available information for particular empirical applications by borrowing a start-up definition from prior research focused on innovation- and growth-oriented start-ups, and offering insights into these specific start-ups (Berger et al., 2025^[12]).

This paper is structured as follows. Section 2 overviews the databases used to study the entrepreneurial ecosystem up to date, commenting on their strengths and limitations. Section 3 presents the building blocks used to create the OECD Start-ups Database. Section 4 outlines the data-linking procedure. Section 5 benchmarks these statistics against those of national and supra-national venture capital associations: Section 6 shows descriptive statistics and trends across locations and time in the entrepreneurial landscape using the OECD Start-ups Database. Finally, Section 7 concludes.

Box 1.1. Publications using the OECD Start-ups Database

Recent publications from the Committee on Industry, Innovation and Entrepreneurship (CIIE) that have used the OECD Start-ups Database. Here is a list of them.

- Berger, Dechezleprêtre, and Verlhac (2025^[13]): This paper provides new empirical evidence on how the COVID-19 crisis affected business dynamics and venture capital activity across OECD countries.
- Berger et al. (2025^[12]): This paper explores whether startup acquisitions by incumbent firms promote or hinder innovation.
- Berger, Criscuolo, and Dechezleprêtre (2025^[14]): This paper introduces a new taxonomy of Government Venture Capital (GovVC) fund designs across OECD countries.
- Dechezleprêtre and Kelly (2025^[15]): This paper examines the role of startups offering products and services relevant to the green transition, i.e. green start-ups, between 2010 and 2022.
- OECD (2025^[16]): This paper provides a comprehensive reference on how the Science, Technology, and Innovation (STI) Directorate at the OECD contribute to environmental sustainability.
- OECD (2025^[17]): This paper reviews Slovenia's "Industrial Strategy 2021-2030 for green, creative and smart development" (SIS) and examines the strategy's design, governance, policy instruments, and their effectiveness in achieving Slovenia's objectives.
- Berger, Dechezleprêtre, and Fadic (2024^[18]): This paper provides evidence on the role of Government Venture Capital (GovVC) supporting innovation-driven firms across OECD countries.
- Dechezleprêtre et al. (2024^[19]): This paper analyses bottlenecks in the renewable energy ecosystem.
- Cervantes et al. (2023^[20]): This paper argues that STI&I policies aimed at low-carbon technologies are essential for achieving carbon neutrality, but current innovation levels fall short of the net-zero challenge due to a focus on deployment over R&D support.
- Dechezleprêtre et al. (2023^[21]): This paper offers a novel and holistic view of the automotive sector and its surrounding ecosystem.
- Berger et al. (Forthcoming^[22]): This paper explores how corporate venture capital (CVC) influences start-up innovation.
- Berger, Dechezleprêtre, and Kirpichev (Forthcoming^[23]): This paper shows how academic start-ups depend more on grants and non-equity funding but have more difficulties transitioning to equity financing.

2 Overview of existing databases

Start-ups play a central role in innovation, jobs, and growth, attracting attention across many research fields (Grimaldi et al., 2011^[24]; Calvino, Criscuolo and Menon, 2018^[25]; Haltiwanger, Jarmin and Miranda, 2013^[1]; Decker et al., 2014^[26]). Yet, no standardised database captures them comprehensively, forcing researchers to rely on diverse sources (Hess, 2025^[27]). Their small size, private ownership, often short life span, and the lack of a universal definition make consistent data collection difficult.³

This section reviews the main databases used to study start-ups: venture capital, administrative, and survey data, before turning to the OECD Start-ups Database. Details on the coverage, variables, strengths, and limitations of each data kind are summarised in Table 2.1.

Venture capital data

A traditional way to track start-up activity has been through venture capital databases such as ThomsonOne and Venture Source, later joined by Preqin and Pitchbook.⁴ These sources have been central to entrepreneurial finance research (Mann and Sager, 2007^[28]; Masulis and Nahata, 2009^[29]; Block et al., 2014^[30]; Hochberg, Serrano and Ziedonis, 2018^[31]; Cunningham, Ederer and Ma, 2021^[32]), but face major limitations (Da Rin, Hellman and Puri, 2013^[5]; Kaplan and Lerner, 2016^[4]). They are US-centric, designed around investors' portfolios rather than start-up firms, and cover only VC-backed ventures, excluding start-ups financed through alternative sources of financing like loans, grants, crowdfunding schemes, incubators and accelerators, or angel investors.⁵

Administrative data

Administrative data provide another way to study start-ups. In the United States, examples include the Startup Cartography Project, which uses business registration records (1988–2016) combined with patent data from the United States Patent and Trademark Office (USPTO) and investment data from Thomson Reuters SDC Platinum to characterise the entrepreneurial ecosystem (Andrews et al., 2022^[33]). Another key source is the Longitudinal Business Database (Jarmin and Miranda, 2002^[6]), a U.S. Census panel tracking establishments since 1975 with detailed information on employment, payroll, industry, and location.

In Europe, the ZEW Foundation and Mannheim Enterprise Panels cover around 8 million German firms since 1992 and over 300 000 Austrian start-ups, using CREDITREFORM data to track firm entry and exit, industry classification, employees, and sales (Almus, Engel and Prantl, 2000^[34]). These panels provide detailed information on firm entry and exit, industry classification, number of employees, and sales, among others, enabling many company-level analyses.

In addition to these country-specific databases, there have also been efforts to gather information that can be compared across countries. The OECD exploits cross-country micro-distributed administrative data through the DynEmp and MultiProd databases (Desnoyers-James, Calligaris and Calvino, 2019^[7]). These databases collect and harmonise company-level data from administrative sources for approximately 20 OECD countries and partner economies.⁶ From this information, aggregate statistics are reported for

different company types, categorised by sector, country, and characteristics. In the context of start-ups, these databases have been used to examine aggregate patterns of young companies, focusing on variables such as exit and entry rates, productivity, and production inputs and outputs (Calligaris et al., 2023^[35]; Calvino, Criscuolo and Menon, 2018^[25]).

Overall, administrative databases provide broad national coverage but have clear limits, as shown in Table 2.1: their geographic scope is restricted, there are fewer available variables, and start-ups' identification is mostly proxied by firm age rather than innovation indicators or VC funding.

Survey data

An alternative to administrative data is the use of survey data. They offer rich information on start-ups, including founder traits, strategies, and financing. Examples include “Système d'information sur les nouvelles entreprises” (SINE) in France, Bottazzi, Da Rin, and Hellmann (2008^[36]) on European VC deals, and Hellmann and Puri (2002^[37]) or Hsu (2004^[38]) on US start-ups.^{7,8} Their main drawbacks are limited representativeness and small sample sizes, as they are often designed for specific research purposes.

Broader initiatives, such as the Kauffman Firm Survey in the US and the IAB/ZEW Start-up Panel in Germany, provide systematic and longitudinal coverage of thousands of firms, tracking aspects like employment, financing, and innovation. Yet, these datasets still face the challenges in scope, coverage, and cross-country comparability, as highlighted in Table 2.1.⁹

Entrepreneurial ecosystem data

In recent years, entrepreneurial ecosystem databases have emerged as an alternative. Unlike venture capital sources, they focus directly on start-ups (whether or not they receive external investment), capturing both VC-backed firms and those financed through alternatives like grants or with no funding at all. These databases do not intend to capture the universe of young firms, but rather focus on capturing growth- and innovation-oriented young companies, as well as the relevant players in their ecosystems.

These databases have become increasingly used to analyse venture capital trends and entrepreneurial ecosystems more broadly. Among them, the OECD Start-ups Database leverages Crunchbase and Dealroom. Compared to competitors such as CB Insights, Tracxn, Capital IQ, or AngelList, Crunchbase has emerged as the preferred dataset among researchers in recent years, offering coverage of US start-ups that is broadly similar to these other proprietary sources (including venture capital and ecosystem datasets) but with greater academic uptake (Retterath and Braun, 2020^[8]; Dalle, den Besten and Menon, 2017^[9]).¹⁰ What sets Dealroom apart is its stronger European focus, making it complementary to Crunchbase's US orientation. Together, the two offer a more balanced global perspective than either could alone. First, their coverage is not limited to VC-financed companies. According to the National Venture Capital Association, less than 0.1% of active companies in the United States received any venture capital (National Venture Capital Association, 2025^[11]), meaning that a large share of innovation and growth-oriented companies do not receive venture capital financing (Catalini, Guzman and Stern, 2019^[39]). This also helps expand coverage beyond the US to other territories where venture capital markets are not as developed.¹¹ Third, some authors have also shown that sectoral coverage of key innovative industries like Biotech or IT is better represented in these data (Wang, 2018^[40]; Retterath and Braun, 2020^[8]) than in venture capital data.¹² Finally, compared with other proprietary competitors like Preqin or Pitchbook, which remain investor-oriented, Crunchbase and Dealroom offer not only comparable funding information but also extensive details on firm characteristics, founders, technologies, and products. As a result of these advantages, they have become increasingly adopted in academic entrepreneurship research (Gugler, Szucs and Wohak, 2025^[41]; Veugelers and Amaral-García, 2025^[42]).

In sum, there is no universal database to analyse the entrepreneurial ecosystem. Each type of data offers specific advantages and drawbacks, as summarised in Table 2.1, and researchers often need to combine them depending on the question at hand. Benchmarking exercises remain scarce and usually limited to VC-backed firms (Matts et al., 2011^[43]; Da Rin, Hellman and Puri, 2013^[5]; Kaplan and Lerner, 2016^[4]; Retterath and Braun, 2020^[8]), which is insufficient to capture global, innovation-oriented entrepreneurship. Against this background, the OECD Start-ups Database is an unprecedented effort to harmonise Crunchbase and Dealroom and link them to additional company-level data, such as Orbis and to intellectual property data, enabling cross-country analysis of start-up innovation, financing, and performance.

Table 2.1. Summary of the types of data used in start-up and entrepreneurial research

Type of data	Examples	Coverage	Available variables	Strengths	Limitations	Typical use cases
Venture capital databases	ThomsonOne (VentureXpert), Venture Source, Preqin, Pitchbook	Mainly US; investor portfolios	Investment rounds, investors, deal size, valuation, exit events	Widely used in entrepreneurial finance literature; detailed deal-level info	Investor-focused, only VC-backed firms; US-centric; issues with consistency and quality over time	Investment trends, exit analysis
Administrative data	Longitudinal Business Database (US Census), Startup Cartography Project, ZEW Foundation & Mannheim Enterprise Panels (Germany & Austria), OECD DynEmp/MultiProd	Country-specific; OECD cross-country initiatives	Firm age, size, employment, payroll, industry, location; sometimes sales	Comprehensive within a country; official records; enables longitudinal analysis	Limited cross-country comparability; fewer variables (esp. on innovation); often must proxy start-ups with “young firms”	Firm dynamics, employment effects
Survey data	Kauffman Firm Survey (US), IAB/ZEW Start-up Panel (Germany), Bottazzi et al. (Europe), Hellmann & Puri (US)	National or geographical samples	Founders' demographics, financing sources, strategies, innovation, growth	Rich info on founders, strategies, financing, innovation	Limited representativeness; small sample sizes; not cross-country comparable	Founder behaviour, innovation strategy
Entrepreneurial ecosystem databases	Crunchbase, Dealroom, Angellist, Tracxn, Capital IQ, CB Insights	Global, with US/Europe focus	Firm profiles, funding events, investors, founders, sectors, technologies (e.g. AI tags)	Cover both VC- and non-VC-funded start-ups; rich info on firms, founders, investors, technologies; complementary geographic scope	Retrospective data less complete; registration lag; reliant on contributed/public info	Mapping ecosystems, network analysis

Note: This table provides an overview of the main types of data used in start-up and entrepreneurial research. It outlines representative examples, geographic coverage, key strengths, and typical uses of each data type.

Source: Authors' elaboration.

3 Data sources

The OECD Start-ups Database leverages information from several sources. As mentioned, the bulk of the data is based on Crunchbase and Dealroom, two proprietary databases with complementary geographical coverage that provide information on the entrepreneurial ecosystem. The OECD Start-ups Database is integrated into the infrastructure of the STI Microdata Lab. In particular, the information on the entrepreneurial ecosystem is further complemented with data on patent filings and registered trademarks from the STI Microdata Lab infrastructure: i.e. patent data from the Worldwide Patent Statistical database (PATSTAT) maintained by the European Patent Office (EPO) and data on trademarks registered at the European Union Intellectual Property Office (EUIPO), at the Japan Patent Office (JPO) and at the USPTO. Balance sheet information from Orbis and exit transactions' data from Orbis M&A are also integrated. This section provides an overview of all the data sources incorporated into the OECD Start-ups Database.

Data on the entrepreneurial ecosystem

Crunchbase

Crunchbase is a proprietary database, considered by investors and researchers as one of the most comprehensive databases on the entrepreneurial ecosystem. It was created in 2007 with a US focus but includes data from start-ups worldwide and retrospective data from as early as the 1950s. While the coverage before its creation is imperfect, it has been improving over time.¹³ The information in the database comes from several sources. First, it relies on an active community of contributors that includes venture capital companies, executives, investors, and entrepreneurs who update the data monthly.¹⁴ Second, Crunchbase uses AI and machine learning algorithms to search and validate data using government filings and over a thousand top news outlets. Finally, Crunchbase also collaborates with third parties like tech stack data or interest signals to complement the available information. In addition, all these sources are continuously manually verified by their group of experts to ensure their quality.

Overall, Crunchbase presents data on approximately 4 million entities from around the world, with roughly 40% of the coverage corresponding to entities based in the United States..¹⁵ It also provides information on approximately 300 thousand investors (both institutional and individual investors) and over 700 thousand funding rounds, distinguishing the financial instruments used to raise capital and the stage at which said rounds took place. Funding rounds are available between 1990 and 2024, and historical data for selected years from 1953 until 1990. Finally, they also provide additional information like tags about the company's activities, products and technologies, founder information, company descriptions, years of incorporation, and headquarters' locations.

Dealroom

Dealroom is also a proprietary database that provides comprehensive information on the entrepreneurial ecosystem. Unlike Crunchbase, it has better coverage of European start-ups.¹⁶ Dealroom was created in 2013 and, similarly to Crunchbase, provides retrospective information for companies born before this year (albeit information before its creation is yet again incomplete). It also collects information from several sources using AI and machine learning algorithms, exploiting publicly available information from company

filings, news, domain and trade registries, job boards, web and app store analytics, and investor portfolios. Likewise, it has partnerships with local governments, and it also outsources data from third parties.¹⁷ Finally, Dealroom users can also incorporate information into the database. All the information is manually verified and validated by their team of experts.

Overall, Dealroom provides information on more than three million entities. Among those entities, it provides information on more than 250 thousand investors involved in more than 900 thousand transactions between 1990 and 2024. The data is similarly structured to Crunchbase's, providing complementary information on companies' activities, products, and technologies, background information on founders, owners, and staff, years of incorporation, and headquarters locations, among others.

Similar to Crunchbase, Dealroom includes start-ups without a venture capital round and is more likely to capture early-stage deals like angel financing, spinouts, and crowdfunding initiatives. Given the European focus of this database, this aspect is particularly relevant due to the higher share of European start-ups financed through alternative financing instruments, like grants (Berger, Dechezleprêtre and Kirpichev, Forthcoming^[23]). Importantly, while the two databases overlap substantially, they exhibit systematic differences in geographic focus and coverage depth. Their integration within the OECD Start-ups Database is therefore motivated by the objective of improving global coverage and reducing regional imbalances present in each source when used in isolation. (more details below).

Complementary information on company financing and innovative performance

While Crunchbase and Dealroom provide extensive information on the entrepreneurial ecosystem, they provide limited information on the firm-level outcomes of the companies over time. Therefore, the resulting merger between Crunchbase and Dealroom is further integrated into the STI Microdata Lab, a sizeable data infrastructure containing more than one billion entries with firm-level information on financial and innovation outcomes. This subsection provides an overview of the sources from the STI Microdata Lab used to complement the information on start-ups with additional firm-level outcomes, notably on their innovative activities. First, it provides an overview of the patent filings and registered trademarks information. Next, it outlines the financial information obtained from Moody's Orbis and Orbis M&A. Finally, it concentrates on the hand-collected complementary databases on corporate and government venture capital.

Worldwide patents from PATSTAT

Start-ups are key actors when it comes to bringing radical innovations to consumer markets (Kortum and Lerner, 2000^[44]). Therefore, characterising their innovation activities is one of the focuses when building the OECD Start-ups Database. To do so, the OECD Start-ups Database exploits patent filings information coming from PATSTAT.¹⁸ PATSTAT is the bibliographic database of the EPO, covering detailed information on patent applications, publications, and legal events. Its main variables, coverage, and value added are summarised in Table 3.1. This information enables the proxy of the innovation behaviour of firms as well as the computation of widely used indicators of patent quality, such as forward and backwards citations, claims, scope, and breakthrough patents (Squicciarini, Dernis and Criscuolo, 2013^[45]). For additional detail on the matching procedure, see the next section.

Trademarks registered at EUIPO, JPO, and USPTO

The OECD Start-ups Database is complemented with trademark data. The intent to incorporate trademark data into the OECD Start-ups Database is to obtain insights into the product market activity of the companies. Trademark data have been argued to provide valuable insights into innovation activities (Castaldi, 2023^[46]), especially in industries that do not rely on patenting as a way to protect their inventions

(Milot, 2009^[47]). The STI Microdata Lab infrastructure incorporates all trademarks registered at the EUIPO, the JPO, or the USPTO.

Orbis

Beyond innovation, the OECD Start-ups Database also tracks start-ups' performance in areas such as employment, productivity, and growth. For this, it leverages Orbis, a commercial database provided by Moody's and one of the largest cross-country firm-level sources. Orbis compiles data from official registries, chambers of commerce, financial statements, stock exchanges, and local providers, and standardises this information across countries and over time. Its coverage, main variables, and value added are summarised in Table 3.1. Orbis covers all the periods included in Crunchbase and Dealroom, and balance sheet information is available for more than 100 countries. In addition, it also includes industry codes which are not available in either Crunchbase or Dealroom.¹⁹ The industry coverage reflects the non-farm business sector, i.e. all industries excluding agriculture and public services. Crucially, it is the only company-level cross-country database that provides balance sheet information for both privately held and publicly listed companies, enabling the tracking of start-ups (typically privately held during the first years) through growth and scaling. For more details on the Orbis coverage, variables, and limitations, see Bajgar et al. (2020^[48]).

Orbis M&A

This transaction-level database, also maintained by Moody's, covers Initial Public Offerings (IPOs), joint ventures, capital increases, venture capital (VC) investments, acquisitions, and other financial transactions of both private and public firms, with global coverage since 2001 (Europe since 1997). It draws on diverse sources such as stock exchanges, company reports, and financial news. Its coverage, variables, and added value are summarised in Table 3.1.

Orbis M&A strengthens the database in several ways: it improves coverage of majority stake acquisitions and enables reconstruction of investor portfolios beyond start-ups. Whereas Crunchbase and Dealroom mainly capture early-stage deals, Orbis M&A adds insights into later stages as firms scale and exit, primarily because acquisition is a common exit strategy among start-ups.

Government Venture Capital (GovVC)

The information reported in Crunchbase and Dealroom on investors provides tags to distinguish among different kinds of investors. Nevertheless, this information is not harmonised across databases, and, therefore, some relevant funding activity from a policy perspective, like government venture capital funding, is hard to identify. In general, Crunchbase and Dealroom do not allow for a proper understanding of what kind of relationship exists between the investing company and the government (for example, direct or indirect ownership or participation). To address this, the OECD surveyed its member countries, validating 308 GovVC entities across 38 OECD economies.²⁰ For further details on how these data are constructed, see Berger, Dechezleprêtre, and Fadic (2024^[18]).

Corporate Venture Capital (CVC)

Similar to the government venture capital case, Crunchbase and Dealroom have tags that allow identifying the corporate investors in the data. However, these categories only allow the identification of direct investments by corporations and not those executed through their VC arms. Therefore, a list of the top CVC entities in North America and Europe was manually compiled from Thomson Reuters (Eikon), which was then linked with Orbis M&A to identify both CVC-backed start-ups and their corporate parents (Berger et al., Forthcoming^[22]).²¹ As shown in Table 3.1, CVC data adds precision by distinguishing corporate venture arms from general corporate or financial investors.

Table 3.1. Data sources integrated into the OECD Start-ups Database

Source	Coverage	Main variables	Value added to the database
Crunchbase	~4M entities	Funding rounds, investors, founders, company profiles	Captures early-stage and non-VC firms; widely adopted in academic research
Dealroom	~3M entities	Similar to Crunchbase; firm profiles, funding, sectors	Complements Crunchbase geographically; stronger EU/UK representation
Patents (EPO PATSTAT)	~130M patents worldwide	Patent applications and all the information within (abstract, title, fields), inventors, citations made or received	Measures technological innovation; links start-ups to global patenting activity
Trademarks (EUIPO, JPO, USPTO)	~16M trademarks; US, EU, JP markets	Trademark filings, classes, goods and services description	Captures non-technological innovation and product commercialisation
Orbis	~450M firms, +100 countries	Firm demographics, industry codes, ownership, financial accounts	Tracks firm performance; provides size, age, and balance sheet data
Orbis M&A	~3M global transactions	IPOs, acquisitions, minority stakes	Identifies exit events and corporate transactions
GovVC/ CVC	+300 GovVC agencies, +200 CVC investors, OECD-wide	Investor type and funding activity	Identifies public and corporate venture capital participation

Note: This table presents the core data sources from the STI Microdata Lab integrated into the OECD Start-ups Database. It includes coverage of each source, its main variables, and the value it adds to the OECD Start-ups Database.

Source: Authors' elaboration.

4 Methodology

This section outlines the process of combining the various data sources that underpin the OECD Start-ups Database. The integration begins with Crunchbase and Dealroom, which undergo extensive cleaning and harmonisation to enable their merge. Once these two sources are combined, complementary information on company performance is added through string-matching with other databases.

Cleaning, harmonising, and linking the entrepreneurial ecosystem databases

Crunchbase and Dealroom are the two main sources of information for the database.²² While more detailed information is available in each source, three main types of information are used to build the OECD Start-ups Database:

1. **Entities:** information on organisations active in the entrepreneurial ecosystem, including name, country, year of incorporation, description of activities, website, organisational type (company, university, investor, etc.), and patent activity. This information is available cross-sectionally, i.e. at the moment of data extraction or last updated, and does not allow tracking entities over time.
2. **Funding:** details on financing activity, including the volume raised, the timing and types of funding rounds (equity, debt, grants, etc.), and the kinds of investors involved (venture capital, private equity, angels, corporates, government, etc.).
3. **People:** information on founders, staff, and investors, such as their roles within entities, educational background, and gender.

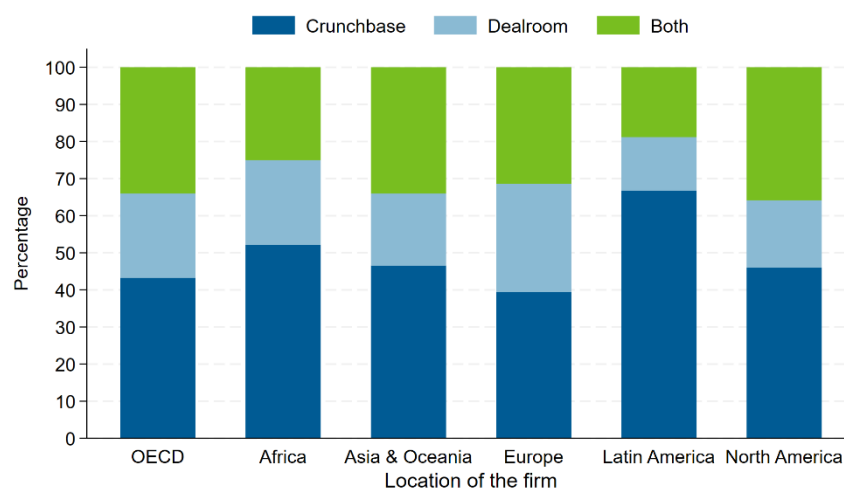
Throughout this section, the paper uses the term *entity* to refer to any company or organisation active in the entrepreneurial ecosystem and included in Crunchbase or Dealroom. The same methodology is applied to harmonise and merge start-ups, their competitors, investors, and other relevant actors. The following section provides further insights into the different kinds of entities included in the datasets.

A key step in the construction of the OECD Start-ups Database is to harmonise information across the two sources and ensure that entities appearing in both are not double-counted. Linking is conducted primarily through names, URLs, and other basic characteristics such as country or foundation year. In cases of ambiguity, additional information is used to disambiguate entities. The outcome of this process is a consolidated list of companies, investors, and other institutions, including both those present in one source only and those matched across the two. Technical details of the linking procedure are provided in Annex A.

Figure 4.1 shows the amount of overlap between Crunchbase and Dealroom, showcasing their geographical complementarity. It shows the percentage of companies that appear in Crunchbase, in Dealroom, or in both data sources, by location. On average, around 30% of the entities in the OECD Start-ups Database are identified in both databases, except for Latin American entities, where the overlap is below 20%.²³ Crunchbase provides the best coverage of North America, with more than 40% of entities in this location being identified uniquely from this source. For Europe, both Crunchbase and Dealroom each uniquely cover around 35% of European entities. While Crunchbase is broader in scope and also captures a substantial number of European entities, clearly Dealroom identifies a substantial portion in this region

that is not identified in Crunchbase. Furthermore, Crunchbase also accounts for most of the observations in Latin America, Africa, and Asia & Oceania. These patterns demonstrate that combining both sources enables the OECD Start-ups Database to present a more comprehensive and accurate view of the global entrepreneurial ecosystem than either source alone.

Figure 4.1. Distribution of entities in the OECD Start-ups Database, by data source and location



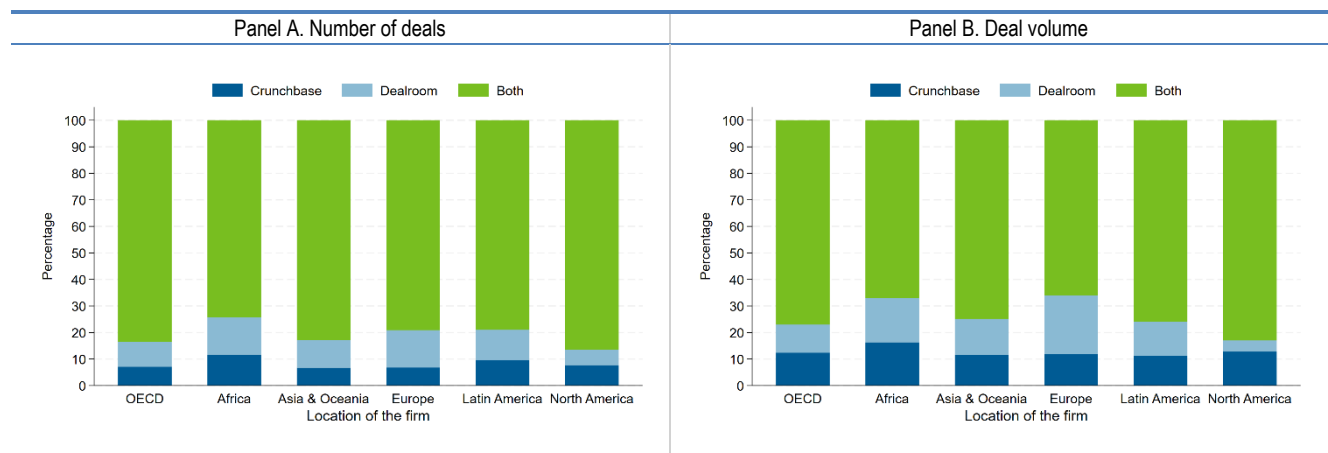
Note: This figure shows the share of entities by their data source (Crunchbase or Dealroom) across locations. Only entities with identifiable country information are included in the analysis.

Source: OECD Start-ups Database, June 2025.

After creating a unique list of entities, their funding records must also be cleaned, harmonised, and linked. A first challenge is that Crunchbase and Dealroom use different classifications of deal types, which are standardised before integration (see Table A A.1 in Annex C for more details). A second issue is complementarity: some funding rounds are reported in both databases, while others appear in only one. By comparing deal-level characteristics (such as round dates and type), overlaps can be identified, and missing information can be filled. In cases where both databases report the same deal, Crunchbase is used as the primary source.²⁴ For more details on the funding-level matching, see Annex A.

Figure 4.2 presents the results of this matching exercise, showing the share of transactions by number (Panel A) and by deal volume (Panel B) coming from Crunchbase, Dealroom, or both. Overall, the overlap in funding data is considerably higher than for the number of start-ups themselves. Non-matching transactions account for less than 30% across locations, with North America showing the highest overlap (almost 85% of deals captured in both sources). By contrast, overlaps are lower in Europe and Africa, where Dealroom contributes more unique information, particularly in terms of deal volume. Dealroom also provides stronger coverage in Africa and Latin America. These results highlight that while the two databases are complementary in terms of entity coverage (Figure 4.1), they show high consistency when it comes to funding activity, reflecting the fact that start-ups that raise financing are more likely to be tracked across sources.

Figure 4.2. Number and volume of start-up funding deals in the OECD Start-ups Database, by data source and location



Note: This figure shows the distribution of start-up funding across data sources. Panel A shows the number of deals, while Panel B displays the total deal volume in USD billions. Data are sourced from Crunchbase, Dealroom, or both.

Source: OECD Start-ups Database, June 2025.

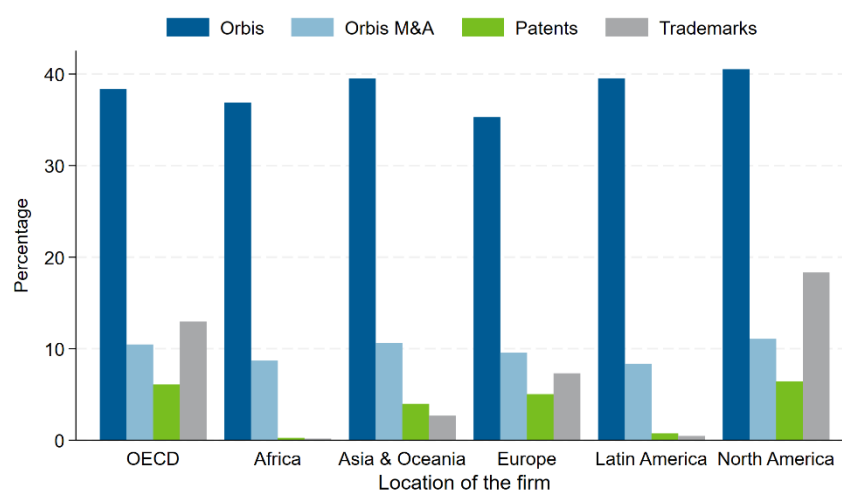
Linking the entrepreneurial ecosystem data with firm-level outcomes

As mentioned above, the resulting combination of Dealroom and Crunchbase is further linked to other databases from the STI Microdata Lab: Orbis, Orbis M&A, and patents and trademarks data. The link between databases is executed using a similar string-matching procedure to the one described above and the one outlined in Tarasconi and Menon (2017^[49]). First, the entity names across databases are harmonised using a set of country-level dictionaries, notably to correct for special characters and variations in the way legal entity types are spelled out (e.g. “Ltd.,” “Corp.,” “GMBH,” “kabushiki kaisha”). Then, the harmonised entity names are string-matched on a country-by-country basis. Finally, the matches are triangulated across databases (i.e. matches between PATSTAT and the OECD Start-ups Database were identified through the links established between PATSTAT and Orbis, and between Orbis and the OECD Start-ups Database). For further information on the matching procedure, see Annex A.

Figure 4.3 shows the percentage of entities by location that have linkages to any of these additional databases (i.e. Orbis, Orbis M&A, patents, and trademarks data). Some Orbis information is available for approximately 40% of the entities from North America. The coverage in locations like Africa, Latin America, and Asia & Oceania is similar, but for Europe is slightly lower. It is worth highlighting that being recorded in the Orbis database and having balance sheet information are not equivalent. Some companies are recorded in Orbis, and some cross-sectional information is provided, like age or industries, but the firm's balance sheet information might be missing or might not overlap with the period of analysis. Therefore, these percentages represent upper bounds on the available information on balance sheet variables.

As expected, linkages to Orbis M&A, patents, and trademarks are inevitably less frequent than to Orbis, as not all entities engage in financial, patenting, or trademarking activities.²⁵ On average, slightly more than 10% of entities in OECD economies have data on financial transactions from Orbis M&A. Patents and trademarks present even lower linkages: around 5% of entities in Europe, North America, and Asia & Oceania have patent information, while Latin America and Africa present negligible amounts of patenting entities. Trademarks are more commonly used than patents to protect the entity's intellectual property, with approximately 15% of entities in OECD countries having at least one registered trademark, mainly driven by North American entities, followed by Europe and Asia & Oceania.

Figure 4.3. Share of entities from the OECD Start-ups database linked with external databases, by external database and location



Note: This figure shows the share of entities linked to external databases, Orbis, Orbis M&A, Patents and Trademarks, across locations. These linkages provide insights into firm-level outcomes and connections within the entrepreneurial ecosystem. Trademarks percentages are based on an earlier version of the OECD Start-ups Database (2023) and may not reflect the most recent updates.

Sources: OECD Start-ups Database, June 2025; Orbis and Orbis M&A, Bureau van Dijk, March 2025.

5

Benchmarking against other datasets

To assess its quality, coverage, and representativeness, the OECD Start-ups Database is compared with several external sources at the international and national levels, which represent well-established sources for aggregate start-up activity. These sources include other proprietary vendors, aggregate data from institutional sources, and national or supra-national venture capital associations. Where possible, data were verified through direct correspondence with the respective organisations. In other cases, data were drawn from official publications and databases made available on the Organisations' websites. The analysis concludes that the OECD Start-ups Database consistently captures more deals and greater investment volumes than the benchmark data, representing a step ahead in entrepreneurial research data.

Data

The different database used for the benchmarking exercise covers different reporting periods and collection methods. Table 5.1 summarises the key benchmark information used in this analysis, along with the source organisations and the countries included.

Each different benchmark source might differ in the way data are collected, i.e. which companies and the types of financing are considered. To address this concern, the benchmarking exercise restricts the funding instruments considered only to equity-related venture capital investments. In other words, only those deals classified as seed, early-stage, late-stage or growth in Table A C.2 are reported in all the graphs below.²⁶ Note that while this approach enhances comparability across sources, it is likely that the benchmarking sources do not cover some transactions included in the OECD Start-ups Database. For instance, some of these sources rely on surveys completed by their members, and, in general, may overlook angel investments, which are instead considered in the OECD Start-ups Database.

To ensure international comparability in real terms, all benchmark data were converted to US dollars and adjusted for inflation using the National Consumer Price Indices (CPIs), with 2015 as the base year. The CPI and exchange rate data used for these adjustments were sourced from the OECD data portal.²⁷

Table 5.1. Sources of benchmark database, by country and organisation

Organisation	Country	Time Period	Source
Preqin and the Australian Investment Council	Australia	2010-2023	Downloaded from the "Australian Private Capital Market Overview: A Preqin and Australian Investment Council Yearbook 2024"
Canadian Venture Capital & Private Equity Association	Canada	2013-2024	Direct contact
Invest Europe	Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Moldova, Montenegro, Netherlands, North Macedonia (Republic of), Norway, Poland, Portugal, Romania, Serbia, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Ukraine, and United Kingdom.	2007-2023	Direct contact
Central Bureau of Statistics	Israel	2014-2023	Direct contact
Venture Enterprise Center	Japan	2010-2023	Direct contact
Latin America Venture Capital Association	Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, Venezuela	2015-2024	Direct contact
Korean Venture Capital Association	Korea	2014-2024	Downloaded from the "KVCA Statistics Summary Report (2024)"
startups.watch platform, in collaboration with the Presidency of the Republic of Türkiye Investment Office, Financial Investments Unit	Türkiye	2007-2025	Direct contact
PitchBook–NVCA	United States	2014-2024	Downloaded from the "Q42024 PitchBook–NVCA Venture Monitor" Datapack

Note: This table shows multiple sources used to create the benchmark data, the countries the data come from, the time period, and the method of collection.

Source: Authors' elaboration.

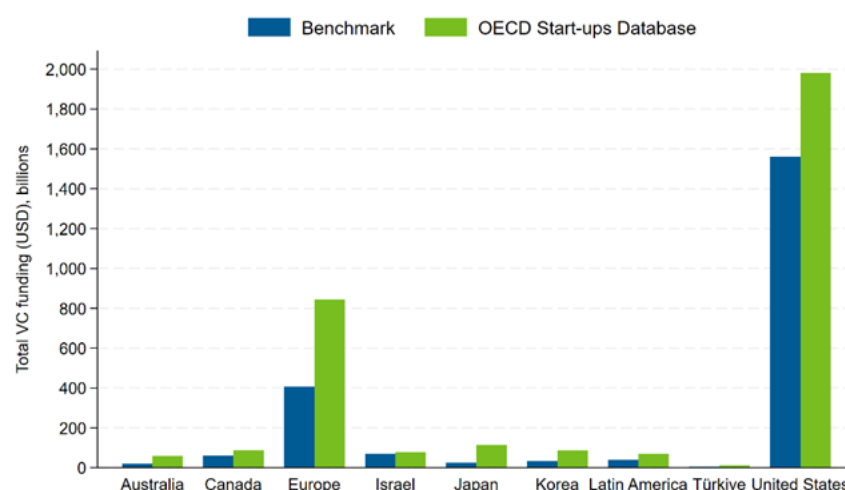
Benchmarking

The benchmarking exercise is carried out for three main variables: number of VC deals, number of VC-funded companies, and total VC deal volumes. Comparisons across databases for the deal volume are reported in the main text, as it is the only variable consistently reported across all sources; instead, the number of VC deals and the number of VC-funded companies are not consistently present across benchmark databases and are therefore provided in Annex B.

Overall, the OECD Start-ups Database consistently captures more deals and greater investment volumes than the benchmark data. Figure 5.1 compares the VC deal volumes from the OECD Start-ups Database with benchmark data across different sources. Across all benchmark databases, the OECD Start-ups Database consistently reports higher levels of venture capital funding. This is largely attributable to its construction from Crunchbase and Dealroom, which aim to capture newly founded companies and funding comprehensively. By contrast, benchmark data sources could underreport funding for several reasons. First, many seed and early-stage deals can be difficult to track. Second, some benchmark data rely on

surveys sent only to companies that are members of the collecting associations, leaving out the non-responders and potentially many newly founded investing companies. Lastly, some of the benchmark sources are private initiatives that focus on large VC funding, excluding smaller VC funding, resulting in an incomplete representation of the VC funding landscape in the country.²⁸

Figure 5.1. Comparison of VC investment coverage: benchmark data vs. OECD Start-ups Database



Note: This figure compares the total VC funding in the OECD Start-ups Database with multiple benchmark databases.

Source OECD Start-ups Database, June 2025, Preqin and the Australian Investment Council; Canadian Venture Capital & Private Equity Association; Invest Europe; Israel Central Bureau of Statistics; Venture Enterprise Center, Japan; Latin America Venture Capital Association; Korean Venture Capital Association; startups.watch platform (in collaboration with the Presidency of the Republic of Türkiye Investment Office, Financial Investments Unit); and PitchBook-NVCA.

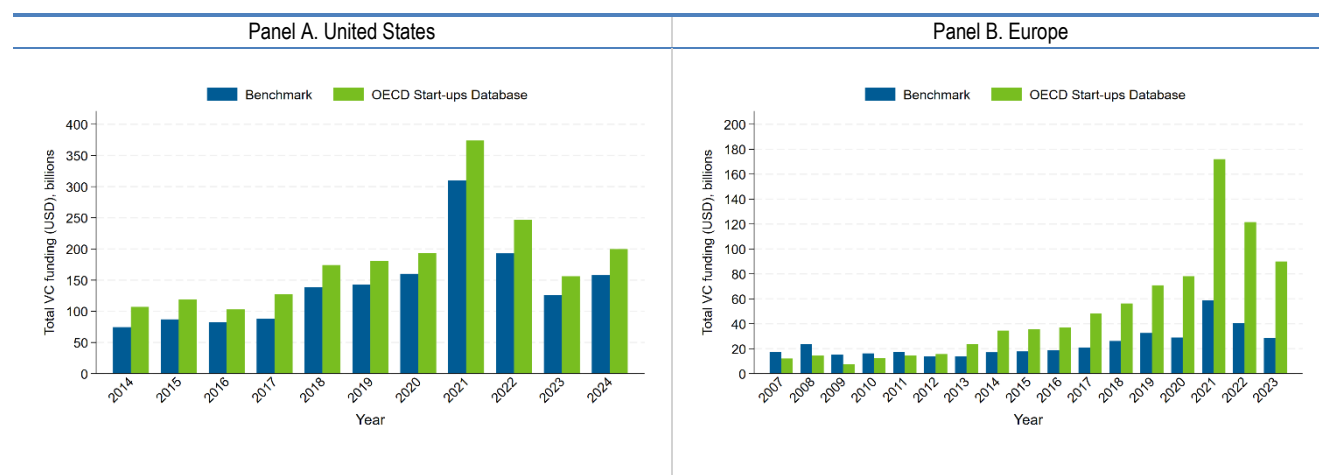
Table A B.1 in Annex B shows these differences across locations in more detail. It shows that the OECD Start-ups Database generally reports higher deal volumes, with particularly large differences in Australia (200.40%), Europe (104.43%), Japan (364.23%), Korea (165.63%), and Türkiye (164.03%). These differences confirm that the OECD Start-ups Database captures a broader share of funding activity across locations, especially where benchmarking sources may have gaps.

Figure 5.2 provides a more detailed, year-by-year comparison of the OECD Start-ups Database with the benchmark sources in the United States and Europe, the two locations that are the main focus of both source databases.²⁹ In the United States, the difference in coverage between the OECD Start-ups Database and the benchmark data remains consistently low across years. This overlap is among the largest observed across all benchmark databases, with the OECD Start-ups Database covering nearly 27% more volume than the benchmark. The reasons might be the following. First, the National Venture Capital Association sources its data from Pitchbook. As mentioned above, Crunchbase and Pitchbook are likely to offer similar coverage of funding deals. However, Dealroom and Crunchbase offer highly complementary funding deal data, as mentioned above, which helps explain the broader coverage of the OECD Start-ups Database.

Europe presents a larger difference than the US, with deal volumes 104.43% higher in the OECD Start-ups Database. One reason might be that InvestEurope (formerly European Venture Capital Association) collects data through a survey among its members. Investments from non-members, including non-member VC firms and different actors, such as angel investors or family offices, are therefore less likely to be captured compared to the OECD Start-ups Database. In addition, while coverage in Europe was initially lower than in the benchmark data, it has significantly surpassed it since 2013. This could be due to the

creation of Dealroom in 2013, which enabled a more accurate record of companies' information in Europe from that year onwards. Nevertheless, the trends in funding are remarkably similar across the benchmark and the OECD Start-ups Database.

Figure 5.2. Comparison of VC investment coverage: benchmark data vs. OECD Start-ups Database by year



Note: This figure compares annual venture capital investment data reported by benchmark sources with that in the OECD Start-ups Database for the United States (Panel A) and Europe (Panel B).

Source: OECD Start-ups Database, June 2025, PitchBook-NVCA, and Invest Europe.

Figure A B.1 in Annex B compares the OECD Start-ups Database with other data sources for companies in Australia (Panel A), Canada (Panel B), Israel (Panel C), Japan (Panel D), Korea (Panel E), Latin America and the Caribbean (Panel F), and Türkiye (Panel G). In all cases, the OECD Start-ups Database captures a larger share of deal volume. The differences observed across benchmark sources reflect not only variations in coverage but also the distinct patterns and trajectories of funding activity over time in each location.³⁰

6 Descriptive statistics and trends

To allow for a better understanding of all the data in the OECD Start-ups Database and its potential uses, this section presents descriptive statistics and trends in the entrepreneurial ecosystem over the last two decades. It provides an overview on trends in the number of start-ups covered by the data, their funding rounds, the kinds of investors that engage with them, and their innovation activities.

Crunchbase and Dealroom allow distinguishing between “companies” and “investors.” In Section 4, these were referred to as “entities.” In what follows, the section treats “companies” and “investors” separately.

Regarding the entities labelled as “investors,” Crunchbase and Dealroom aim at capturing those providing funds to start-ups. However, some of these investors may be simultaneously labelled as “investors” and “companies,” and thus appear as both providing and receiving financing rounds. This may happen for several reasons. For instance, a start-up that scaled up at some point during the period covered by the dataset may later engage in acquisitions and/or funding of other start-ups (e.g. OpenAI). For consistency, the paper includes all entities labelled as “companies” in the analysis of companies, even if they are also labelled as investors. Likewise, the analysis of investors consists of all entities labelled as “investors,” even if they are also labelled as “companies”.³¹

As mentioned extensively throughout the paper, Crunchbase and Dealroom offer information on what is broadly known as the entrepreneurial ecosystem. Entities labelled as “companies” are likely to capture mostly innovation- and growth-oriented young companies, which are commonly referred to as “start-ups”. The focus on more innovation- and growth-oriented young companies into the OECD Start-ups Database is reasonable for at least two reasons. First, the number of companies in the dataset is far below the aggregate number of yearly business registrations in each country, showing that not all newly established companies are included in the database.³² Second, as previously mentioned, Crunchbase and Dealroom obtain information on companies from different sources that are relevant to the entrepreneurial ecosystem.

While start-ups are central players in the entrepreneurial ecosystem, they are not the only companies relevant to it. For instance, the data may include relevant competitors in the same market that may not necessarily be young firms but rather more established companies. Some companies may also provide complementary services to the entrepreneurial ecosystem (e.g. legal, financial, or consulting services). When the objective of the analysis is to study innovation- and growth-oriented start-ups, an important step is to fine-tune the definition using available information in the data.

This section starts with descriptive statistics for all entities labelled as “companies” in the data. Throughout this section, the analysis refers to these entities as “companies” or “start-ups.” After presenting the descriptives for all companies in the data, it then illustrates how to leverage the available information to narrow the scope on ‘innovative start-ups’, defined as young companies born from 2000 onwards that filed patents or received venture capital funding, along the lines of prior literature. An advantage of the OECD Start-ups Database is that it allows leveraging many of the definitions laid down in previous literature to capture innovative and growth-oriented start-ups.

Descriptive statistics

Table 6.1 provides an overview of the data, including the number of entities in the OECD Start-ups Database, distinguishing between investors and companies. The database includes information on

approximately 6.2 million entities. Approximately 94% of companies are classified as such, and 7% of these provide some form of financing to other companies. As implied by these percentages, approximately 1% of companies and investors overlap.

Table 6.2 focuses on companies alone, showing the available information on their characteristics, funding status, intellectual property, exit events, geographic distribution, and founder demographics.³³ The OECD Start-ups Database features almost six million companies, the vast majority of which were founded after 1990 (88%) and over 80% from 2000 onward. Almost 67% of companies are located in OECD countries, based on the country information reported for each company, with a high concentration in North America (37%) and Europe (25%), as well as a relatively large share in other economies.³⁴ Approximately 10% of companies have some external funding, out of which more than half is venture capital, but an important percentage comes from other instruments.

Regarding exits, start-up exits are more frequent in the form of acquisitions (3.5%) than in an IPO (1%), in line with the general trend in OECD countries (Ederer and Pellegrino, 2023^[50]). Although these figures may seem small, they highlight the extensive global reach of the OECD Start-ups Database. For example, the database captures 58 498 IPOs worldwide, and the World Federation of Exchanges reports approximately 58 thousand publicly listed companies in the world as of 2024. Likewise, Fons-Rosen, Roldán-Blanco, and Schmitz (2022^[51]) report that only 0.02% of new registered businesses in the United States are acquired. In contrast, the OECD Start-ups Database shows an acquisition rate of 3.5%, which reflects its focus on VC-backed and innovative start-ups, which are more likely to be acquired.

Around 17% of the companies in the database have information related to their founder(s). This percentage varies by location and by the company's founding year, with newer companies being more likely to include information about their founder(s) (see Figure A C.2 in Annex C). It also differs based on funding status: among companies founded since 2010, about 60% of those with VC funding had founder information available, compared to around 40% for all companies in that cohort (see Figure A C.3 in Annex C). Among those companies having information on founders, 15% have at least one female founder, and over 40% of the founders have a university degree. This includes roughly 22% with a master's degree and nearly 5% with a PhD.

Regarding patents, almost 5% of the companies in the data filed a patent. Generally, this percentage increases with the amount of information available about the company. For example, among start-ups in OECD countries, 25% of those with funding information and around 10% of those with founder information have filed a patent. This suggests that, on average, more innovative companies are also more likely to be covered across multiple dimensions in the OECD Start-ups Database. See Figure A C.1 in Annex C for more details.

Table 6.1. Entities in the OECD Start-ups Database

	Number	Percentage (%)
Entities	6 250 148	100.00
Companies	5 894 247	94.31
Investors	459 996	7.36

Note: This table summarises the composition of the OECD Start-ups Database, showing the total number of entities and their breakdown into companies and investors, along with their share of the total.

Source: OECD Start-ups Database, June 2025.

Table 6.2. Key characteristics of the OECD Start-ups Database

	Number	Percentage (%)
From 1990	5 217 357	88.52
From 2000	4 750 583	80.60
Location*		
Africa	88 142	1.50
Asia & Oceania	871 929	14.79
Europe	1 500 467	25.46
Latin America	246 023	4.17
North America	2 154 112	36.55
Not available	1 033 574	17.54
Performance and founder information*		
Received financing	573 351	9.73
VC-backed	345 364	5.86
Patents	270 409	4.59
IPO	58 498	0.99
Acquired	208 035	3.53
Companies with founder information	978 364	16.60
Founders' information**		
At least one female founder	139 777	14.29
Has a university degree	476 401	48.69
Has a master's degree	247 059	25.25
Has a PhD degree	54 260	5.55

Note: This table provides descriptive statistics on the OECD Start-ups Database, including temporal coverage, geographical distribution, funding and innovation outcomes, and founder characteristics. Europe refers to all countries in the European continent. Table A C.1 shows the country aggregation. *Figures on location and performance and founder information, only include companies with country information (82.46% of the total). **Founder-related percentages are calculated as a share of companies with available founder information (16.60% of the total).

Source OECD Start-ups Database, June 2025.

Trends

After an overview of the available information in the data and its limitations, the section provides trends in the number of companies in the data, the types and volumes of financing they have received, the investors that provide the financing for these companies, and their innovation activities over the last two decades.

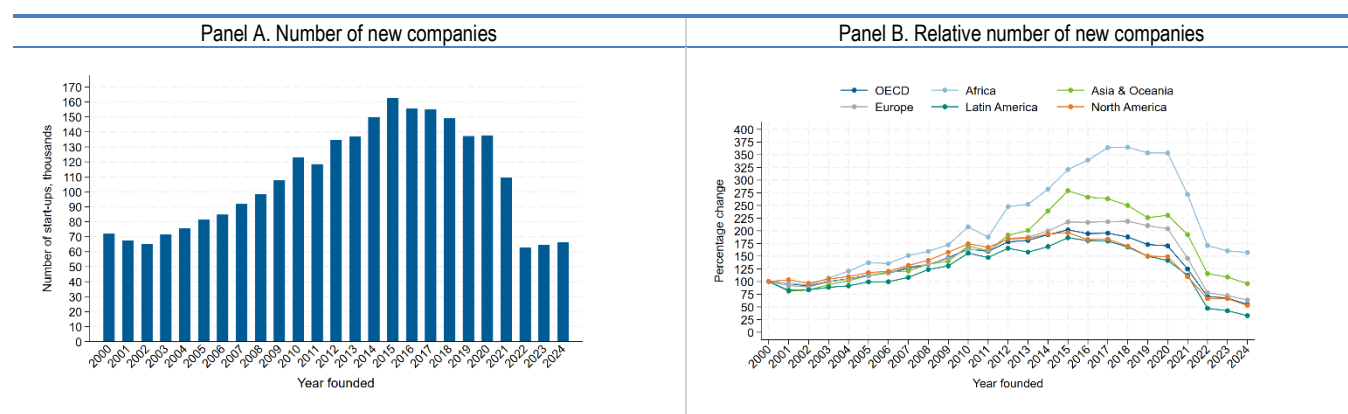
Number of companies

Figure 6.1 shows trends in company formation by founding year. Panel A shows the annual number of companies founded each year, while Panel B presents relative growth by location, indexed to 100 in 2000. Overall, the number of new companies increased steadily over the sample period until 2015. From that year onward, the number of new companies decreased, with a sharp drop in 2022. When looking across locations, Panel B shows some heterogeneity in the expansion of company formation. The number of new companies rose to a peak between 2015 and 2018 across all regions before declining. North America has the largest share of companies, while Asia & Oceania, Africa, and Europe have seen higher relative growth recently.

In addition to reflecting the underlying business dynamics, the patterns in Figure 6.1 may also be influenced by how companies enter the database. On average, it takes between three and four years for a company to appear in the data (see Box 6.1). To understand this registration lag, it is important to recall that neither Crunchbase nor Dealroom has access to administrative data from business registrations. Unlike

administrative records that register companies when they are born, Crunchbase and Dealroom likely register companies when they reach a milestone that attracts attention, for example, raising funding or launching a product or service. Similarly, data preceding the creation of the respective databases (for Crunchbase 2007, and 2013 for Dealroom) may be incomplete: many companies founded earlier might not have survived, and those that went out of business are unlikely to be recorded. Also, the heterogeneity across locations might reflect changes over time in the ability of the data providers to track companies in each location, rather than true start-up entry rates. Consequently, registration lags, potential survivorship bias, and location bias should be considered when interpreting company formation trends over time. These limitations are common to all non-administrative sources. Nevertheless, the datasets provide broad and timely coverage of innovative and high-growth start-ups across countries, making them particularly valuable for international analysis. In addition, as shown below, the sudden drop in 2022 does not appear among start-ups with funding information or those classified as “innovative start-ups”, suggesting that focusing on externally financed start-ups might alleviate these concerns.

Figure 6.1. Trends in company registration in the OECD Start-ups Database, by year and location



Note: This figure shows the trend in the number of new companies registered in the OECD Start-ups Database, by year and location. Panel A shows the annual number of new companies by founding year, while Panel B presents the relative growth in new company formation by location, indexed to 2000 (2000 = 100).

Source: OECD Start-ups Database, June 2025.

Funding

The OECD Start-ups Database includes information on the funding rounds of start-up companies. For each deal, the database contains information about the date, currency, deal value, deal type (e.g. venture capital, grants, convertibles, debt), and details about the company or individual investor involved in the deal, among other information. Even though trends in funding may also be subject to registration lags, they are less likely to capture this, as information on funding rounds is usually more readily available in public sources than business registrations.

Figure 6.3 shows the trends in start-up deals (Panel A) and deal volume (Panel B) by year and by location. Panel A shows that North American start-ups have the highest number of deals in the database, followed by Europe and Asia & Oceania. Although Europe has the second-highest number of deals, the figure shows that its deal volume starts falling below that of Asia & Oceania since the early 2020s. While Latin America and Africa begin from very low levels in both deal numbers and volumes, their growth is rapid and consistent, aligning with the increase in the number of start-ups covered. Overall, both the number of deals and the volume increase steadily across locations since the early 2000s, with large gains during the 2010s and a slowdown in recent years. Importantly, as mentioned in the previous paragraphs, the registration lag

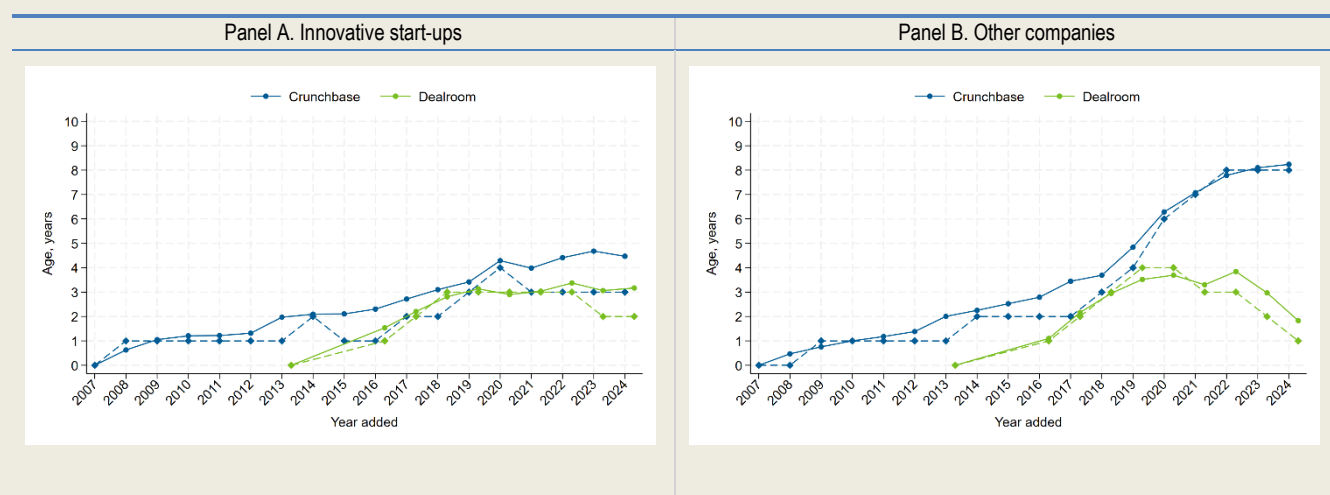
is less relevant for companies with funding information; therefore, the sharp decline observed in the broader sample in recent years is in the number of deals but not in the deal volume.

Box 6.1. Registration lag

As mentioned in the main text, companies are typically recorded in Crunchbase and Dealroom with a delay relative to their founding, creating a registration lag. Coverage is also incomplete for firms founded before the launch of each platform (2007 for Crunchbase; 2013 for Dealroom), which introduces survivorship bias and limits the completeness of backwards-looking data. Figure 6.2 presents the average (dot) and median (diamond) age at which companies were registered in Crunchbase or Dealroom, by year of registration and distinguishing between innovative start-ups (Panel A) and other companies (Panel B). Both databases tend to register innovative start-ups at an average age of about three years. For other companies, Dealroom follows a similar pattern, while Crunchbase tends to include companies with a higher average age at registration, suggesting that Crunchbase retrospective adds observations to a greater extent than Dealroom.

To sum up, in line with the average registration lag, both Crunchbase and Dealroom predominantly add young firms, with Crunchbase more actively incorporating older companies retroactively in recent years.

Figure 6.2. Average age of companies at registration, by innovative status

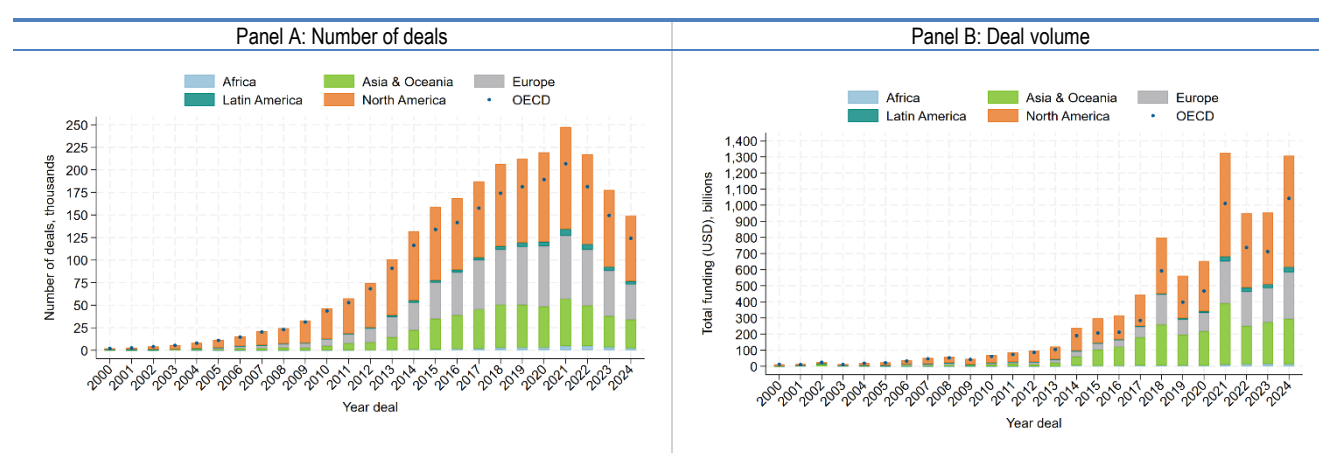


Note: This figure shows the average (dot) and median (diamond) age of companies at the time they were registered in Crunchbase or Dealroom, based on the year they were added to the database. Panel A focuses on innovative start-ups, defined as companies that received funding or filed a patent between 2000 and 2024, while Panel B covers all other companies.

Source: OECD Start-ups Database, June 2025.

As shown in Figure 4.2, most deals are present in both Crunchbase and Dealroom, especially the largest ones in terms of volume. However, combining both databases yields substantial gains, especially for European capital markets. Figure A C.4 in Annex C shows the number of deals and deal volume, respectively, by location over the last two decades. Panel A shows the funding by companies uniquely identified in Crunchbase, while Panel B shows the funding by companies uniquely identified by Dealroom. Clearly, Crunchbase has a comparative advantage in tracking funding activity in North America, while Dealroom provides stronger coverage of the European funding. In 2024 alone, companies identified exclusively in Crunchbase raised 160 billion USD in external financing worldwide, mostly from North America, while those identified only in Dealroom raised 80 billion USD, primarily from Europe and Asia & Oceania. These differences highlight the importance of utilising both sources and expanding coverage beyond North America.

Figure 6.3. Global trends in start-up deals and deal volumes, by location and year



Note: This figure presents global trends in start-up activity across locations and years. Panel A shows the number of financing deals, while Panel B displays total deal volumes in USD billions. All types of financing are included (e.g. venture capital, grants, and other forms). Each bar represents the annual totals segmented by location: Africa, Latin America, Asia & Oceania, Europe, and North America. OECD countries are additionally marked with small blue dots.

Source: OECD Start-ups Database, June 2025.

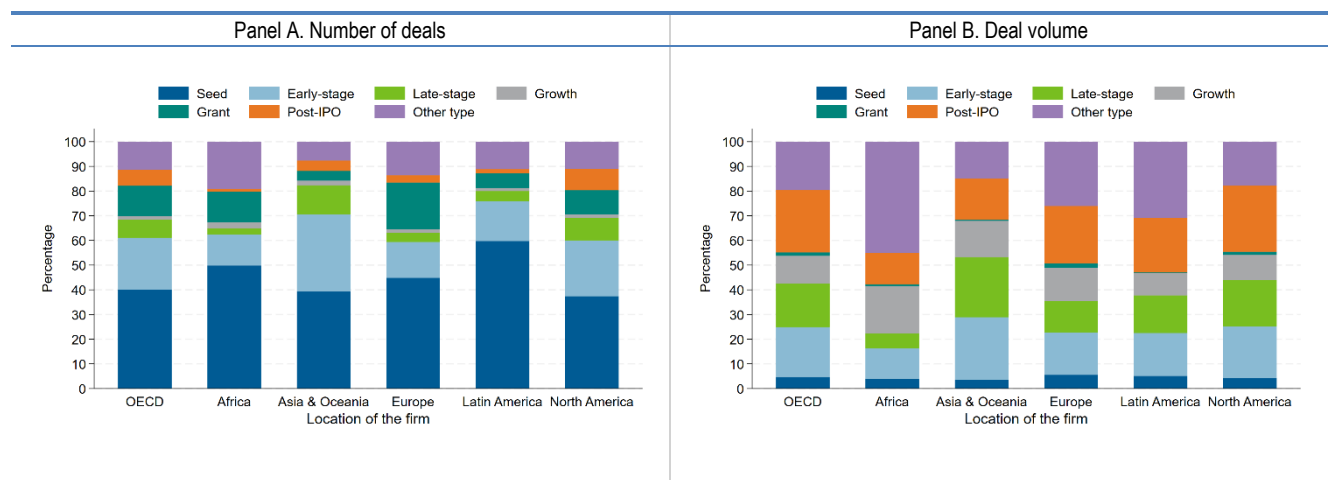
Not all deals in the OECD Start-ups Database include information on funding amounts. Figure A C.5 in Annex C shows the percentage of companies with missing information about their deal volume, by location and year. On average, 70% of deals in North America and Asia & Oceania have reported a funding amount, compared with slightly less than 60% in Europe. This may reflect the greater use of funding instruments other than venture capital in Europe, whose amounts are more difficult to identify. Encouragingly, the percentage of deals with reported deal volume has increased over time, reaching approximately 70% of deals by the end of the sample.

Zooming into the types of external financing included in the data, Figure 6.4 shows the percentage distribution of deal types within each location. In particular, the figure distinguishes between venture capital funding, which is further divided into the stage when the company receives the deal (i.e. seed, early-stage, growth, and late-stage), and other kinds of funding, divided into grants, post-IPO funding, and other kinds of funding (e.g. debt, convertibles, assistance).³⁵ Panel A shows the distribution of the number of deals, and Panel B shows the distribution of the deal volume by type.

The composition of deal types varies notably across locations. Venture capital deals account for approximately 70 to 80% of all deals in the database, except in Europe, where the percentage is close to 65%, and with higher percentages in Asia & Oceania and Latin America (Panel A). Within venture capital,

North America and Asia & Oceania present higher percentages of late-stage deals compared to other locations. While seed and early-stage deals are key for kick-start companies, later-stage deals are particularly important when it comes to the scale-up phase. These deals are also much larger than seed or early-stage deals in terms of the deal volume (Panel B). The differences between Panel A and Panel B illustrate the financing dynamics along the start-up's life cycle. Seed and early-stage deals are numerous but typically involve smaller amounts, whereas growth and late-stage financing are concentrated among fewer start-ups with substantially larger investments. This pattern reflects the increasing capital needs of more mature firms, combined with lower perceived risk as companies demonstrate traction and scalability (Arnold, 2024^[52]).

Figure 6.4. Distribution and volume of start-up deal types, by location



Note: This figure shows the composition of start-up financing by deal type across locations. Panel A shows the number of deals, while Panel B displays the corresponding deal volumes in USD billions. Deal types include seed, grant, early-stage, growth, late-stage, post-IPO, and other forms (e.g. debt, secondary market). Each bar represents 100% of deals in a given location, segmented by type.

Source: OECD Start-ups Database, June 2025.

The importance of alternative funding instruments also varies across locations. For instance, Panel A shows that in Europe, grants account for approximately 20% of the number of deals, showcasing the importance of accounting for start-ups relying on alternative sources of funding to capture the full entrepreneurial landscape in Europe. However, Panel B also shows that their deal volume is relatively small compared to other sources like venture capital, making them less suitable for financing the large amounts start-ups need to scale-up (Quas et al., 2022^[53]).

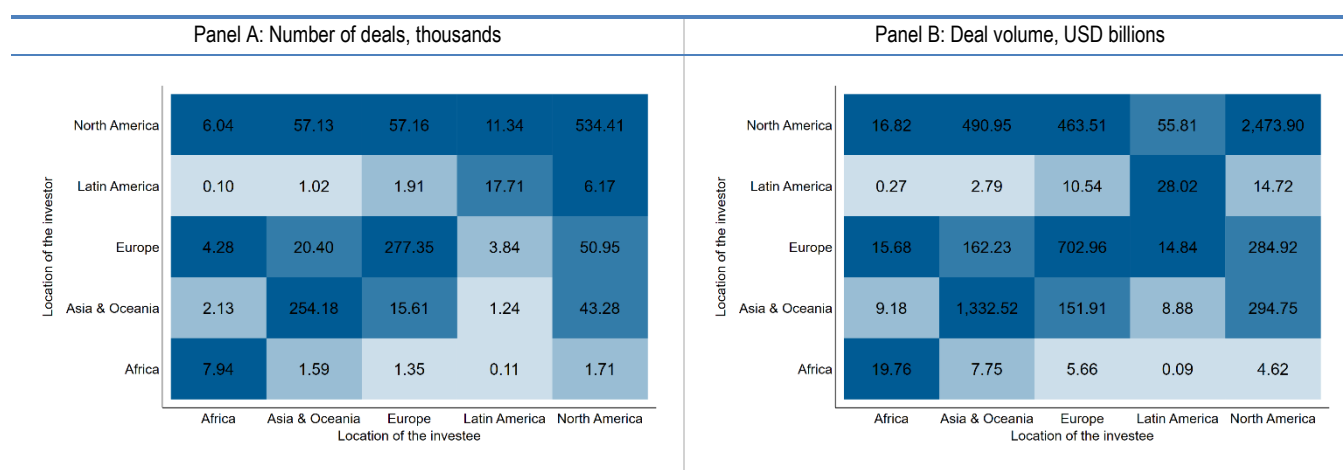
Investors

The OECD Start-ups Database also includes information regarding the investors. On average, each investor invests in 3.2 deals, with most investors taking part in one to three deals, and around a small minority (about 5%) investing in ten or more deals. This concentration suggests that few investors account for most deals in the data. However, investor information is available for approximately 73% of deals on average (more than six hundred thousand). As shown in Figure A C.6 in Annex C, the availability of investors' information varies by location and the year of the deal, with North America presenting the lowest percentage of deals with investors' information. There is no change in the percentage of deals with investors' information over the years.

Figure 6.5 shows the distribution of the number and volume of deals by the location of the investor and the investee in Panels A and B, respectively. In particular, each panel displays a matrix of investment flows,

where the rows represent the locations of investors and the columns represent the locations of companies. Each cell in Panel A shows the number of deals between the two, and Panel B shows the deal volume.³⁶ For instance, the first cell in Panel A would read as “6.04 thousand deals took place between investors in North America and companies in Africa.” Likewise, the first cell in Panel B would read as “Investments from North American investors in African companies amounted to 16.82 billion USD.” In particular, Panel A shows that deals typically occur within specific locations, with those in North America, Europe, and Asia & Oceania dominating the entrepreneurial landscape. There are 534.41 thousand deals involving both investors and investees based in North America, totalling USD 2 473.90 billion. In Europe, 277.35 thousand deals occurred between European investors and European companies. However, flows across locations are also sizeable. Approximately 14% of deals by European investors (50.95 thousand deals) target North American companies. Instead, North American investors directed 8% of their deals to companies in Asia & Oceania (57.13 thousand deals) and European companies (57.16 thousand deals).

Figure 6.5. Domestic and foreign investment: number and volume of deals, by location



Note: This figure presents the distribution of domestic and foreign investment activity across investor and investee locations. Panel A shows the number of deals (in thousands), and Panel B displays the corresponding deal volumes (in billions USD), distinguishing between domestic and foreign deals. Domestic investment refers to transactions where the investor and the investee company are based in the same location, while foreign investment refers to transactions where the investor is located in a different location. Each square reflects the level of activity, with colour intensity based on percentile ranges (20th, 40th, 60th, 80th, and 100th). Darker shades indicate higher percentiles. In Panel B, deal volume amounts are allocated proportionately to the number of listed investors in each location per deal, as individual investor contributions are not observed.

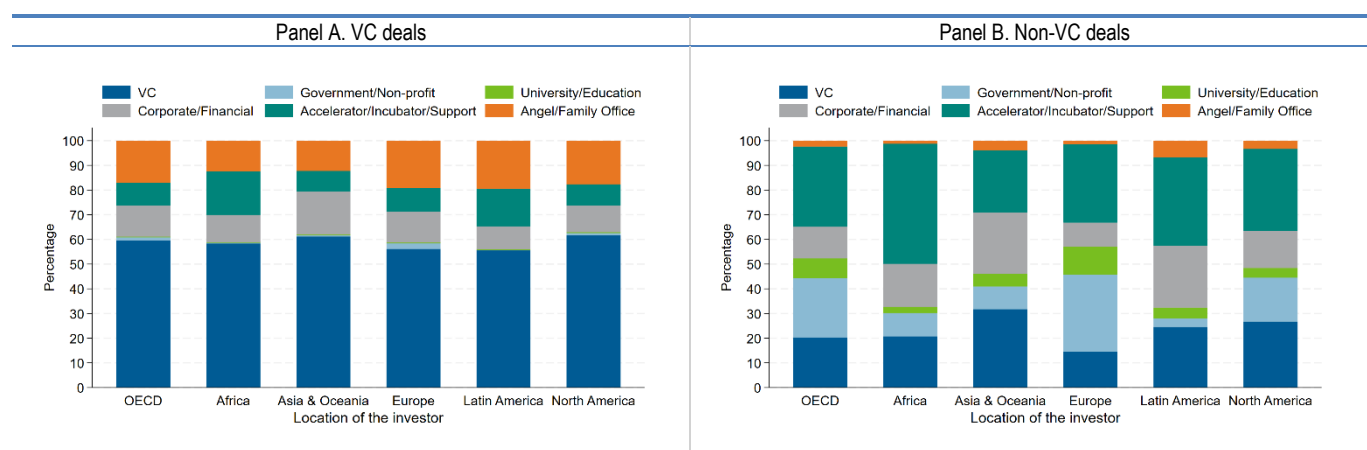
Source: OECD Start-ups Database, June 2025.

Several changes relative to the differences between Panel A and Panel B are worth highlighting. First, there is a shift in importance across geographical players between Panels A and B. While European investors and investees are ranked second in terms of the number of deals (Panel A), they are third when measured by deal volume (Panel B). Transactions within Asia & Oceania are approximately twice as large as those within Europe when considering deal volume. Second, cross-location deals become more significant when considering deal volume rather than the number of deals. While deals across locations represent a smaller proportion of the total number of deals, deals are, on average, larger. This pattern is especially pronounced among North American investors: only 8% of their deals involve companies in Europe or Asia & Oceania, but these locations account for approximately 13% and 14% of their total deal volume, respectively. North American investors also represent a significant share of investment in Africa and Latin America, surpassing domestic investors in those locations when measured by volume. Likewise, North American companies attract substantial inbound investment, receiving approximately 24% of flows from Europe and 17% from Asia & Oceania (including domestic deals).

Finally, the types of investors in the OECD Start-ups Database are analysed. As mentioned in Section 3, the types of investors used in Crunchbase and Dealroom use different conventions. Several manual checks are performed to assess the quality of these data and complement them. In general, having information on investor type is both valuable and relatively unique, but it should be used with caution. While major investors are typically identified correctly, the classification systems used in the source databases often blur the line between a company's ownership structure and its investment activity. For example, corporate or government venture capital entities are commonly categorised as venture capital firms in Dealroom. Crunchbase, however, sometimes labels them as traditional venture capital firms, and sometimes as government agencies (in the case of government VC) or corporates (in the case of CVC subsidiaries). This happens because these kinds of companies are both a venture capital firm and a government agency or a corporate arm. For these reasons, the OECD Start-ups Database has been augmented with information on government and corporate venture capital from other sources. In Annex C, further details are provided on the Crunchbase and Dealroom classification of investors, and how they are leveraged to build the aggregate investor types reported in Figure 6.6.

Figure 6.6. shows the distribution of investor types across locations by type of deal.³⁷ Venture capital deals are shown in Panel A, while the rest of the funding instruments are shown in Panel B.³⁸ Panel A shows that VC firms play a crucial role in financing start-ups, accounting for approximately 60% of all funds provided. Angel investors are, on average, the second-highest investors, closely followed by corporates.³⁹ Accelerators and incubators also play a relevant role, particularly in emerging locations. VC firms lose importance when it comes to alternative instruments of financing, as shown in Panel B, particularly in some locations like Europe. In this case, actors who play almost a negligible role in providing venture capital, like government agencies and universities, gain importance. Lastly, VC investors not only participate in traditional equity rounds but also provide non-VC funding through a variety of instruments. For example, VC investors are involved in debt financing, grants, and convertible notes, as well as other mechanisms like support programs.

Figure 6.6. Distribution of investor types, by location and deal type



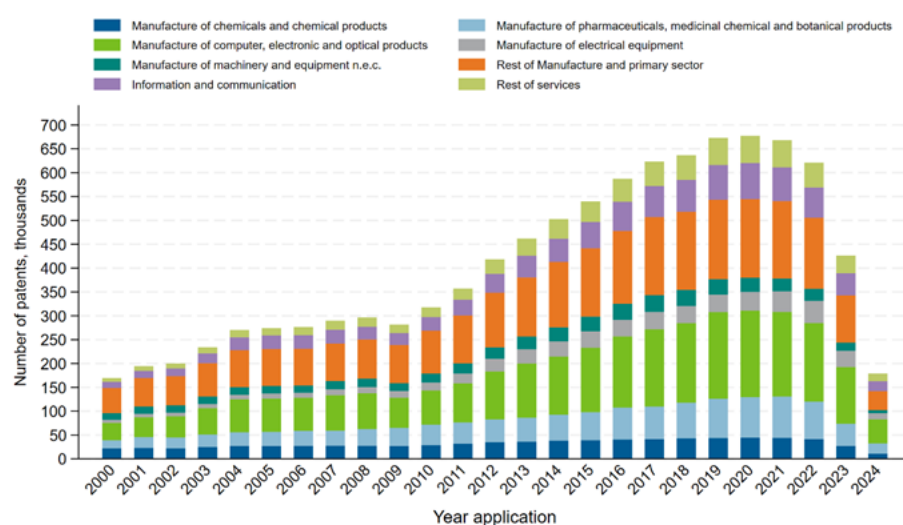
Note: This figure presents the share of investor types across locations, distinguishing between venture capital (VC) deals (Panel A) and non-VC deals (Panel B). A deal is classified as venture capital (VC) if it involves seed funding, early-stage investment, growth capital, or late-stage financing. Non-VC deals include those related to debt instruments, post-IPO transactions, grants, and other similar financial mechanisms. Investor categories include VC firms, universities and educational institutions, accelerators/incubators/support organisations, government and non-profits, corporate and financial investors, and angel/family offices. The percentage of companies receiving some funding represents 10% of all companies in the data and covers approximately 600 000 deals with available information on investor type. The total deal volume of non-VC deals and VC deals is 2 152 billion USD and 4 470 billion USD, respectively. Source: OECD Start-ups Database, June 2025.

Innovation

The OECD Start-ups Database is linked with Intellectual property data (patents and trademarks), offering a unique database that connects start-up companies and their patenting activity. The distribution of patent ownership among companies is highly skewed, with a small number of firms holding a disproportionately large share of patents. While the average number of patents per company is 14, the median is only three. Among companies that have filed at least one patent since 2000, roughly 30% hold just a single patent, whereas nearly 17% own ten or more.

Due to the detailed patent information in the OECD Start-ups Database, it is also possible to track how patent activity has evolved among the companies included. For example, Figure 6.7 shows the evolution of patenting activity over time, decomposing the overall figures according to the ISIC Rev. 4 industries, using the concordances between patent technology classes and industries created by Lybbert and Zolas (2014^[54]).⁴⁰ It shows that the number of patents filed by companies in the database has increased steadily over time, especially starting in 2010. In terms of relative importance, patents in ICT and other services have increased their weight from 10% to 20% from 2000 to 2024.⁴¹ This likely reflects (i) a shift of start-ups away from manufacturing towards services, (ii) the influence of the ICT revolution in the technological paradigm, and (iii) the Alice Corp jurisprudence in 2014, and its interpretation on software patentability in the US. Related to this, patents related to the manufacture of computers, electronic equipment, and optical products also increased substantially from 20% of total patents in 2000 to 30% by 2024. Nevertheless, a decline in the share of patents classified in manufacturing has been observed during the last two decades.

Figure 6.7. Patent filings, by ISIC Rev.4 classification and by filing year



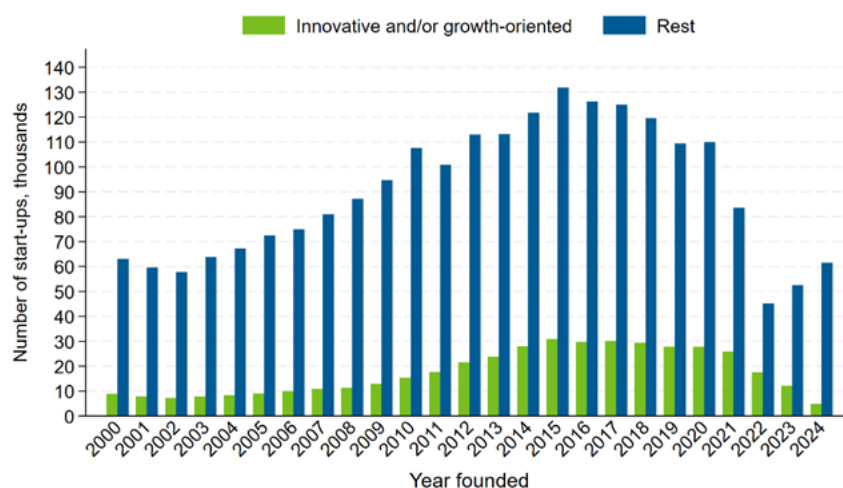
Note: This figure shows the number of patents filed annually from 2000 to 2024. Data from 2022 onwards are truncated. Each bar reports the probability-weighted sum of patent filings, with counts allocated across categories according to the IPC–ISIC concordance. The categories correspond to ISIC Rev. 4 classifications as follows: Manufacture of chemicals and chemical products (Industry 20, Section C), Manufacture of pharmaceuticals, medicinal chemical and botanical products (Industry 21, Section C), Manufacture of computer, electronic and optical products (Industry 26, Section C), Manufacture of electrical equipment (Industry 27, Section C), Manufacture of machinery and equipment n.e.c. (Industry 28, Section C), and Information and communication (Section J). “Rest of Manufacturing and primary sector” comprises all other divisions within Section C (Manufacturing) together with Sections A, B, D, E, and F. “Rest of services” covers all remaining service activities across Sections G–U (excluding Section J).

Source: OECD Start-ups Database, June 2025 and OECD IP. Database, August 2025.

Innovative start-ups

This section presents summary statistics based on a similar definition of innovative start-ups as in Berger et al. (2025^[12]) to illustrate the analytical potential of the OECD Start-ups Database. Berger et al. (2025^[12]) define an innovative start-up as a company that either received its first round of venture capital financing or filed its first patent during the sample period, and was less than ten years old at the time of that event. Similarly, this section defines innovative start-ups as those that received any type of funding (not limited to venture capital) or obtained a patent during the period from 2000 to 2024.⁴² They represent approximately 16% of all companies founded after 2000. shows an increase in the number of innovative start-ups in the last decade. The decay after the 2015 peak is still palpable for innovative start-ups, and it is relatively similar in proportion to that of the rest of the companies. However, the drop of 2022 shown for all companies is smoother when it comes to these innovative start-ups. As Box 6.1 shows, the abovementioned registration lag seems to affect the pattern of innovative start-ups less. As innovative start-ups are more active in the market due to their funding and/or innovative outputs, they are likely to be added to these databases more rapidly than other companies.

Figure 6.8. Number of new innovative start-ups registered in the OECD Start-ups Database, by year founded

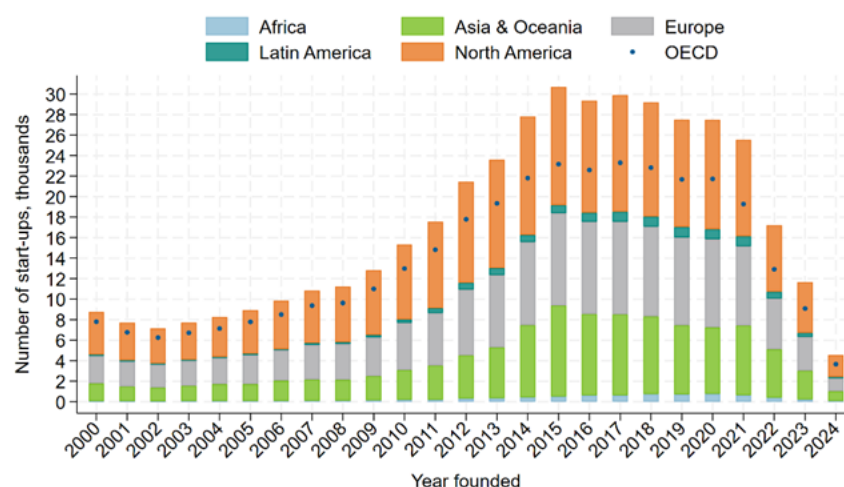


Note: This figure shows the annual number of start-ups founded between 2000 and 2024, distinguishing between those classified as innovative and/or growth-oriented and the rest. Innovative start-ups are defined as those that received any funding (not limited to venture capital) or received a patent over the period 2000 to 2024.

Source: OECD Start-ups Database, June 2025.

Table 6.2. shows that the majority of the companies in the OECD Start-ups Database are located in North America. Nevertheless, the distribution of companies is more dispersed across locations when zooming in on innovative start-ups. Figure 6.9 shows indeed that in the last years innovative start-ups are almost equally divided between North America, Europe, and Asia & Oceania. Additionally, most of the funding shown in Figure 6.3 is raised by innovative start-ups, as shown in Figure 6.10.

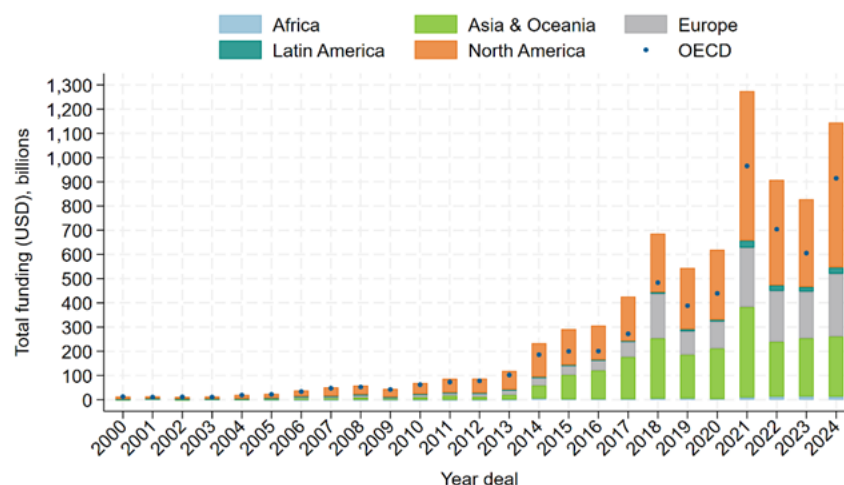
Figure 6.9. Number of innovative start-ups, by foundation year and by location



Note: This figure shows the annual number of innovative start-ups founded globally between 2000 and 2024, disaggregated by location. Innovative start-ups are defined as those that received any funding (not limited to venture capital) or received a patent over the period 2000 to 2024. Each bar represents the annual totals segmented by location: Africa, Latin America, Asia & Oceania, Europe, and North America. OECD countries are additionally marked with small blue dots.

Source: OECD Start-ups Database, June 2025.

Figure 6.10. Deal volume of innovative start-ups, by deal year and by location



Note: This figure shows the evolution of the deal volume received by innovative start-ups across years and locations. Innovative start-ups are defined as those that received any funding (not limited to venture capital) or received a patent over the period 2000 to 2024. Each bar represents the annual totals segmented by location: Africa, Latin America, Asia & Oceania, Europe, and North America. OECD countries are additionally marked with small blue dots.

Source: OECD Start-ups Database, June 2025.

7 Conclusion

The OECD Start-ups Database represents one of the richest and most comprehensive databases to analyse the global entrepreneurial ecosystem up to date. It overcomes many of the hurdles presented by traditional databases used in prior entrepreneurial literature. First, it overcomes the focus on venture capital-backed companies, which represent only a fraction of all entrepreneurial activity (National Venture Capital Association, 2025^[11]). Second, it extends the geographical coverage beyond the countries traditionally covered by prior literature (mostly the US). Finally, it provides researchers with a wide array of variables that enable an in-depth analysis of the worldwide entrepreneurial ecosystem.

The database's building blocks are Crunchbase and Dealroom, two proprietary data sources focused on providing insights into the entrepreneurial ecosystem with complementary geographical coverage, as shown in this paper. The rich data on start-up companies and their funding events are further complemented with information on their innovation indicators from patents, trademark data from the STI Microdata Lab, balance sheet and financial information from Orbis, transaction data from Orbis M&A, corporate venture capital links from Thomson Reuters, and, government venture capital activity based on a survey among the Delegates of the Committee on Industry, Innovation, and Entrepreneurship at the OECD. Overall, the OECD Start-ups Database provides researchers with the tools to study the determinants and outcomes of entrepreneurship over space, industries, and time.

The OECD Start-ups Database is benchmarked against the data of national and supra-national venture capital associations of the OECD countries and partner economies. This exercise shows that the OECD Start-ups Database consistently captures more venture capital activity than what is reported in aggregate statistics of these associations. While striking, this finding can be rationalised by differences in methodologies, criteria, and sources.

Finally, the descriptive statistics and trends presented in the report showcase the ability of the data to study these questions and shed light on important issues about the worldwide entrepreneurial ecosystem. The global entrepreneurial ecosystem is dominated by companies located in North America and Europe, but other locations are catching up at a fast pace. Among these locations, companies in Asia are standing out, both in terms of the number of start-ups and funding attracted. In terms of funding, venture capital provided by venture capital funds and other financial institutions remains among the most relevant sources of financing for innovative start-ups. However, this dominance presents heterogeneities across locations, and across other sources of funding, like grants or assistance, are also relevant, especially outside North America. Furthermore, the entities providing these alternative sources of funding are more varied than in the case of venture capital, and are less concentrated in financial institutions, but are split among non-financial corporations, governments, and universities. Nevertheless, investors providing venture capital and alternative sources of funding remain strongly concentrated within the country of the start-ups.

To conclude, this report highlights the exceptional coverage and quality of the OECD Start-ups Database, which succeeds in capturing the dynamics of entrepreneurial ecosystems across locations and over time with a level of detail and consistency unmatched by other sources.

Endnotes

¹ Cunchbase: <https://www.crunchbase.com/>; Dealroom: <https://dealroom.co/>

² For example, other entities that provide venture capital funding include corporate and financial institutions, government and non-profit agencies, angel investors, accelerators, and incubators, among others.

³ There is no universally accepted definition of a start-up. Researchers typically relied on the available data variables to define them.

⁴ In addition to the micro-level sources mentioned in this section, several sources provide aggregate statistics on venture capital activity at the national or locational level. For example, the National Venture Capital Association provides aggregate trends and statistics for the United States. InvestEurope (formerly the European Venture Capital Association) provides aggregate statistics for European countries. The Latin America Venture Capital Association provides these statistics for Latin American countries, and the Asian Venture Capital Journal provides statistics for Asian economies. While these statistics are useful for capturing trends and are usually used as motivating facts, many of these rely on the aforementioned databases for their elaboration and have a limited scope to study company-level phenomena. For more details on these data, see Section 5, where the OECD Start-ups Database is benchmarked against these aggregate statistics.

⁵ These funding types are relevant for prominent start-ups, including green start-ups (Dechezleprêtre and Kelly, 2025^[15]; Hottenrott, 2024^[55]) or academic start-ups (Berger, Dechezleprêtre and Kirpichev, 2026^[23]).

⁶ 20 countries in MultiProd and 16 countries in DynEmp, currently expanding to another 8 members, but both excluding the US.

⁷ <https://www.insee.fr/fr/metadonnees/source/serie/s1271>.

⁸ Bottazzi, Da Rin, and Hellmann (2008^[36]) collected data over the period 1998-2001 for venture capital deals in 17 European countries, covering 119 venture capital companies, 503 partners, and 1 652 portfolio companies. Hellmann and Puri (2002^[37]) and Hsu (2004^[38]) obtained data on 173 Silicon Valley technology start-ups and 149 MIT start-ups, respectively.

⁹ The Kauffman Firm Survey was a panel study that tracked approximately 5 thousand US-based businesses that were founded in 2004 – drawn from a stratified random sample – over the first 8 years of their activity with an annual survey (Farhat and Robb, 2017^[57]). The goal of the Kauffman Firm survey was to provide new insights on the nature of new business formation, the characteristics and strategies of companies, their employment patterns, their financial arrangements, and the characteristics of their

founders (Robb et al., 2009^[60]). In a similar vein, the IAB/ZEW Start-up Panel is a longitudinal and representative survey of German businesses, designed to provide insights into the founders and their start-ups in the early stages of the life cycle. Unlike the Kauffmann Firm Survey, the IAB/ Start-up Panel is organised as a cohort-based replacement sample that adds newly founded companies each year. Established in 2008, it includes information on approximately 40 000 companies founded between 2005 and 2024, and covers various dimensions relevant to the founding process, such as founder characteristics, entrepreneurial strategies, financing arrangements, and innovation activities (Rodepeter, Hottenrott and Gottschalk, 2025^[56]).

¹⁰ Angellist and Tracxn have been much less exploited in the literature, and existing comparative studies indicate that their coverage is worse than that of the competitor databases. Capital IQ is a much broader database that provides information on financials, and whose aim is not only to focus on start-ups. CB Insights has a similar focus to Crunchbase, but Crunchbase has dominated in the academic field over the last years (Retterath and Braun, 2020^[8]).

¹¹ Nevertheless, these datasets do not capture fully the universe of start-ups either. They focus on relevant innovation- and growth-oriented companies (i.e. start-ups) and the main entities in their ecosystems (i.e. competitors, corporates, universities, investors, etc.), and rely largely on public sources, which may result in incomplete coverage of relevant start-ups. Despite these limitations, the OECD Start-ups Database represents a step ahead of previous sources by capturing a broader and more comprehensive set of relevant companies.

¹² In particular, ThomsonOne and Venture Source are heavily biased towards the manufacturing and computers and software sectors, as these were the most relevant technologies at the time of inception of the databases.

¹³ The dataset includes firms retroactively, meaning it captures companies that were founded several years before being added to the database. For example, the average age of entities incorporated in the dataset is eight years old in 2024.

¹⁴ Venture capital companies that are part of Crunchbase's Venture Programme update monthly information to Crunchbase in exchange for access to the database.

¹⁵ In Crunchbase 40% of entities come from the US, while 26% of entities come from.

¹⁶ 24% entities come from Europe, while 28% from the US. Although the number of start-ups in Europe is lower in the data, this likely reflects the smaller overall start-up universe in Europe compared to the US.

¹⁷ The data exchange programme is called Ecosystem Platform, and it consists of partnerships with local governments.

¹⁸ The current version of the OECD Start-ups Database is based on EPO PATSTAT Global – Spring 2025 edition (see <https://www.epo.org/en/searching-for-patents/business/patstat> for further information on the database and its coverage).

¹⁹ The industry codes are NACE Rev. 2 from the Statistical Classification of Economic Activities in the European Community, in its second revision.

²⁰ An initial list of institutions providing government assistance to innovation-oriented start-ups was collected from different commercial databases (Crunchbase, Dealroom, Preqin, and Pitchbook), open-source data available from previous academic literature, and data from Invest Europe. To ensure the quality of the list, delegates of OECD Member countries to the Committee on Innovation, Industry and Entrepreneurship (CIIE) were invited to validate the results. Delegates could review the institutions, correct any information, and add funds that were not included in the previous list. The validation exercise was carried out between July 2020 and December 2021 and was completed by all OECD Member countries.

²¹ Thomson Reuters' Eikon platform, which was part of Refinitiv, Thomson Reuters' former Financial & Risk business. Refinitiv was acquired by the London Stock Exchange Group (LSEG) in 2021, and Eikon has since been phased out in favour of LSEG Workspace.

²² For the current version of this paper, Crunchbase and Dealroom data refer to data downloaded on 22 May 2025.

²³ Although Mexico is geographically part of North America, it is included in Latin America. This is because the data for Mexico in the benchmarking exercise are collected by the Latin American Venture Capital Association (LAVCA), and the benchmarking comparisons are conducted at the Latin American level. See Table A C.1 for the aggregation of countries.

²⁴ The reason to prioritise Crunchbase as the primary data source is due to its widespread adoption in academic research, making it a well-established benchmark in the literature. See Dalle, den Besten, and Menon (2017^[9]).

²⁵ For reference, aggregate data from the US Economic Census show that, on average, only 5.5% of companies in the manufacturing sector own patents (Natarajan and Sivadas, 2011^[58]). Focusing on start-ups, Fons-Rosen, Roldán-Blanco, and Schmitz (2022^[51]) show in a sample covering the universe of new registered businesses in the US that patenting firms represent 0.2% of the total population of new businesses. Regarding the number of companies that engage in financial transactions with other companies, Didier, Levine, and Schmukler (2016^[59]) show that in the median country, only 20 listed firms per year issue securities (out of a sample of 50 thousand firms). Likewise, Fons-Rosen, Roldán-Blanco and Schmitz (2022^[51]) show that only 0.06% of all new registered businesses get acquired by age 6. Likewise, the North American Venture Capital Association states that only 0.1% of all companies obtain VC funds. For this reason, the percentages shown in Figure 4.3 seem relatively high compared to the universe of occurrences of patenting firms and financial transactions in capital markets.

²⁶ Growth equity and late-stage venture capital are included in the benchmark sources.

²⁷ OECD Data Explorer [Consumer price indices](#) and [Annual Purchasing Power Parities and exchange rates](#).

²⁸ Another possibility is that Crunchbase and Dealroom may occasionally reflect announced deals that were never materialised. Importantly, this limitation is not unique to these sources, as other commonly used benchmarking databases – such as PitchBook in the United States and Preqin in Australia – face similar challenges.

²⁹ The list of countries included in Europe is presented in Figure 6.1.

³⁰ Figure A B.2 in Annex B shows the equivalent exercise for the number of deals. When comparing both figures, the number of deals recorded in the data is mostly higher in the benchmark data. Nevertheless, the funding amounts were fairly similar. This entails that in some countries, like Japan and Korea, the creation of Dealroom allowed for a better collection of deals that were hardly detected by Crunchbase.

³¹ To ensure that companies also tagged as investors are true corporates, several checks are carried out. First, a manual check is carried out on a random sample of companies tagged as companies and investors. More than 90% were actual corporates, while the rest were VC firms. Second, the industry tags provided in Dealroom and Crunchbase are analysed, showing that these companies are not concentrated in the financial sector.

³² For instance, the number of new registered businesses in the United States in 2024 alone approaches the overall number of US companies included in the data.

³³ Table A C.4 and Table A C.5 provide more statistics, by location and by the year founded.

³⁴ See Table A C.1 for the aggregation of countries.

³⁵ Deal types are classified according to the stage of venture financing and the deal volume. “Other type” comprises deals outside the traditional venture capital stages, such as “Debt Financing”, “Grant”, “Post-IPO Debt”, “Private Placement”, and other non-equity or hybrid instruments. See Annex C for more details.

³⁶ The aggregate number of deals may not add up to the one reported above because each deal can involve multiple investors from several countries, which will be counted in all the relevant cells. For instance, one deal that features a North American company and one European and one African investor will be counted twice, first in the Europe-North America cell and another time in the Africa-North America cell. For Panel B, the cells are filled using fractional counting. The OECD Start-ups Database reports funding amounts but does not distinguish how much was provided by each investor participating in the deal.

³⁷ A deal is classified as venture capital (VC) if it involves seed funding, early-stage investment, growth capital, or late-stage financing. Non-VC deals include those related to debt instruments, post-IPO transactions, grants, and other similar financial mechanisms.

³⁸ The percentage of companies receiving some funding represents 10% of all companies in the data and covers approximately 600 000 deals with available information on investor type. The total deal volume of non-VC deals and VC deals is 2 152 billion USD and 4 470 billion USD, respectively.

³⁹ It is worth highlighting that financial institutions other than investment companies are classified as corporates.

⁴⁰ The International Standard Industrial Classification of All Economic Activities (ISIC) consists in a coherent and consistent classification structure of economic activities based on a set of internationally agreed concepts, definitions, principles and classification rules. It provides a comprehensive framework within which economic data can be collected and reported in a format that is designed for purposes of economic analysis, decision-making and policy-making. It comprises 21 sections (A–U) and 88 industries (two-digit codes). More information can be found in: <https://unstats.un.org/unsd/classifications/Econ/isic>. Lybbert and Zolas (2014^[54]) assign each IPC code (technological sub-classes of the patent) to at least one

ISIC Rev.4 category, with an associated probability weight, and the weights for a given IPC code sum to one.

⁴¹ Within services, industries 59, 61, 74, and 86 have been the main drivers of growth. Industry 59: Motion picture, video and television programme production, sound recording and music publishing activities (Section J); Industry 61: Telecommunications (Section J); Industry 74: Other professional, scientific and technical activities (Section M); Industry 86: Human health activities (Section Q).

⁴² Excluding post-IPO.

References

- Almus, M., D. Engel and S. Prantl (2000), "The ZEW Foundation Panels and the Mannheim Enterprise Panel (MUP) of the Centre for European Economic Research (ZEW)", *Journal of Applied Social Science Studies*, Vol. 120/2, pp. 301-308, <https://doi.org/10.3790/schm.120.2.301>. [34]
- Andrews, R. et al. (2022), "The Startup Cartography Project: Measuring and mapping entrepreneurial ecosystems", *Research Policy*, Vol. 51/2, p. 104437, <https://doi.org/10.1016/j.respol.2021.104437>. [33]
- Arnold, N. (2024), "Stepping Up Venture Capital to Finance Innovation in Europe", *IMF Working Papers*, Vol. 2024/146, p. 1, <https://doi.org/10.5089/9798400280771.001>. [52]
- Bajgar, M. et al. (2020), "Coverage and representativeness of Orbis data", *OECD Science, Technology and Industry Working Papers*, No. 2020/06, OECD Publishing, Paris, <https://doi.org/10.1787/c7bdaa03-en>. [48]
- Berger, M. et al. (2025), "Acquisitions and their effect on start-up innovation: Stifling or scaling?", *OECD Science, Technology and Industry Working Papers*, Vol. 2025/10, <https://doi.org/10.1787/b4efd3ab-en>. [12]
- Berger, M. et al. (Forthcoming), "Corporate Venture Capital and Start-up Innovation in the Digital Age", *OECD Science, Technology and Industry Working Papers*, OECD Publishing, Paris. [22]
- Berger, M., C. Criscuolo and A. Dechezleprêtre (2025), "Understanding Government Venture Capital: A primer and a taxonomy", *OECD Science, Technology and Industry Working Papers*, No. 2025/06, OECD Publishing, Paris, <https://doi.org/10.1787/2157a5b3-en>. [14]
- Berger, M., A. Dechezleprêtre and M. Fadic (2024), "What is the role of Government Venture Capital for innovation-driven entrepreneurship?", *OECD Science, Technology and Industry Working Papers*, No. 2024/10, OECD Publishing, Paris, <https://doi.org/10.1787/6430069e-en>. [18]
- Berger, M., A. Dechezleprêtre and D. Kirpichev (Forthcoming), "Academic start-ups' funding and the role of alternative funding and support instruments", *OECD Science, Technology and Industry Working Papers*, OECD Publishing, Paris. [23]
- Berger, M., A. Dechezleprêtre and R. Verlhac (2025), "Did the COVID-19 crisis uplift business dynamics and venture capital financing?", *OECD Science, Technology and Industry Working Papers*, No. 2025/15, OECD Publishing, Paris, <https://doi.org/10.1787/95c2c9cd-en>. [13]
- Block, J. et al. (2014), "Trademarks and Venture Capital Valuation", *Journal of Business Venturing*, Vol. 29/4, pp. 525-542, <https://doi.org/10.1016/j.jbusvent.2013.07.006>. [30]

- Bottazzi, L., M. Da Rin and T. Hellmann (2008), "Who are the active investors? Evidence from venture capital?", *Journal of Financial Economics*, Vol. 89/3, pp. 488-512, <https://doi.org/10.1016/j.jfineco.2007.09.003>. [36]
- Breschi, S., J. Lassébie and C. Menon (2018), "A portrait of innovative start-ups across countries", *OECD Science, Technology and Industry Working Papers*, No. 2018/2, OECD Publishing, Paris, <https://doi.org/10.1787/f9ff02f4-en>. [10]
- Calligaris, S. et al. (2023), "Is there a trade-off between productivity and employment?: A cross-country micro-to-macro study", *OECD Science, Technology and Industry Policy Papers*, No. 157, OECD Publishing, Paris, <https://doi.org/10.1787/99bede51-en>. [35]
- Calvino, F., C. Criscuolo and C. Menon (2018), "A cross-country analysis of start-up employment dynamics", *Industrial and Corporate Change*, Vol. 27, pp. 677-698. [25]
- Castaldi, C. (2023), "Off the mark? What we (should) know about the bright and dark sides of corporate trademark practices", *Industrial and Corporate Change*, Vol. 32/5, pp. 1046-1062, <https://doi.org/10.1093/icc/dtad011>. [46]
- Catalini, C., J. Guzman and S. Stern (2019), *Hidden in Plain Sight: Venture Growth with or without Venture Capital*, National Bureau of Economic Research, Cambridge, MA, <https://doi.org/10.3386/w26521>. [39]
- Cervantes, M. et al. (2023), "Driving low-carbon innovations for climate neutrality", *OECD Science, Technology and Industry Policy Papers*, No. 143, OECD Publishing, Paris, <https://doi.org/10.1787/8e6ae16b-en>. [20]
- Commission, E. (ed.) (2024), *Green Start-ups*, Publications Office of the European Union, <https://doi.org/10.2777/965670>. [55]
- Cunningham, C., F. Ederer and S. Ma (2021), "Killer Acquisitions", *Journal of Political Economy*, Vol. 129/3, pp. 649-702, <https://doi.org/10.1086/712506>. [32]
- Dalle, J., M. den Besten and C. Menon (2017), "Using Crunchbase for economic and managerial research", *OECD Science, Technology and Industry Working Papers*, No. 2017/08, OECD Publishing, Paris, <https://doi.org/10.1787/6c418d60-en>. [9]
- Dechezleprêtre, A. et al. (2024), "A comprehensive overview of the renewable energy industrial ecosystem", *OECD Science, Technology and Industry Working Papers*, No. 2024/11, OECD Publishing, Paris, <https://doi.org/10.1787/94dce592-en>. [19]
- Dechezleprêtre, A. et al. (2023), "How the green and digital transitions are reshaping the automotive ecosystem", *OECD Science, Technology and Industry Policy Papers*, No. 144, OECD Publishing, Paris, <https://doi.org/10.1787/f1874cab-en>. [21]
- Dechezleprêtre, A. and P. Kelly (2025), "The New Green Economy: Venture Capital, Innovation and Business Success in Cleantech Startups", *OECD Science, Technology and Industry Working Papers*, No. 2025/13, OECD Publishing, Paris, <https://doi.org/10.1787/ba73f647-en>. [15]
- Decker, R. et al. (2014), "The role of entrepreneurship in US job creation and economic dynamism", *Journal of Economic Perspectives*, Vol. 28, pp. 3-24. [26]

- Desnoyers-James, I., S. Calligaris and F. Calvino (2019), "DynEmp and MultiProd: Metadata", *OECD Science, Technology and Industry Working Papers*, No. 2019/03, OECD Publishing, Paris, <https://doi.org/10.1787/3dcde184-en>. [7]
- Didier, T., R. Levine and S. Schmukler (2016), "Capital market financing, firm growth, and firm size distribution", *European Systemic Risk Board Working Paper*, Vol. 4, <https://doi.org/10.2139/ssrn.3723351>. [59]
- Ederer, F. and B. Pellegrino (2023), "The Great Start-up Sellout and the Rise of Oligopoly", *AEA Papers and Proceedings*, Vol. 113, pp. 274-278, <https://doi.org/10.1257/pandp.20231024>. [50]
- Farhat, J. and A. Robb (2017), "Analyzing complex survey data: the Kauffman Firm Survey", *Small Business Economics*, Vol. 50/3, pp. 657-670, <https://doi.org/10.1007/s11187-017-9913-3>. [57]
- Fons-Rosen, C., P. Roldán-Blanco and T. Schmitz (2022), "The effect of startup acquisitions on innovation and economic growth", *CEPR Discussion Papers*, Vol. 17752, <https://doi.org/10.2139/ssrn.3785485>. [51]
- George M. Constantinides, M. (ed.) (2013), *A Survey of Venture Capital Research*, Elsevier, <https://doi.org/10.1016/B978-0-44-453594-8.00008-2>. [5]
- Grimaldi, R. et al. (2011), "30 years after Bayh-Dole: Reassessing academic entrepreneurship", *Research Policy*, Vol. 40, pp. 1045-1057. [24]
- Gugler, K., F. Szucs and U. Wohak (2025), "Start-up acquisitions, venture capital and innovation: A comparative study of Google, Apple, Facebook, Amazon and Microsoft", *International Journal of Industrial Organisation*, Vol. 99, pp. 103-148, <https://doi.org/10.1016/j.ijindorg.2025.103148>. [41]
- Haltiwanger, J. et al. (eds.) (2017), *High Growth Young Firms: Contribution to Job, Output, and Productivity Growth*, University of Chicago Press. [2]
- Haltiwanger, J., R. Jarmin and J. Miranda (2013), "Who creates jobs? Small versus large versus young", *Review of Economics and Statistics*, Vol. 95, pp. 347-361. [1]
- Hellmann, T. and M. Puri (2002), "Venture Capital and the Professionalization of Start-Up Firms: Empirical Evidence", *The Journal of finance*, Vol. 57/1, pp. 169-197, <https://doi.org/10.1111/1540-6261.00419>. [37]
- Hess, S. (2025), "Empirical entrepreneurial ecosystem research: A guide to creating multilevel datasets", *Journal of Business Venturing Insights*, Vol. 23, p. e00511, <https://doi.org/10.1016/j.jbvi.2024.e00511>. [27]
- Hochberg, Y., C. Serrano and R. Ziedonis (2018), "Patent collateral, investor commitment, and the market for venture lending", *Journal of Financial Economics*, Vol. 130/1, pp. 74-94, <https://doi.org/10.1016/j.jfineco.2018.06.003>. [31]
- Hsu, D. (2004), "What Do Entrepreneurs Pay for Venture Capital Affiliation?", *The Journal of Finance*, Vol. 59/4, pp. 1805-1844, <https://doi.org/10.1111/j.1540-6261.2004.00680.x>. [38]
- Jarmin, R. and J. Miranda (2002), "The Longitudinal Business Database", *Centre for Economic Studies Discussion Paper CES-WP-02-17*, <https://doi.org/10.2139/ssrn.2128793>. [6]

- Kaplan, S. and J. Lerner (2016), "Venture Capital Data: Opportunities and Challenges", *NBER Working Paper* 22500, <https://doi.org/10.3386/w22500>. [4]
- Kortum, S. and J. Lerner (2000), "Assessing the Contribution of Venture Capital to Innovation", *RAND Journal of Economics*, Vol. 31/4, pp. 674-692, <https://doi.org/10.2307/2696354>. [44]
- Lybbert, T. and N. Zolas (2014), "Getting patents and economic data to speak to each other: An 'Algorithmic Links with Probabilities' approach for joint analyses of patenting and economic activity", *Research Policy*, Vol. 43/3, pp. 530-542, <https://doi.org/10.1016/j.respol.2013.09.001>. [54]
- Mann, R. and T. Sager (2007), "Patents, Venture Capital, and Software Start-ups", *Research Policy*, Vol. 36/2, pp. 193-208, <https://doi.org/10.1016/j.respol.2006.10.002>. [28]
- Masulis, R. and R. Nahata (2009), "Financial contracting with strategic investors: Evidence from corporate venture capital backed IPOs", *Journal of Financial Intermediation*, Vol. 18/4, pp. 599-631, <https://doi.org/10.1016/j.jfi.2009.06.001>. [29]
- Matts, F. et al. (2011), "On the Consistency and Reliability of Venture Capital Databases", *Unpublished manuscript*. [43]
- Millot, V. (2009), "Trademarks as an Indicator of Product and Marketing Innovations", *OECD Science, Technology and Industry Working Papers*, No. 2009/6, OECD Publishing, Paris, <https://doi.org/10.1787/224428874418>. [47]
- Natarajan, B. and J. Sivadas (2011), "What happens when firms patent? New evidence from U.S. Economic Census data", *The Review of Economics and Statistics*, Vol. 93/1, pp. 126-146, https://doi.org/10.1162/REST_a_00058. [58]
- National Venture Capital Association (2025), *2025 Yearbook*. [11]
- OECD (2025), "A review of Slovenia's industrial strategy: Policies for a green, innovative, and smart economy", *OECD Science, Technology and Industry Policy Papers*, No. 177, OECD Publishing, Paris, <https://doi.org/10.1787/f10e4ef9-en>. [17]
- OECD (2025), *Measuring Science and Innovation for Sustainable Growth*, OECD Publishing, Paris, <https://doi.org/10.1787/3b96cf8c-en>. [16]
- Quas, A. et al. (2022), "The scale-up finance gap in the EU: Causes, consequences, and policy solutions", *European Management Journal*, Vol. 40/5, pp. 645-652, <https://doi.org/10.1016/j.emj.2022.08.003>. [53]
- Retterath, A. and R. Braun (2020), "Benchmarking Venture Capital Databases", *Unpublished manuscript*, <https://doi.org/10.2139/ssrn.3706108>. [8]
- Robb, A. et al. (2009), "An Overview of the Kauffman Firm Survey: Results from the 2004-2007 Data", <https://doi.org/10.2139/ssrn.1392292>. [60]
- Rodepeter, E., H. Hottenrott and S. Gottschalk (2025), "The IAB/ZEW Start-up Panel", *Journal of Economics and Statistics*, <https://doi.org/10.1515/jbnst-2025-0023>. [56]
- Schnitzer, M. and M. Watzinger (2022), "Measuring the Spillovers of Venture Capital", *The Review of Economics and Statistics*, Vol. 104/2, pp. 276-292, https://doi.org/10.1162/rest_a_00937. [3]

- Squicciarini, M., H. Dernis and C. Criscuolo (2013), “Measuring Patent Quality: Indicators of Technological and Economic Value”, *OECD Science, Technology and Industry Working Papers*, No. 2013/3, OECD Publishing, Paris, <https://doi.org/10.1787/5k4522wkw1r8-en>. [45]
- Tarasconi, G. and C. Menon (2017), “Matching Crunchbase with patent data”, *OECD Science, Technology and Industry Working Papers*, <https://doi.org/10.1787/15f967fa-en>. [49]
- Veugelers, R. and S. Amaral-García (2025), *Financing EU Health Innovation: The Role of Venture Capital*. [42]
- Wang, X. (2018), “Catering Innovation: Entrepreneurship and the Acquisition Market”, *Kenan Institute of Private Enterprise Research Paper No. 18-27*, <https://doi.org/10.2139/ssrn.3247274>. [40]

Annex A. Additional details on the link between data

Linking Crunchbase and Dealroom

Linking company-level information

Crunchbase and Dealroom entities were linked using commonly available information on both sides, relying mainly on the company name and other information such as internet links of the company, addresses, or foundation dates. The following steps were performed to link and disambiguate companies and investors across the two sources:

1. **Linking records using similar URLs:** The internet address (URL) of a given company is likely to remain stable, as a unique entry point to which customers and partners can refer over time. Both Crunchbase and Dealroom databases provide lists of URLs that point to the company's homepage in over 96% of cases, and/or to LinkedIn, Twitter, or Facebook accounts.

First, conditional on having the same country code across databases, the records are linked based on the homepage address. The homepage addresses are cleaned to identify the domain name (e.g. removing the "http://" or trailing parts of the homepage addresses). Second, Crunchbase links are sometimes provided in Dealroom (22% of records). Conditional on having the same country code, these are used for the observations not linked using the homepage URLs. Third, social media URLs (LinkedIn, Facebook, and X) are used to link the rest of the entities, again after undergoing standard URL-cleaning and conditional on reporting to the same country code.

Once the URL-based linkages have been established, overlapping records were retained in the database if and only if each company on one side is not linked to more than three different companies (to prevent relying on very generic URLs). In case of many-to-many matches, string-matching comparison of the company's name was performed to help disambiguate duplicated linkages (more below). The address information (city and, when available, postal code), as well as the foundation date, were used to de-duplicate records.

2. **Linking records using name similarity:** Among all records that were not matched using the available URLs, data were linked using string-matching algorithms, after a short harmonisation step (capitalisation, removal of punctuation, use of dictionaries to remove endings like "Ltd.," "Corp.," etc.). For this exercise, the algorithm relies on two Levenshtein edit distances ("Indel" and "Discounted Levenshtein") provided in Python's "name-matching" package. For each company or investor name available in Dealroom, the algorithm was set up to retrieve up to 10 possible matched pairs identified in Crunchbase, using a threshold similarity score, on a country-by-country basis. Complementary information, such as city names, postal codes, and foundation dates, was also used to disambiguate the match.
3. **Linking records using website similarity (for observations without country codes):** The previous steps relied on the companies to be from the same country. For those companies without country codes, the matching algorithm is similar to that of step 2. However, instead of using name

similarity, the algorithm relies on URL similarity for the matching procedure, and controls for the similarity of names.

Linking funding-level information

After creating a unique list of entities, the funding information surrounding these institutions also needs to be cleaned, harmonised, and linked. The procedure to link funding information across databases needs to consider several concerns. First, the types of deals recorded in Dealroom and Crunchbase are different. Therefore, a standardisation of these classifications is required to link the funding information provided by both data providers. Second, the funding information is complementary. In other words, it might be the case that the funding information for a given entity is collected in both databases, or some funding rounds only appear on one or another. Deal records are linked using the Python “recordlinkage” library. In what follows, the different steps of the record linkage are described:

1. **Harmonisation of deal types across databases:** As mentioned above, Dealroom and Crunchbase use different classifications for the types of deals. In a first step, the information on funding types across the two databases is brought to a harmonised format. For more details on the harmonisation, see Table A A.1 in Annex C.
2. **Creating candidate deals to be matched:** In a second step, we use the company-level linkage to build matching candidates for deals. For each unique company identifier, the matching candidates are tuples of all potential deal matches between Crunchbase and Dealroom. For example, if there are $N = 3$ deals in Dealroom and $M = 2$ deals in Crunchbase for a given company, there are $N \times M = 2 \times 3 = 6$ matching candidates for that company.
3. **Comparing candidate deals:** For all matching candidates, the deal-level information between the two databases is compared based on the following features of funding records: funding round dates, deal type, funding amounts, the currency of the deal, an indicator if funding amounts are missing, the number of investors, and the investor identifiers from prior matched entities. The comparison of the funding information across the databases results in a comparison vector that contains normalised values for each of the nine matching features.
4. **Classifying matching funding deal information:** Using the comparison vector, the matching candidate pairs are classified to assess which of them are considered a predicted match. To do so, a machine learning classifier is trained with a sample of manually labelled true matching pairs (“golden pairs”).
5. **Canonicalisation of funding deal information:** After classifying deals into matches and non-matches, the information is further canonicalised from the two data sources. We use both databases to fill in missing information for matches (i.e. a deal that shows up in both databases). In case that information for a deal is available in both Crunchbase and Dealroom, the information from Crunchbase is used by default.¹

Table A A.1. Deal type harmonisation across Dealroom and Crunchbase

	Crunchbase	Dealroom
References	https://support.crunchbase.com/hc/en-us/articles/115010458467-Glossary-of-Funding-Types accessed 20-10-2023 Methodology CB news: https://news.crunchbase.com/methodology/ accessed 20-10-2023 Note that the “funding stage” definitions Crunchbase News uses for different funding stages is expansive by design.	https://intercom-help.eu/dealroom/en/articles/22311-taxonomy-definitions accessed 20-10-2023
Deal Type	Description	
ACQUISITION & MERGERS	In Crunchbase M&A is recorded in different tables, sometimes with more detailed information. Based on some examples, the ‘BUYOUT’ category in Dealroom seems to be an ‘ACQUISITION’ in Crunchbase. Examples: GRAM POWER acquired by I Squared Capital SIRTl acquired by Mutaes AG WOODSAGE acquired by Milton Street Capital Also, important to understand how mergers are recorded in the data, and how this might differ from Dealroom. https://support.crunchbase.com/hc/en-us/community/posts/16575051291027-How-are-mergers-and-Acquisitions-shown-on-Crunchbase	ACQUISITION <i>“majority stake (50-100%), acquired, controlling stake, (acquisition amount = valuation if no mention of stake %). Generally, we add the transaction when the acquisition is announced. If it does not go through, we remove it. Once it’s finalized we can update the transaction date.”</i> BUYOUT <i>“30-100% Acquisition by Private Equity companies, BUYOUT mentioned”</i> MERGER <i>“is a fusion of two companies into one new legal entity agreed on generally equal terms. For example, Daum Communications and Kakao Inc. merged in 2014. Once added, a merger transaction will be displayed on both companies’ profiles.”</i>
ANGEL FINANCING	ANGEL <i>“An angel round is typically a small round designed to get a new company off the ground. Investors in an angel round include individual angel investors, angel investor groups, friends, and family.”</i>	ANGEL <i>“when only angel investors made the investment”</i>
CONVERTIBLES	CONVERTIBLE NOTE <i>“A convertible note is an ‘in-between’ round funding to help companies hold over until they want to raise their next round of funding. When they raise the next round, this note ‘converts’ with a discount at the price of the new round. You will typically see convertible notes after a company raises, for example, a Series A round but does not yet want to raise a Series B round.”</i>	CONVERTIBLE <i>“convertible notes, convertible loans”</i>
CORPORATE FINANCING ROUNDS	CORPORATE ROUND <i>“A corporate round occurs when a company, rather than a venture capital company, makes an investment in another company. These are often, though not necessarily, done for the purpose of forming a strategic partnership.”</i>	CORPORATE SPINOUT N.A.
	NOTE: Corporate Round and Corporate Spinout are distinct categories. Corporate Rounds may be classified as “Early VC” in Dealroom (e.g. VeriTread’s financing round from Sumitomo Corporation of Americas) Corporate Spinouts do not seem to be registered in Crunchbase.	

CROWDFUNDING	EQUITY CROWDFUNDING <i>"Equity crowdfunding platforms allow individual users to invest in companies in exchange for equity. Typically on these platforms the investors invest small amounts of money, though syndicates are formed to allow an individual to take a lead on evaluating an investment and pooling funding from a group of individual investors."</i>	
	PRODUCT CROWDFUNDING <i>"In a product crowdfunding round, a company will provide its product, which is often still in development, in exchange for capital. This kind of round is also typically completed on a funding platform."</i>	
DEBT FINANCING	DEBT FINANCING <i>"In a debt round, an investor lends money to a company, and the company promises to repay the debt with added interest."</i>	DEBT <i>"it is defined as money borrowed by one party to another, with the arrangement to be paid back at a later date, usually with interest. Startups & Scaleups choose to opt for this option as it might be cheaper than issuing stock in certain industries (real estate, manufacturing, for example)."</i>
GRANT	GRANT <i>"A grant is when a company, investor, or government agency provides capital to a company without taking an equity stake in the company."</i>	GRANT <i>"it is a financial award given by governments, international institutions, universities etc. It is like a gift as the grant maker won't receive equity or payment back. For example, the European Innovation Council (EIC) has given grants to many startups."</i>
INITIAL COIN OFFERING (ICO)	INITIAL COIN OFFERING <i>"An initial coin offering (ICO) is a means of raising money via crowdfunding using cryptocurrency as capital. A company raising money through an ICO holds a fundraising campaign, and during this campaign, backers will purchase a percentage of a new cryptocurrency (called a "token" or "coin"), often using another cryptocurrency like bitcoin to make the purchase, in the hopes that the new cryptocurrency grows in value."</i>	ICO <i>"more info can be found here"</i>
INITIAL PUBLIC OFFERING (IPO)	In Crunchbase IPOs are recorded in a different table, providing more detailed information about IPOs (e.g. at which stock exchange it occurred, the initial share price etc.)	IPO <i>"more info can be found here"</i>
POST IPO DEBT	POST IPO DEBT <i>"A post-IPO debt round takes place when companies loan a company money after the company has already gone public. Similar to debt financing, a company will promise to repay the principal as well as added interest on the debt."</i>	POST IPO DEBT <i>"A post IPO debt round is a transaction when corporates loan a company money after the company has already gone public. Similar to debt, a company promises to repay the principal as well as added interest on the debt."</i>
		POST IPO CONVERTIBLE <i>"A post IPO convertible round takes place when a company receives a convertible round after the company has already gone public."</i>
	NOTE: Deals that are classified as POST IPO CONVERTIBLE in Dealroom are classified as POST IPO DEBT in Crunchbase. Examples: Oryzon Genomics Jul. 2022 by Nice & Green SA; IsoEnergy Nov. 2022 by Queen's Road Capital Investment	
POST IPO EQUITY	POST IPO EQUITY <i>"A post-IPO equity round takes place when companies invest in a company after the company has already gone public."</i>	POST IPO EQUITY <i>"A post IPO equity round takes place when companies invest in a company after the company has already gone public."</i>

POST IPO SECONDARY	POST IPO SECONDARY <i>"A post-IPO secondary round takes place when an investor purchases shares of stock in a company from other, existing shareholders rather than from the company directly, and it occurs after the company has already gone public."</i>	POST IPO SECONDARY <i>"A post IPO convertible round takes place when a company receives a convertible round after the company has already gone public."</i>
SECONDARY FINANCING	SECONDARY MARKET <i>"A secondary market transaction is a fundraising event in which one investor purchases shares of stock in a company from other, existing shareholders rather than from the company directly. These transactions often occur when a private company becomes highly valuable and early-stage investors or employees want to earn a profit on their investment, and these transactions are rarely announced or publicized."</i>	SECONDARY <i>"0-20% ownership by investing through buying shares from existing investors"</i>
SEED FINANCING	PRE SEED <i>"A Pre-Seed round is a pre-institutional seed round that either has no institutional investors or is a very low amount, often below \$150k." (glossary)</i> <i>"A pre-seed round is a small seed funding, often below \$1 million. Accelerator rounds below \$1 million fit into this funding stage." (methodology CB news)</i>	
	SEED <i>"Seed rounds are among the first rounds of funding a company will receive, generally while the company is young and working to gain traction. Round sizes range between \$10k–\$2M, though larger seed rounds have become more common in recent years. A seed round typically comes after an angel round (if applicable) and before a company's Series A round."</i> <i>"Seed rounds are among the first rounds of funding a company will receive, generally while it is young and working to gain traction. Round sizes typically range between \$1 million and \$3 million, though larger seed rounds have become more common in recent years. A seed round typically comes after an angel round (if applicable) and before a company's Series A round." (methodology CB news)</i>	
SERIES FINANCING	SERIES A and SERIES B <i>"Series A rounds are funding rounds for earlier stage companies and range on average between \$1M–\$30M." (glossary)</i> <i>"These are funding rounds for earlier stage companies and range on average between \$5 million and \$30 million." (methodology CB news)</i>	SERIES A <i>"for \$4-15M deals"</i>
	SERIES C, SERIES D, ... <i>"Series C rounds and onwards are for later-stage and more established companies. These rounds are usually \$10M+ and are often much larger."</i> <i>"For later-stage and more established companies, these rounds are usually at least \$20 million but are often much larger." (methodology CB news)</i>	SERIES B <i>"for \$15-40M deals"</i>
	SERIES UNKNOWN <i>"Venture funding refers to an investment that comes from a venture capital company and describes Series A, Series B, and later rounds. This funding type is used for any funding round that is clearly a venture round but where the series has not been specified."</i>	SERIES C <i>"for \$40-100M deals"</i>
GENERIC TYPES		EARLY VC <i>"when the round type is not mentioned, and the amount is between 2 and 20 million"</i>
		LATE VC <i>"rounds type not mentioned after Series A, B... or Round happening 5 years after launch date"</i>
PRIVATE EQUITY	PRIVATE EQUITY <i>"A private equity round is led by a private equity"</i>	

	company or a hedge fund and is a late-stage round. It is a less risky investment because the company is more company established, and the rounds are typically upwards of \$50M."	
		GROWTH EQUITY VC/ NON-VC "a \$100M+ investment in a fast-growing company mostly a mix of primary+secondary. the investor is a private equity, growth equity, VC and/or corporate. Not every \$100M+ round is GE round. It could be self-reported as Series A, B, C+ etc."
OTHER TYPES	MEDIA FOR EQUITY "when a media group provides a communication/advertising campaign in exchange for shares in the startup"	
	NON-EQUITY ASSISTANCE "Non-equity assistance round occurs when a company or investor provides office space or mentorship and does not get equity in return."	
		PRIVATE PLACEMENT VC / NON VC, SPAC PRIVATE PLACEMENT " add when the private placement type is explicitly mentioned and the company is public. This round type is more common in the US. It has a specific meaning and is never public money: a publicly listed company sells shares privately (not via stock market) to hand-picked individuals (usually family offices, individuals, institutional investors, not government). Private placement is almost always present in the case of SPACs IPOs (Example)."
		SPAC IPO "this round represents the merger between the SPAC vehicle and the company going public and the company public listing. It is therefore implemented as an IPO but with "investor" the SPAC company. (Example). Note: The ownership percentages (e.g. Secondary, Buyout) are indicative."
		SPINOUT "Spinouts are startups where universities have equity and/or royalty/licensing deals. The terms Spinout and Spin-off have the same meaning. We use the term Spinout. We count a startup as Spinout if there's a reliable source, like a university site, the company LinkedIn profile, trusted news source, etc. mentioning it's a "spin-off" / "spinout" or mentioning that the company was created using the technology developed at a University/Research Center. Our definition of Spinout does not cover corporate spinouts (companies that have spun out of large corporates like Samsung, Toyota, etc.). Startups that do not follow this condition, should not be counted as Spinouts. For example, the source mentioning that the company was founded by the Oxford University alumni is not enough to count the startup as a Spinout. Similarly, the source mentioning that the company is a University startup or a startup incubated at the University, is not enough to count this startup as a Spinout."
		PROJECT, REAL ESTATE, INFRASTRUCTURE FINANCE N.A.
		LENDING CAPITAL "working capital for platforms providing lendings and mortgages. These startups require a wide amount of working capital to lend which is often provided by banking partners (ex: LendInvest, Duologi)"

	UNDISCLOSED N.A.	
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Note: This table outlines the harmonisation of deal type classifications between Crunchbase and Dealroom, based on publicly available definitions and platform-specific methodologies. It includes descriptions of various funding instruments such as crowdfunding, debt financing, corporate rounds, and post-IPO deals. Differences in terminology and categorisation between the two sources are noted, including cases where similar deal types are labelled differently (e.g. “Post IPO Convertible” in Dealroom vs. “Post IPO Debt” in Crunchbase).

Source: Authors’ elaboration.

Linking investor information

Both Crunchbase and Dealroom provide the type of investor involved in a deal. However, they use different conventions that may not overlap. In many instances, this happens because both use different conventions to refer to companies or entities that operate in different activities or have private or public ownership. In this section, the categories are analysed and harmonised for their use in the OECD Start-ups Database. More importantly, the limitations of using these categories provided by Crunchbase and Dealroom are also underscored, highlighting the need for the complementary data available in corporate venture capital and government venture capital investments when studying these specific kinds of investments.

Crunchbase provides only one tag for each investor, while Dealroom provide a main tag and a sub-tag. In general, the main tag in Dealroom captures the company's overall activity (i.e. whether it is an investment company or another corporation). At the same time, the sub-tag gives further details about its ownership or activity. To see whether these tags can be used for research, the top 500 VC investors and top 500 non-VC investors are manually categorised and compared against the tags provided by Dealroom and Crunchbase. Because the top investors are concentrated in investment companies and (both financial and non-financial) corporates, the exercise also looks into the top 150 non-equity assistance investors and 50 randomly selected investors behind deals that are categorised as “spinouts”, and another 50 randomly selected behind deals of the type “angel” (see Table A A.2 for more details). The reasoning is to capture investors who use alternative funding instruments or who do not contribute with such large amounts, such as angel groups, family offices, incubators and accelerators, universities, and so on.

The categories inserted manually capture both the ownership and the activity of the company. The first column states the name of the category, while the second column includes a brief description of the kinds of investors included in each category.

Table A A.2. Categories used for the manual checking of investor types

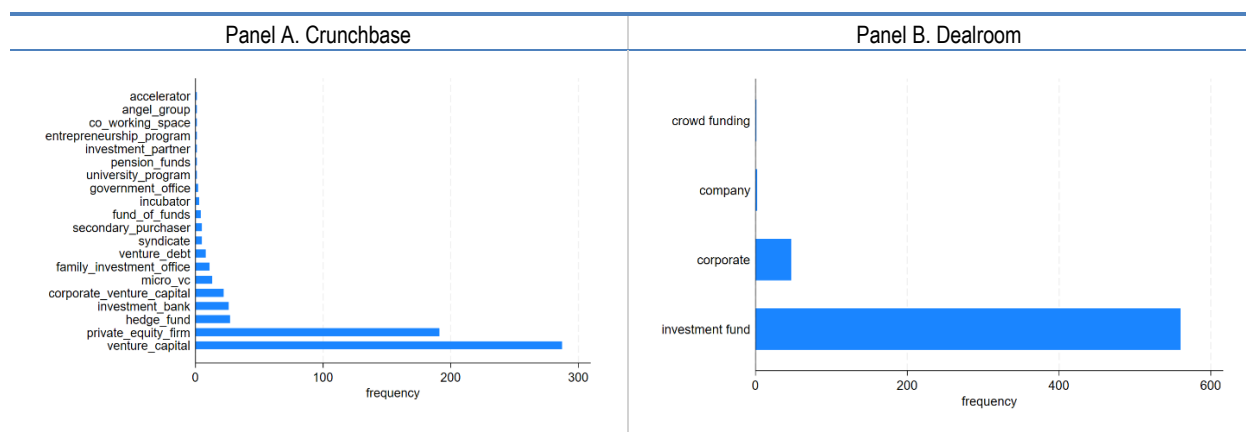
Category	Full description
Angels and other individual or family investors	Angels, angel groups, family offices, and other kinds of smaller-scale investors who provide what is known as pre-seed or angel funding.
Corporates	Corporate investors include all kinds of firms whose purpose is broader than to invest in other companies, particularly those with strategic interests in specific product or service markets. Among corporate investors, four types are further distinguished. First, the classification distinguishes between financial (e.g. HSBC) and non-financial activities (e.g. Google). Second, the classification distinguishes whether there is government involvement in the company in the form of partial or full ownership (e.g. GovVC agencies) or not.

Government	Government units (e.g. the Ministry of Economy). Firms that are owned by the government are included in publicly intervened corporations.
Venture capital and private equity firms, and other institutional investors (e.g. asset management companies, “blank check” companies)	This category includes all companies whose sole purpose is investing and returning value to their owners. If a firm provides other kinds of financial services, such as commercial banking or others, it is classified as a financial corporation.
Incubators and accelerators	This category includes companies that not only finance but also offer networks, workspace, training, competitions, programs, and the like for start-ups.
Universities and their faculties	This category includes universities and their faculties or other units. Incubators and accelerators owned by universities are classified as such.

Note: This table presents the categories used for manually classifying investor types in the OECD Start-ups Database. Each category is defined based on the nature and role of the investor, including individual and family investors, corporate entities, government bodies, institutional investors, incubators and accelerators, and universities.
Source: Authors' elaboration.

The investors in the sample described above are classified into the categories mentioned above, and then the tags in Dealroom and Crunchbase are analysed. Out of the 1250 investors, approximately 750 correspond to investment firms (individuals, firms, or governments can own them). Roughly 700 are private investment firms, and governments somehow intervene in another 50 investment firms. Figure A A.1 shows the Crunchbase and Dealroom tags associated with these kinds of investors. First, Crunchbase primarily assigns these companies to the private equity firms and venture capital categories, while Dealroom classifies an approximately equal number of companies into the investment fund category. Crunchbase also uses other minor categories like hedge fund, corporate venture capital, or micro-VC to classify these companies.

Figure A A.1. Crunchbase and Dealroom categories for private sector investment companies

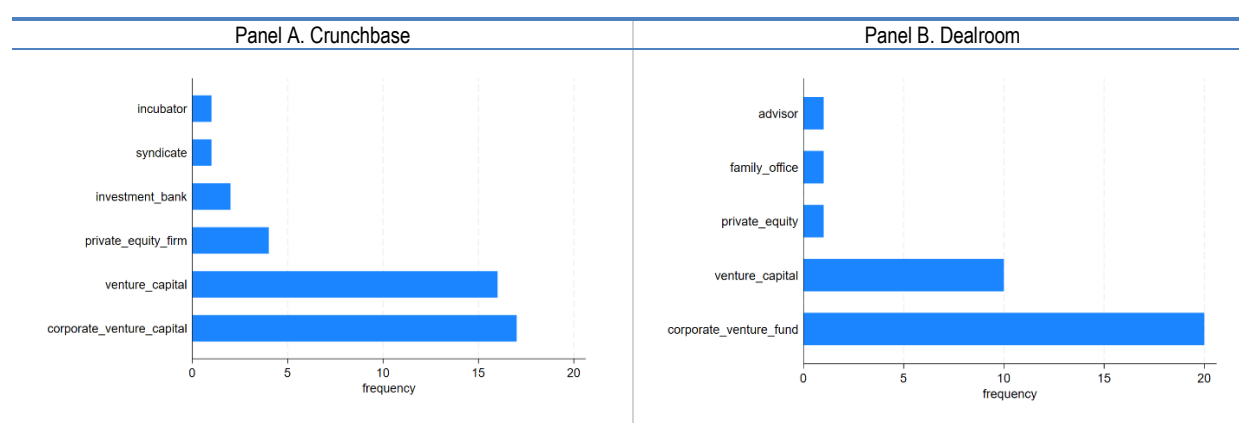


Note: This figure shows a histogram of Crunchbase and Dealroom tags.

Source: OECD Start-ups Database, June 2025.

The analysis continues to deepen into those investment firms classified as corporate venture capital. For this, the investment firm's parent company type was recorded during the manual checking. While corporate venture capital can take many forms in the data, most of the literature on CVCs has focused on those firms set up to handle the venture capital investments of their parent companies. In the manual checking, approximately 40 investment firms are owned by corporations. Figure A A.2 shows the Crunchbase tags and Dealroom sub-tags, as the vast majority are consistently classified into the investment fund category (35 out of 40 cases). As seen, approximately half of these companies are correctly classified into corporate venture capital subsidiaries by Crunchbase and Dealroom.

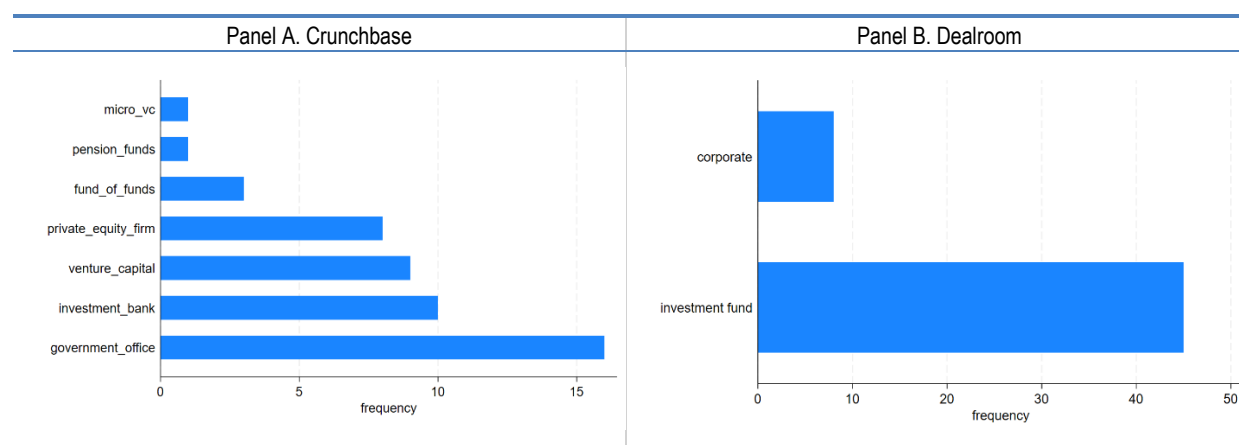
Figure A A.2. Crunchbase and Dealroom categories for corporate venture capital subsidiaries



Note: This figure shows a histogram of Crunchbase tags and Dealroom sub-tags.

Source OECD Start-ups Database, June 2025.

Figure A A.3. Crunchbase and Dealroom categories for government-related investment companies



Note: This figure shows a histogram of Crunchbase and Dealroom government-related investment companies by their tags.

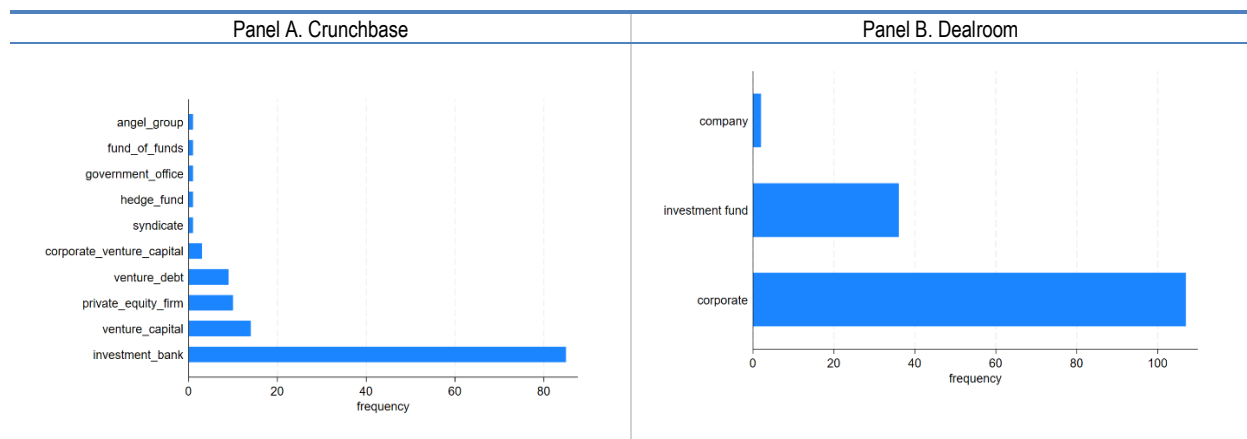
Source: OECD Start-ups Database, June 2025.

In the case of the approximately more than 50 government venture capital agencies identified in the manual check, the results are more mixed, as shown in Figure A A.3. Dealroom classifies all these companies directly as investment funds, similar to private investment companies. However, Crunchbase does identify that some of these agencies are related to government agencies. Nevertheless, it also classifies an important share of these companies as investment banks, private equity, or venture capital firms. Notice

that this classification is not strictly incorrect in that these firms are, for the most part, investment firms. However, only using one tag limits the ability of both databases to capture the activity of the firm (i.e. it is an investment firm) and its ownership (i.e. the investment firm is owned by the government). For this reason, the OECD Start-ups Database is complemented with the survey on government VC explained above.

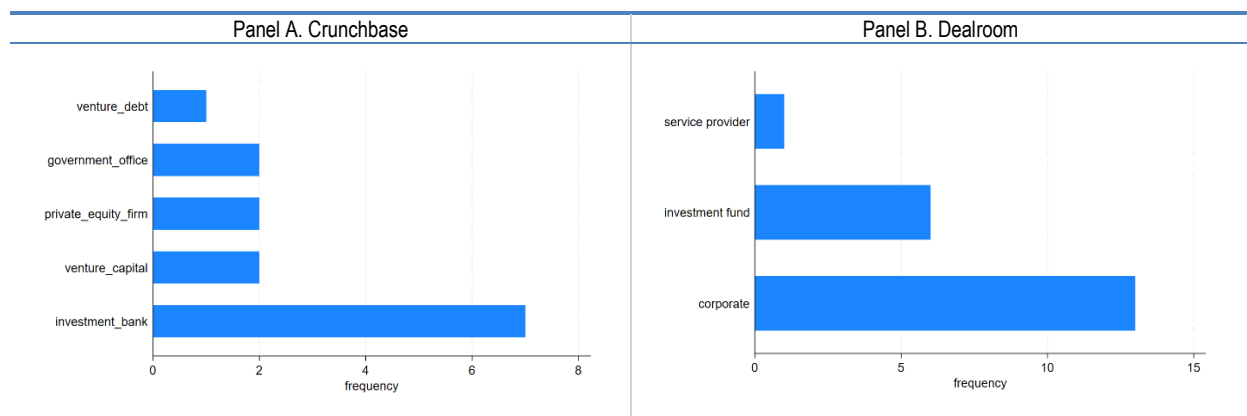
Next, the categories in Crunchbase and Dealroom associated with financial corporations are presented. Out of the 1250 entities analysed, approximately 175 are financial corporate firms from the private sector. Figure A A.4 shows that these financial corporations are mostly classified as investment banks in Crunchbase, while the tag corporate is mostly used in Dealroom. Both categories are followed by the typical tags that both datasets used for investment firms (i.e. private equity and venture capital firms in Crunchbase and investment funds in Dealroom). Additionally, approximately 25 financial corporations that are partially or fully owned by the government are identified. Their tags are Figure A A.5. As shown, most of these financial institutions are simply classified as investment banks in Crunchbase or corporates in Dealroom. On some rare occasions, Crunchbase states that these are government offices (less than 10% of instances).

Figure A A.4. Crunchbase and Dealroom categories for financial corporations in the private sector



Note: This figure shows a histogram of Crunchbase and Deal tags for financial corporations.
Source: OECD Start-ups Database, June 2025.

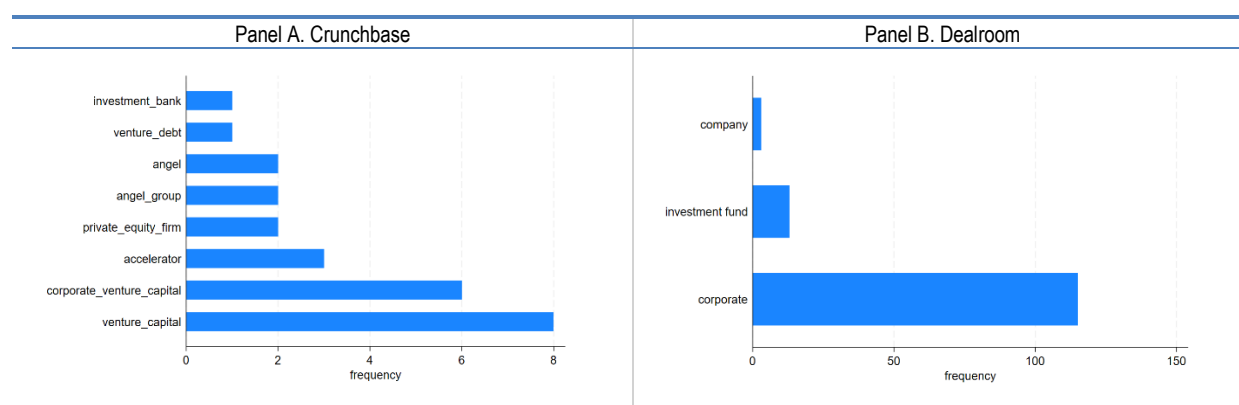
Figure A A.5. Crunchbase and Dealroom categories for government-related financial corporations



Note: This figure shows a histogram of Crunchbase and Deal tags for government-related financial corporations.
Source: OECD Start-ups Database, June 2025.

Next, the analysis follows with non-financial corporations. In this case, the analysis abstracts from the distinction between government-related and non-government-related because the number of non-financial corporates that are publicly intervened in is very limited (4 cases out of the 1250 investors analysed). In total, there are approximately 150 non-financial corporations in the sample. Figure A A.6 shows the distribution of tags across these entities. First, Crunchbase seems to systematically not assign a tag to these corporations. Less than 20% of non-financial corporates have a tag in Crunchbase. Nevertheless, Dealroom correctly classifies almost all of them into the company or corporate categories. Publicly intervened corporates are also classified into these two categories in the latter, similarly to Dealroom's classification of financial corporates.

Figure A A.6. Crunchbase and Dealroom categories for non-financial corporates



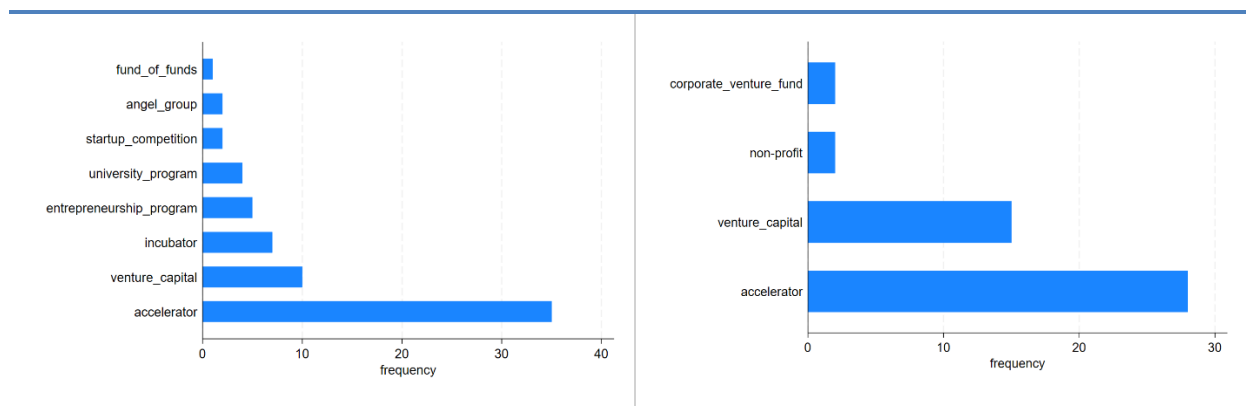
Note: This figure shows a histogram of Crunchbase and Deal tags for non-financial corporations.

Source: OECD Start-ups Database, June 2025.

Lastly, the analysis focuses on less traditional investors like incubators and accelerators, angels, individuals and family offices, universities, or agencies directly controlled by the government (e.g. ministries or similar government units). These entities mainly come from the alternative funding instruments categorised as “spinouts,” “angels,” and “non-equity assistance.” In total, there are approximately 75 incubators and accelerators. Their tags in Crunchbase and Dealroom are shown in Figure A A.7. Similar to corporate venture capital, Dealroom consistently classifies most of these companies as investment funds and provides further details in their sub-tags. Therefore, Dealroom's sub-tags are used in this analysis. As shown in Panel A, Crunchbase classifies most of these entities that correspond to incubators and accelerators like “accelerator,” “incubator,” “entrepreneurship program,” “start-up competition,” or “university program.” Meanwhile, Dealroom's sub-tag also classifies approximately two-thirds of incubators with a sub-tag as such. Notice that the categories associated with investment firms, like venture capital, are also used frequently, and most incubators and accelerators also provide financing to these companies.

Figure A A.7. Crunchbase and Dealroom categories for accelerators and incubators

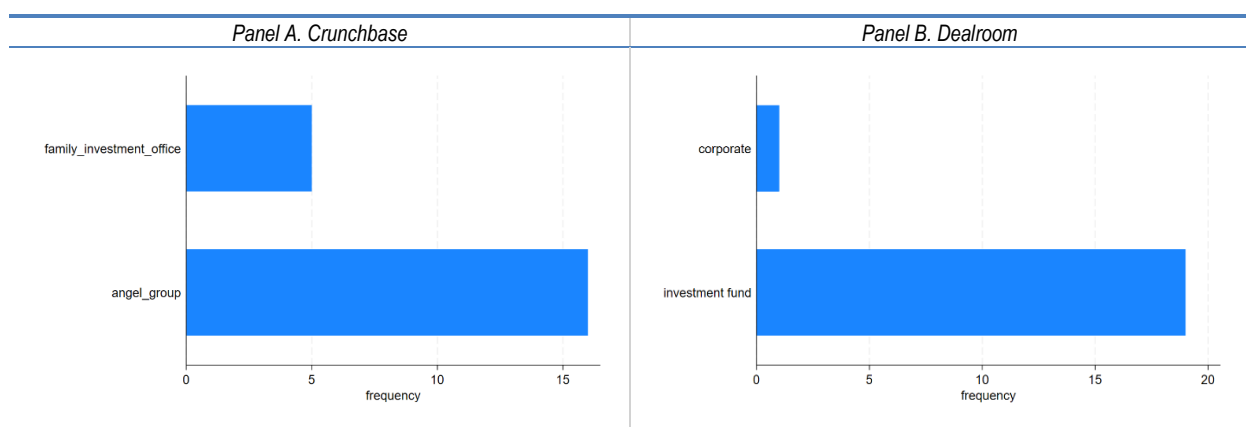
Panel A. Crunchbase	Panel B. Dealroom
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Note: This figure shows a histogram of Crunchbase and Deal for accelerators and incubators tags.
Source: OECD Start-ups Database, June 2025.

Next, the analysis focuses on angel groups, individual investors, family offices, and other minor investors, which are more typical in pre-seed or angel financing rounds. There are 25 investors of this kind in the manual checking. Figure A A.8 shows the distribution of tags for these investors, similar to before. Crunchbase correctly classifies all of them into angel groups or family offices. Meanwhile, Dealroom continues to refer to them as simply investment firms.

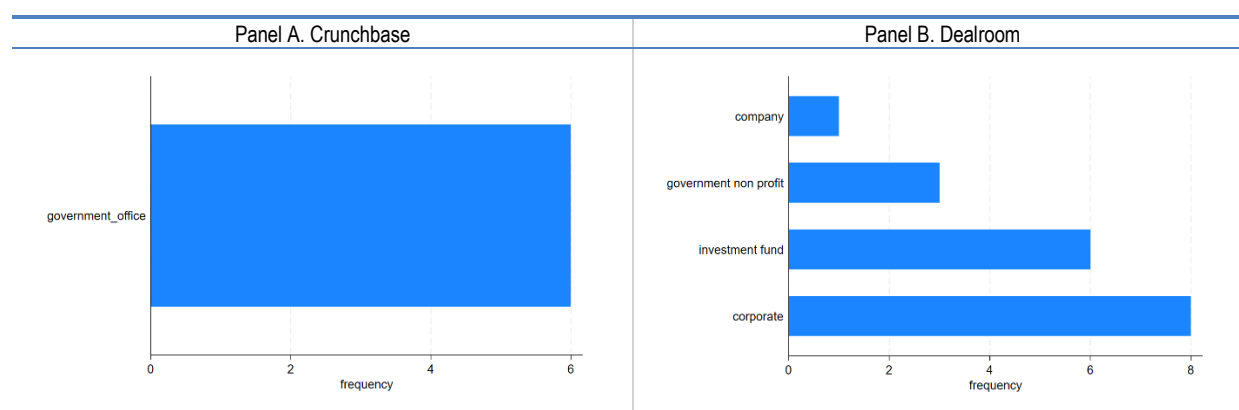
Figure A A.8. Crunchbase and Dealroom categories for angel groups, family offices, and individual investors



Note: This figure shows a histogram of Crunchbase and Deal for angel, family offices, and individual investors tags.
Source: OECD Start-ups Database, June 2025.

Finally, the analysis focuses on universities and institutions or agencies directly controlled by the government (e.g. ministries, public funds). There are approximately 25 units directly controlled by the government in the manual checking exercise. As seen in Figure A A.9, most of them either do not have a tag, as Crunchbase only classifies six of them correctly into government offices, or are classified as something other than government, like corporates or investment funds in Dealroom. While it seems that these government units are harder to identify, many of the universities are correctly identified, as shown in Figure A A.10. Out of the 20 institutions of this kind, the vast majority are identified as universities according to Dealroom or Crunchbase. It is worth highlighting, however, that spinouts and incubators, and accelerators owned by universities are usually classified into the incubators and accelerators category.

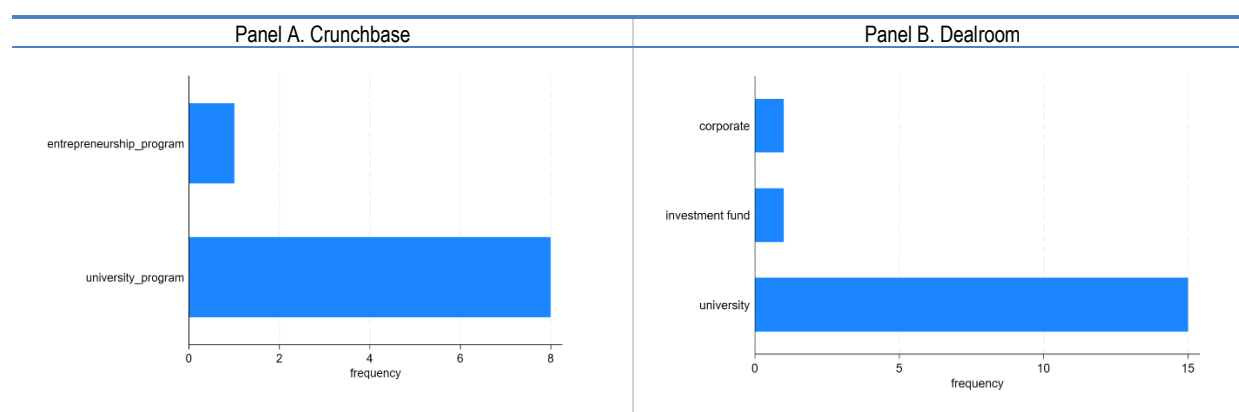
Figure A A.9. Crunchbase and Dealroom categories for government units



Note: This figure shows a histogram of Crunchbase and Deal for government tags.

Source: OECD Start-ups Database, June 2025.

Figure A A.10. Crunchbase and Dealroom categories for universities



Note: This figure shows a histogram of Crunchbase and Deal for the university's tags.

Source: OECD Start-ups Database, June 2025.

Linking the OECD Start-ups Database to complementary sources

The matching between the company names and applicants of IP assets was performed using the procedure embedded in the IMALINKER system that was developed for the OECD in 2013, in the following steps:

1. **Name harmonisation:** Names in the OECD Start-ups Database and the names in Orbis, Orbis M&A, and the OECD IP Database are harmonised using a set of country-level dictionaries, notably to correct for special characters and variations in the way legal entity types are spelled out (e.g. "Ltd.", "Corp.", "GMBH", "kabushiki kaisha").
2. **String-matching of company names:** Harmonised names are linked using a set of string-matching algorithms that exploit token-based measures (weighted frequency of tokens) and distance measures (Jaro-Winkler, Levenshtein distances, Cosine similarity) on a country-by-country basis.
3. **Selection of matches:** High scores of similarities were set to select matched pairs (more than 93% similarity for Levenshtein, Jaro-Winkler or Cosine similarity algorithms, 90 % for the token

weighted measures, conditional on the length of the strings, as short strings tend to increase the number of false positive matches).

4. **Triangulation of matches:** Since the same entity may appear in several of the different databases to which the matches are performed, a final step is taken to propagate matches from one database to another (e.g. a match is propagated if a company in the OECD Start-ups Database has been matched to PATSTAT but not to Orbis, but the company was matched between PATSTAT and Orbis).

Annex B. Additional figures and tables of the benchmarking analysis

Table A B.1. Frequencies and percentage difference of the OECD Start-ups Database and benchmark data, by source

Source	Number of companies			Number of deals (thousands)			VC funding (USD billion)		
	OECD Start-ups Database	Benchmark	Percentage difference	OECD Start-ups Database	Benchmark	Percentage difference	OECD Start-ups Database	Benchmark	Percentage difference
Australia	8 295			5.92	2.82	110.30	58.59	19.51	200.41
Canada	17 188	6 070	183.16	11.64	6.90	68.82	88.40	60.63	45.81
Europe	130 855	104 308	25.45	99.49			844.43	407.31	107.32
Israel	8 458	4 054	108.63	6.91			76.90	70.17	9.59
Japan	18 834			16.37	19.69	-16.86	114.47	24.66	364.23
Korea	13 879			8.46	18.33	-53.84	86.05	32.40	165.63
Latin America	15 615			10.69	6.05	76.87	70.35	40.36	74.30
Türkiye	3 361			2.79	2.69	3.83	11.13	4.22	164.03
United States	196 144			149.21	153.11	-2.54	1 980.91	1 560.39	26.95

Note: This table shows the funding amount, number of companies, and number of deals in the OECD Start-ups Database, along with the benchmark data and their percentage difference, by source. The percentage difference is calculated as the difference between the OECD/ST Start-up Database value and the benchmark value, divided by the benchmark value, and multiplied by 100.

Sources: Elaborated from the OECD Start-ups Database, June 2025, Preqin and the Australian Investment Council; Canadian Venture Capital & Private Equity Association; Invest Europe; Israel Central Bureau of Statistics; Venture Enterprise Center, Japan; Latin America Venture Capital Association; Korean Venture Capital Association; startups.watch platform (in collaboration with the Presidency of the Republic of Türkiye Investment Office, Financial Investments Unit); and PitchBook-NVCA.

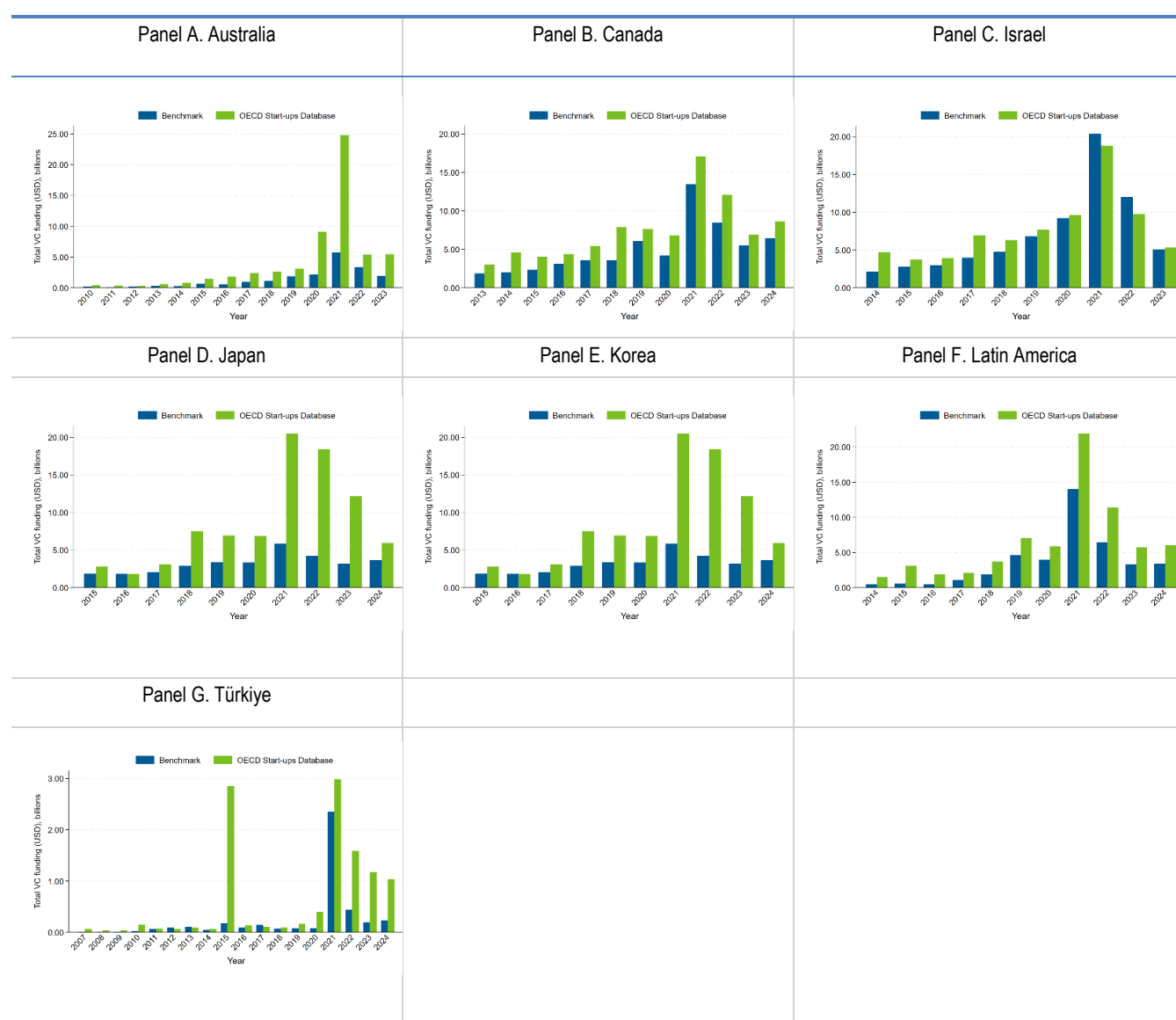
Table A B.2. Frequencies and percentage difference of the OECD Start-ups Database and benchmark data, by source, by country

Country	Number of companies			Number of deals (thousands)			VC funding (USD billion)		
	OECD Start-ups Database	Benchmark	Percentage difference	OECD Start-ups Database	Benchmark	Percentage difference	OECD Start-ups Database	Benchmark	Percentage difference
Australia	8 295			5.92	2.82	110.30	58.59	19.51	200.41
Austria	1 754	1 384	26.73	1.14			5.55	3.84	44.51
Baltic countries	3 160	1 038	204.43	2.63			13.42	2.77	383.72
Belgium	2 867	3 137	-8.61	1.93			12.19	11.64	4.79
Brazil	7 570			4.95	2.84	74.28	37.56	20.26	85.43
Canada	17 188	6 070	183.16	11.64	6.90	68.82	88.40	60.63	45.81
Chile	1 457			1.12	0.56	100.18	3.58	1.83	95.26
Colombia	1 422			1.03	0.54	91.65	7.71	5.00	54.13
Costa Rica	126			0.09	0.04	107.14	0.45	0.04	955.39
Czechia	1 299	257	405.45	0.94			5.22	2.32	125.64
Denmark	2 746	1 433	91.63	2.00			16.28	7.44	118.63
Finland	3 294	3 431	-3.99	2.62			11.24	8.01	40.30
France	14 482	22 795	-36.47	11.57			98.54	94.88	3.86
Germany	14 234	18 233	-21.93	8.78			115.32	53.90	113.95
Greece	511	264	93.56	0.40			1.55	0.81	90.47
Hungary	1 465	1 747	-16.14	1.24			1.79	1.73	3.38
Ireland	3 300	2 452	34.58	2.63			27.60	7.64	261.13
Israel	8 458	4 054	108.63	6.91			76.90	70.17	9.59
Italy	5 032	3 357	49.90	3.98			20.41	17.50	16.63
Japan	18 834			16.37	19.69	-16.86	114.47	24.66	364.23
Korea	13 879			8.46	18.33	-53.84	86.05	32.40	165.63
Luxembourg	492	317	55.21	0.39			4.76	2.60	83.25
Mexico	2 604			1.78	1.22	45.74	11.62	8.33	39.54
Netherlands	6 757	5 615	20.34	4.89			60.51	22.29	171.53
Norway	2 185	2 089	4.60	1.79			12.46	8.16	52.71
Other CEE countries	895	653	37.06	0.70			3.08	1.46	111.63
Other countries	4 988	776	542.78	3.85	0.78	392.96	16.98	6.98	143.32
Poland	2 680	1 089	146.10	1.94			4.67	3.60	29.67
Portugal	1 558	1 369	13.81	1.08			7.02	2.26	210.23
Spain	8 344	7 965	4.76	6.66			40.37	24.39	65.51
Sweden	5 586	5 866	-4.77	4.63			36.63	14.76	148.22
Switzerland	5 298	1 954	171.14	3.93			33.74	13.00	159.67
Türkiye	3 361			2.79	2.69	3.83	11.13	4.22	164.03
United Kingdom	40 042	17 087	134.34	31.32			299.10	98.63	203.25
United States	196 144			149.21	153.11	-2.54	1 980.91	1 560.39	26.95

Note: This table displays the funding amount, number of companies, and number of deals in the OECD Start-ups Database, along with the benchmark data and their percentage difference, by source. The percentage difference is calculated as the difference between the OECD/ST Start-up Database value and the benchmark value, divided by the benchmark value, and multiplied by 100. The Baltic countries include Estonia, Latvia, and Lithuania. Other CEE countries include Bosnia and Herzegovina, Croatia, North Macedonia (Republic of), Moldova, Montenegro, Serbia, Slovak Republic, and Slovenia.

Sources: Elaborated from the OECD Start-ups Database, June 2025, Preqin and the Australian Investment Council; Canadian Venture Capital & Private Equity Association; Invest Europe; Israel Central Bureau of Statistics; Venture Enterprise Center, Japan; Latin America Venture Capital Association; Korean Venture Capital Association; startups.watch platform (in collaboration with the Presidency of the Republic of Türkiye Investment Office, Financial Investments Unit); and PitchBook-NVCA.

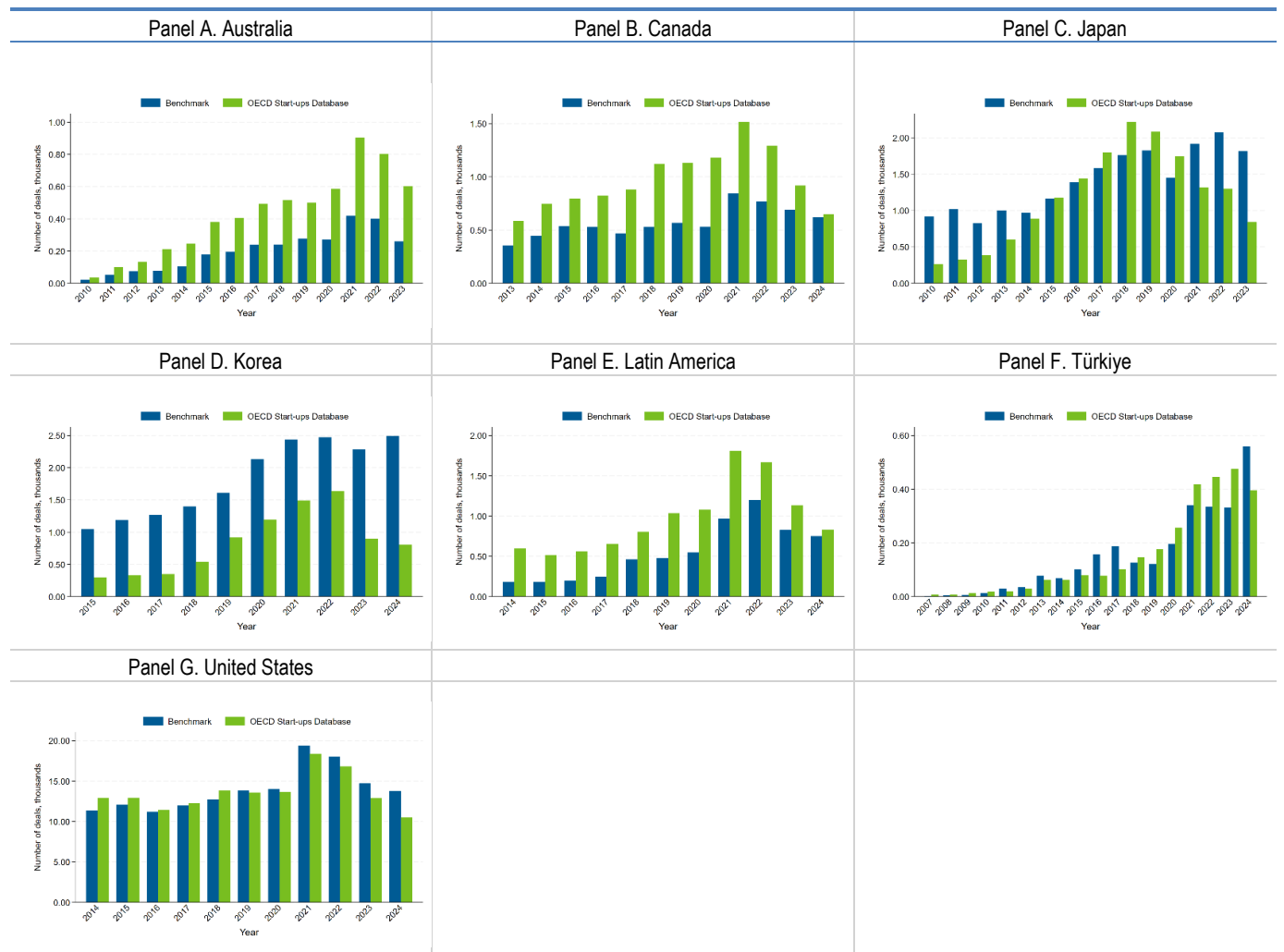
Figure A B.1. Comparison of VC financing in multiple countries and locations, by year



Note: This figure shows the total VC funding from the OECD Start-ups Database and from the benchmark data for multiple countries and locations across years.

Sources: Elaborated from the OECD Start-ups Database, June 2025, Preqin and the Australian Investment Council; Canadian Venture Capital & Private Equity Association; Israel Central Bureau of Statistics; Venture Enterprise Center, Japan; Latin America Venture Capital Association; Korean Venture Capital Association; and startups.watch platform (in collaboration with the Presidency of the Republic of Türkiye Investment Office, Financial Investments Unit).

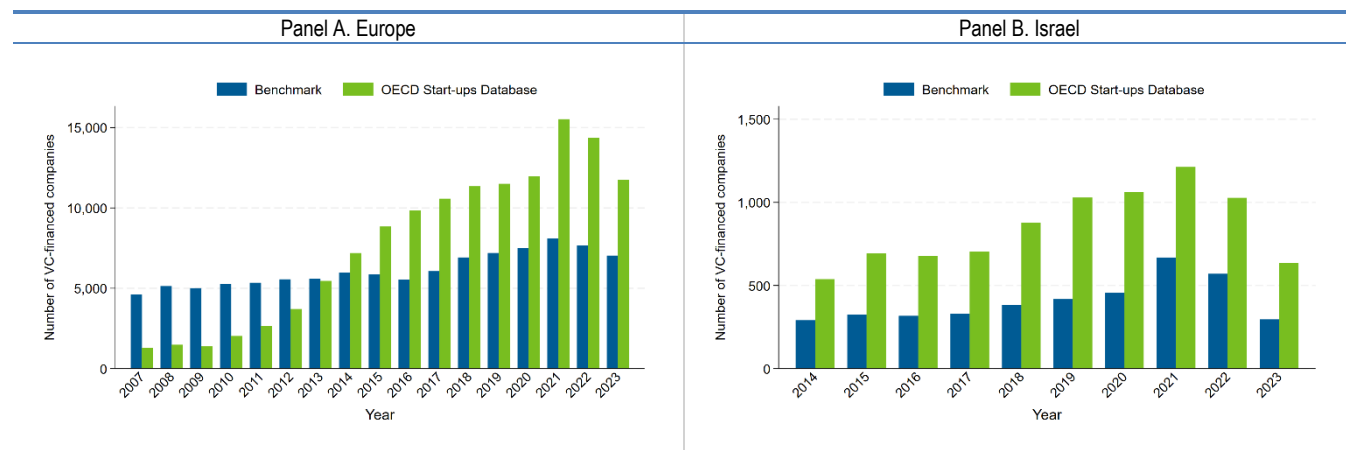
Figure A B.2. Comparison of the number of VC deals in multiple countries and locations, by year



Note: This figure shows the number of VC deals from the OECD Start-ups Database and from the benchmark data for multiple countries and locations across years.

Sources: Elaborated from the OECD Start-ups Database, June 2025, Preqin and the Australian Investment Council; Canadian Venture Capital & Private Equity Association; Venture Enterprise Center, Japan; Latin America Venture Capital Association; Korean Venture Capital Association; startups.watch platform (in collaboration with the Presidency of the Republic of Türkiye Investment Office, Financial Investments Unit); and PitchBook-NVCA.

Figure A B.3. Comparison of the number of VC-funded companies in multiple countries and locations, by year



Note: This figure shows the number of VC-funded companies from the OECD Start-ups Database and from the benchmark data for multiple countries and locations across years.

Source: OECD Start-ups Database, June 2025, Invest Europe, and the Israel Central Bureau of Statistics.

Annex C. Additional figures and tables of descriptive statistics and trends

Table A C.1. Allocation of economies to geographical locations

Location	Country names
Africa	Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, Democratic Republic of the Congo, Côte d'Ivoire, Djibouti, Egypt, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Libya, Madagascar, Malawi, Mali, Mauritania, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, South Sudan, Sudan, Tanzania, Togo, Tunisia, Uganda, Western Sahara, Zambia, Zimbabwe
Asia & Oceania	Afghanistan, Samoa, Armenia, Australia, Azerbaijan, Bahrain, Bangladesh, Bhutan, Brunei Darussalam, Cambodia, China (People's Republic of), Fiji, India, Indonesia, Iran, Iraq, Israel, Japan, Jordan, Kazakhstan, Kiribati, Kuwait, Kyrgyzstan, Lao People's Democratic Republic, Lebanon, Malaysia, Maldives Marshall Islands, Micronesia (Federated States of), Mongolia, Myanmar, Nauru, Nepal, New Zealand, Democratic People's Republic of Korea, Oman, Pakistan, Palau, Papua New Guinea, Philippines, Qatar, Samoa, Saudi Arabia, Singapore, Solomon Islands, Korea, Sri Lanka, Syria Arab Republic, Tajikistan, Thailand, Timor-Leste, Tonga, Turkmenistan, Tuvalu, United Arab Emirates, Uzbekistan, Vanuatu, Viet Nam, Yemen
Europe	Albania, Andorra, Austria, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Guernsey, Hungary, Iceland, Ireland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Moldova, Monaco, Montenegro, Netherlands, North Macedonia (Republic of), Norway, Poland, Portugal, Romania, Russian Federation, San Marino, Serbia, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Türkiye, Ukraine, United Kingdom
Latin America	Antigua and Barbuda, Argentina, Bahamas, Barbados, Belize, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Ecuador, El Salvador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, Uruguay, Venezuela
North America	Canada, United States

Note: This table shows the allocation of countries into the different geographical locations used in the analysis. Although Mexico is geographically part of North America, it is included in Latin America. This is because the data for Mexico in the benchmarking exercise are collected by the Latin American Venture Capital Association (LAVCA), and the benchmarking comparisons are conducted at the Latin American level.

Source: Authors' elaboration.

Table A C.2. Deal type aggregation in OECD Start-up database

Aggregated funding stages	Seed & Angel	Early-stage	Late-stage	Growth
Crunchbase				
Deal types I	Angel Pre-seed Seed Equity Crowdfunding	Series A Series B	Series C – J	Private Equity
Deal types II	Convertible Note Corporate Round Series Unknown < 3 million USD	Convertible Note Corporate Round Series Unknown 3 to 15 million USD	Convertible Note Corporate Round Series Unknown > 15 million USD	
Dealroom				
Deal types I	Angel Seed	Series A Series B	Series C - I	Growth Equity VC Growth Equity Non-VC
Deal types II	Convertible Early VC Late VC < 3 million USD	Convertible Early VC Late VC 3 to 15 million USD	Convertible Early VC Late VC > 15 million USD	

Note: Deal types I are directly used as listed in the primary databases. Deal types II use additional information on deal volumes. For funding rounds that show up in both databases, the deal type provided by Crunchbase is used.

The following funding types from Dealroom are not used in the analysis: *corporate spinout, debt, grant, ICO, lending capital, media for equity, post IPO convertible, post IPO debt, post IPO equity, post IPO secondary, private placement non-VC, private placement VC, project, real estate, infrastructure finance, secondary, SPAC IPO, SPAC private placement, spinout, support program*. The following funding types from Crunchbase are not used in the analysis: *debt financing, grant, initial coin offering, non-equity assistance, post IPO debt, post IPO equity, post IPO secondary, product crowdfunding, secondary market, and undisclosed*.

Source: Authors' elaboration.

Table A C.3. Funding information, by country

Country	Number of companies	Number of VC-backed companies	Total VC funding in USD millions	Mean VC funding per year in USD millions
OECD	669 537	588 988	6 523 928	13.15
Australia	14 966	11 524	127 366	8.06
Austria	3 083	2 632	8 698	5.77
Belgium	4 429	3 877	17 416	5.96
Brazil	10 090	9 351	79 983	8.88
Canada	36 012	27 433	270 396	11.05
Chile	2 256	2 132	5 229	8.36
China (People's Republic of)	54 400	52 930	1 064 478	27.83
Colombia	1 921	1 783	12 940	5.86
Costa Rica	179	159	551	5.37
Czechia	2 318	1 741	9 759	11.55
Denmark	4 793	4 031	27 797	10.70
Estonia	2 409	2 149	6 804	2.47
Finland	4 935	4 538	17 392	6.05
France	21 869	19 232	144 445	10.14
Germany	24 654	20 728	213 348	14.90
Greece	899	676	5 220	4.78
Hungary	1 767	1 556	2 745	1.97
Iceland	1 178	735	5 291	5.62

India	31 431	29 694	396 297	16.60
Ireland	5 470	4 756	68 424	13.29
Israel	13 868	13 074	128 325	10.76
Italy	8 906	6 487	49 852	9.01
Japan	25 235	24 868	173 664	34.93
Korea	16 894	16 504	100 988	8.60
Latvia	857	743	2 354	2.84
Lithuania	1 340	1 198	9 924	12.50
Luxembourg	966	778	12 603	24.49
Mexico	3 564	3 318	25 639	10.19
Netherlands	10 842	9 481	212 243	20.93
New Zealand	1 871	1 696	8 870	6.87
Norway	6 112	4 139	32 381	7.27
Poland	3 928	3 432	9 762	3.05
Portugal	2 881	2 270	12 897	8.02
Russian Federation	3 229	3 038	21 190	14.20
Slovak Republic	506	440	2 154	5.58
Slovenia	473	314	2 994	7.98
South Africa	2 484	2 058	23 889	13.33
Spain	18 878	13 290	57 976	5.03
Sweden	9 701	8 322	80 454	9.49
Switzerland	11 666	9 763	77 348	8.13
Türkiye	4 010	3 702	10 279	5.71
United Kingdom	65 450	55 751	533 838	10.93
United States	328 451	299 736	4 035 561	14.61
Other countries	56 121	50 212	521 094	10.56

Note: This table shows descriptive statistics of the number of companies, the number of VC-backed companies, the total VC deal volume, and the average VC funding per year, by country.

Source OECD Start-ups Database, June 2025.

Table A C.4. Number and share of companies, by variables and by location

Location	Number of companies	Received financing	Received financing (%)	VC-backed	VC-backed (%)	Patents	Patents (%)	IPO	IPO (%)	Acquired	Acquired (%)	With founder information	With founder information (%)
OECD	2 011 525	336 361	16.72	234 351	11.65	122 346	6.08	15 020	0.75	92 275	4.59	605 803	30.12
Africa	53 908	8 146	15.11	5 756	10.68	134	0.25	174	0.32	1 033	1.92	14 531	26.96
Asia & Oceania	535 778	103 680	19.35	84 441	15.76	21 220	3.96	12 002	2.24	13 515	2.52	154 311	28.80
Europe	819 909	128 581	15.68	81 527	9.94	41 182	5.02	3 951	0.48	35 131	4.28	227 344	27.73
Latin America	117 111	13 095	11.18	10 295	8.79	870	0.74	241	0.21	2 492	2.13	23 337	19.93
North America	1 029 448	175 232	17.02	126 546	12.29	66 173	6.43	8 114	0.79	51 459	5.00	339 649	32.99
Missing	135 054	4 441	3.29	3 180	2.35	1 298	0.96	266	0.20	1 548	1.15	31 559	23.37

Note: This table provides the number of companies founded from 2000 and their share for multiple variables in the OECD Start-ups Database, by location.

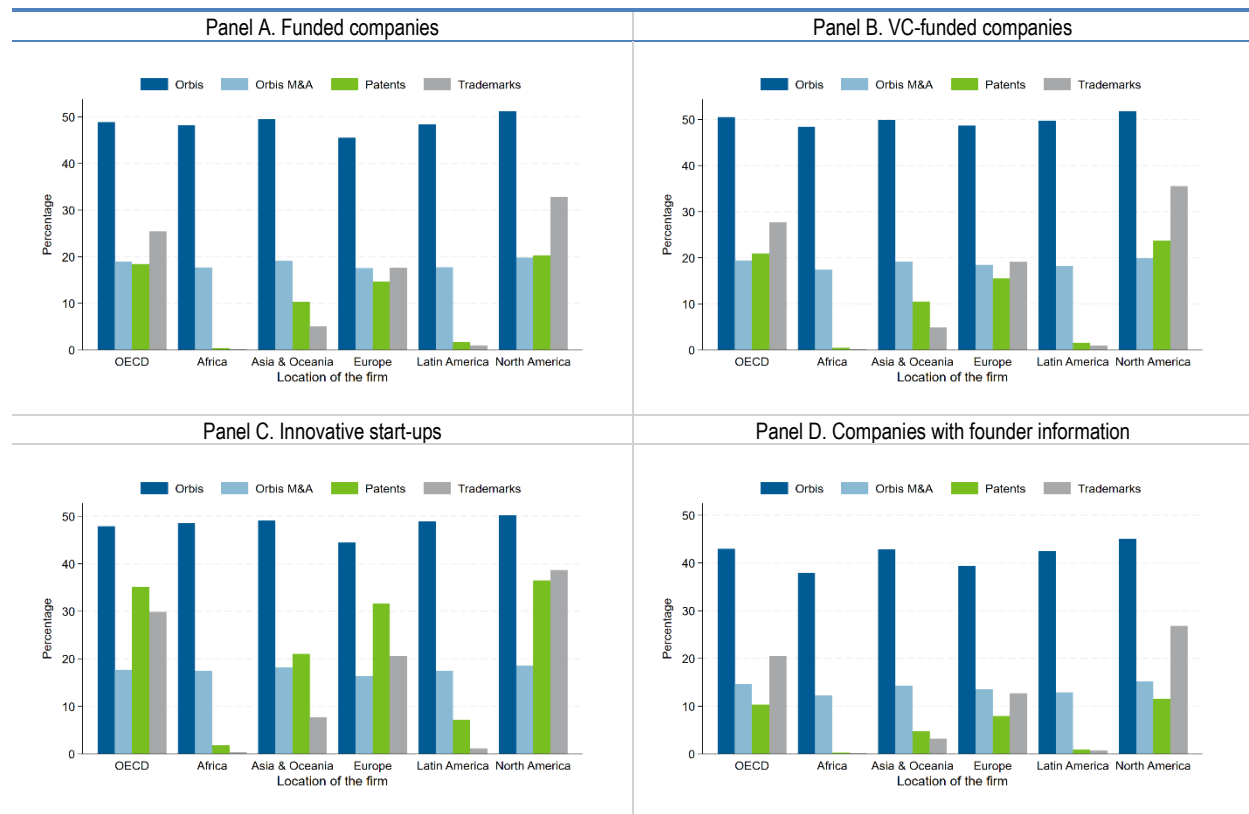
Source: OECD Start-ups Database, June 2025.

Table A C.5. Number and share of companies, by variables and by year founded

Year founded	Number of companies	Received financing	Received financing (%)	VC-backed	VC-backed (%)	Patents	Patents (%)	IPO	IPO (%)	Acquired	Acquired (%)	With founder information	With founder information (%)
Before 2000	1 143 664	111 243	9.73	23 801	2.08	115 914	10.14	31 472	2.75	79 688	6.97	126 182	11.03
2000	71 921	8 468	11.77	3 887	5.40	5 368	7.46	1 593	2.21	5 440	7.56	11 088	15.42
2001	67 479	7 441	11.03	2 962	4.39	4 956	7.34	1 342	1.99	4 995	7.40	10 965	16.25
2002	65 061	6 813	10.47	2 798	4.30	4 580	7.04	1 230	1.89	4 317	6.64	10 539	16.20
2003	71 604	7 454	10.41	3 222	4.50	4 681	6.54	1 290	1.80	4 476	6.25	11 852	16.55
2004	75 563	8 064	10.67	3 729	4.93	4 921	6.51	1 462	1.93	4 612	6.10	12 750	16.87
2005	81 503	8 944	10.97	4 312	5.29	4 968	6.10	1 526	1.87	4 777	5.86	14 277	17.52
2006	84 784	10 036	11.84	5 196	6.13	5 129	6.05	1 642	1.94	5 183	6.11	15 964	18.83
2007	91 866	11 061	12.04	6 004	6.54	5 330	5.80	1 592	1.73	5 226	5.69	18 475	20.11
2008	98 459	11 268	11.44	6 394	6.49	5 149	5.23	1 208	1.23	5 092	5.17	21 091	21.42
2009	107 654	12 767	11.86	7 715	7.17	5 552	5.16	1 144	1.06	5 362	4.98	25 014	23.24
2010	123 039	15 291	12.43	9 756	7.93	5 962	4.85	1 360	1.11	5 772	4.69	29 295	23.81
2011	118 499	17 588	14.84	12 166	10.27	5 945	5.02	1 285	1.08	5 639	4.76	32 053	27.05
2012	134 553	21 146	15.72	15 627	11.61	6 607	4.91	1 087	0.81	5 910	4.39	38 551	28.65
2013	136 904	23 331	17.04	17 666	12.90	6 769	4.94	980	0.72	5 790	4.23	41 505	30.32
2014	149 811	27 407	18.29	21 328	14.24	7 464	4.98	942	0.63	5 755	3.84	47 777	31.89
2015	162 805	30 167	18.53	23 852	14.65	7 956	4.89	871	0.53	5 588	3.43	53 389	32.79
2016	155 846	28 874	18.53	22 694	14.56	7 664	4.92	760	0.49	4 940	3.17	52 289	33.55
2017	155 140	29 526	19.03	23 001	14.83	7 438	4.79	802	0.52	4 385	2.83	54 072	34.85
2018	149 075	28 898	19.38	22 628	15.18	6 902	4.63	662	0.44	3 678	2.47	52 344	35.11
2019	137 230	26 953	19.64	21 503	15.67	6 059	4.42	431	0.31	2 929	2.13	50 770	37.00
2020	137 664	27 636	20.07	22 124	16.07	5 113	3.71	586	0.43	2 357	1.71	53 949	39.19
2021	109 495	26 152	23.88	21 455	19.59	3 566	3.26	597	0.55	1 588	1.45	39 104	35.71
2022	62 791	17 996	28.66	14 980	23.86	1 694	2.70	177	0.28	790	1.26	25 721	40.96
2023	64 560	13 032	20.19	10 879	16.85	781	1.21	102	0.16	448	0.69	31 295	48.47
2024	66 380	6 315	9.51	5 391	8.12	289	0.44	73	0.11	123	0.19	32 240	48.57
2025	11 522	547	4.75	476	4.13	24	0.21	4	0.03	6	0.05	4 362	37.86
Missing	2 059 375	28 933	1.40	9 818	0.48	23 628	1.15	2 278	0.11	23 169	1.13	61 451	2.98

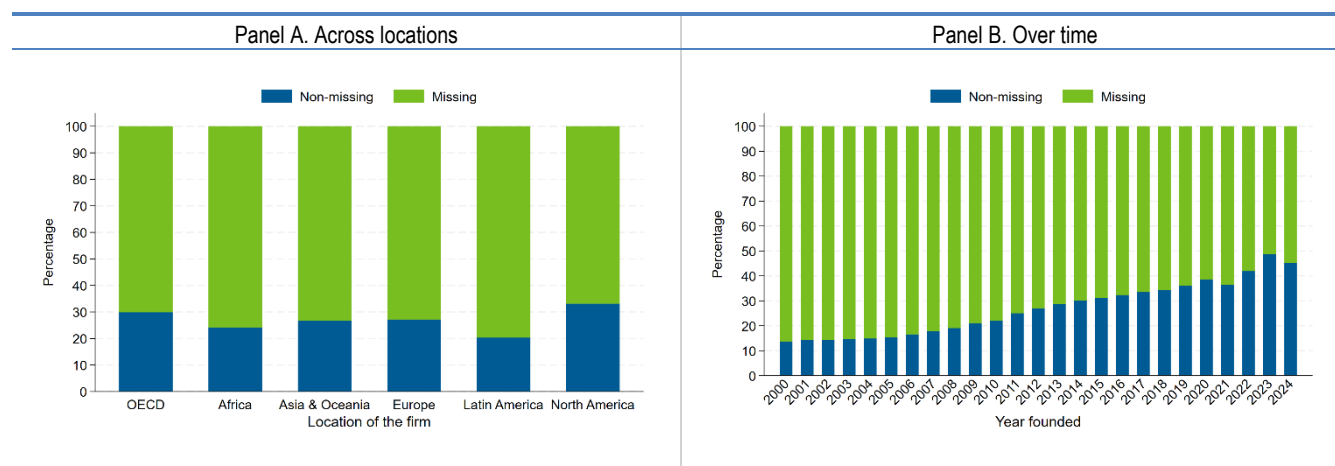
Note: This table provides the number of companies and their share for multiple variables in the OECD Start-ups Database, by the founding year.
Source: OECD Start-ups Database, June 2025.

Figure A C.1. Links to Orbis, Orbis M&A, patents, and trademarks



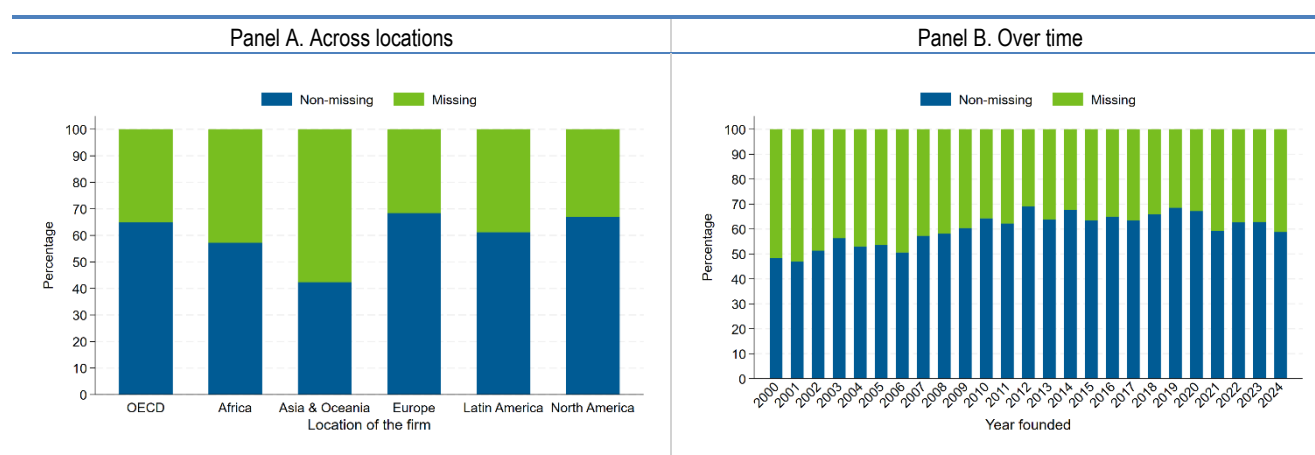
Note: This figure replicates Figure 4.3 for different groups of companies. Panel A shows the linkages to Orbis, Orbis M&A, patents and trademarks for companies that received some financing round. Panel B shows the equivalent for only VC-funded companies. Panel C restricts to those companies that had either a VC financing round or filed a patent when they were younger than 10 years old. Panel D shows companies with founder information. Trademarks percentages are based on an earlier version of the OECD Start-ups Database and may not reflect the most recent updates.

Source: OECD Start-ups Database, June 2025; Orbis and Orbis M&A, Bureau van Dijk, March 2025, and the OECD IP Database, August 2025.

Figure A C.2. Percentage of companies with founders' information across locations and over time

Note: This figure shows the percentage of companies with and without any information on the founder. Panel A shows the distribution of missing founder information across locations, while Panel B shows the distribution of missing information over time.

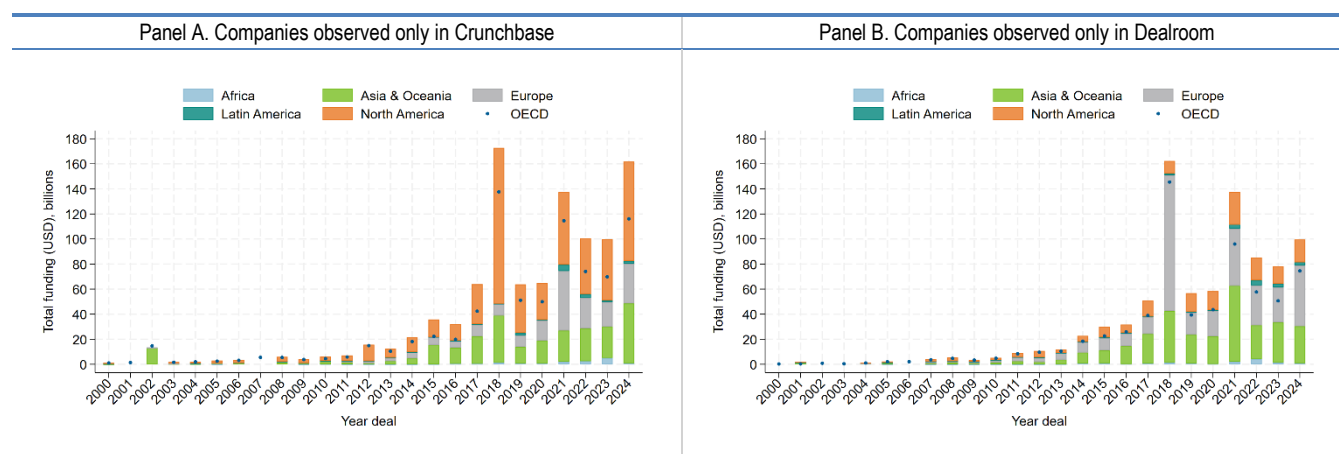
Source: OECD Start-ups Database, June 2025.

Figure A C.3. Percentage of companies with founders' information that had some VC financing across locations and over time

Note: This figure shows the percentage of companies with and without any information on the founder among those that received some VC financing round. Panel A shows the distribution of missing founder information across locations, while Panel B shows the distribution of missing information over time.

Source: OECD Start-ups Database, June 2025.

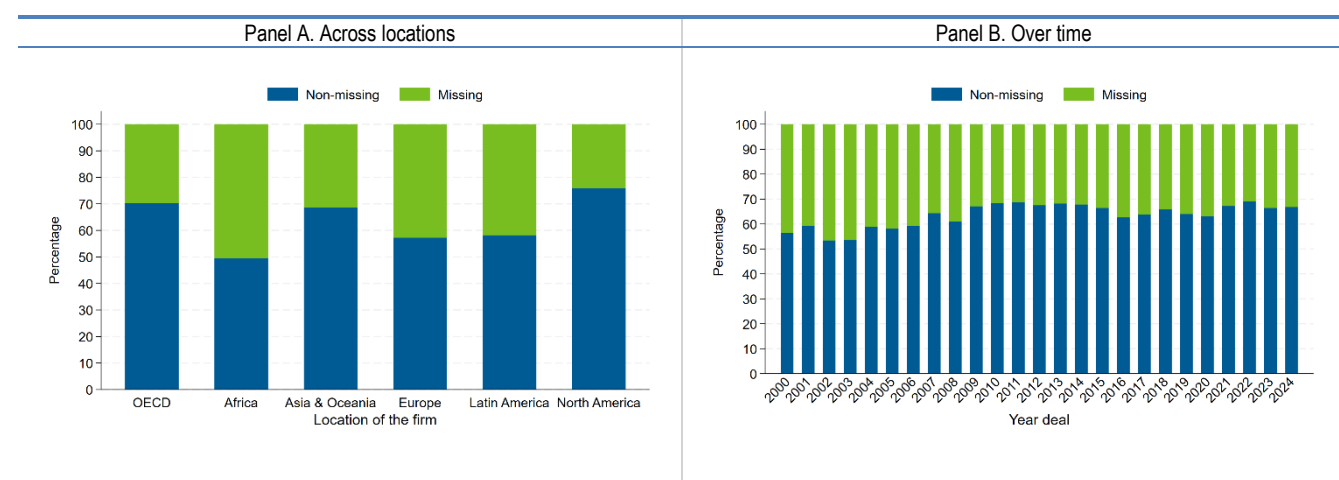
Figure A C.4. Volume of funding, by location, year, and by source



Note: This figure shows the number of deals by location and by year of the deal for deals occurring from 2000 onward. All kinds of financing are included (e.g. venture capital, grants, other kinds of financing). Panel A shows the number of deals for companies only observed in Crunchbase, while Panel B shows the number of deals for companies only observed in Dealroom.

Source: OECD Start-ups Database, June 2025.

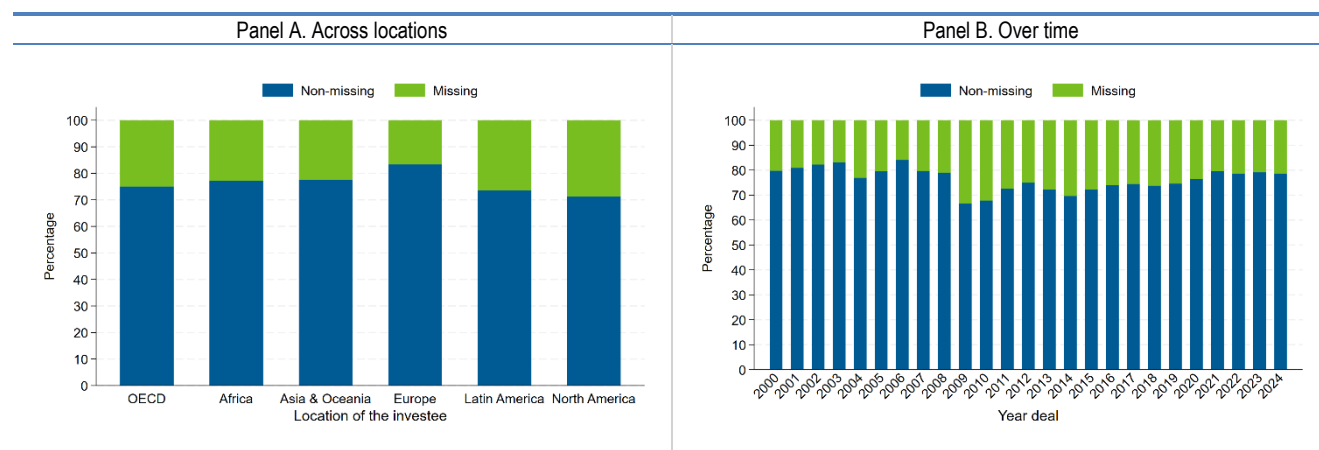
Figure A C.5. Percentage of deals with missing information on funding amounts across locations and over time



Note: This figure shows the percentage of companies with and without information on the funding amount among those that received some form of financing. Panel A shows the distribution of missing funding amounts across locations, while Panel B shows the distribution of missing funding amounts over time.

Source: OECD Start-ups Database, June 2025.

Figure A C.6. Percentage of deals with information on investors across locations and over time



Note: This figure shows the percentage of companies with and without any information on the investor among those that received some financing round. Panel A shows the distribution of missing investor information across locations of the investee, while Panel B shows the distribution of missing investor information over time.

Source: OECD Start-ups Database, June 2025.